

Part III

Monetary Policy Framework

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Chapter 6

Monetary Policy Strategic Framework

6.1. Introduction

Fundamentally, monetary policy encompasses the regulation of money and credit (in terms of volume and/or interest rates) in the economy to achieve the final target of price (and exchange rate) stability and sustainable economic growth. Through implementation, as elaborated in Chapter 3, monetary policy in the long term has a stronger effect on price stability, contrasting the short-term trade-off between inflation and economic growth. In an open economy, as explained in Chapter 4, the ability of monetary policy to achieve the target of domestic price stability and sustainable economic growth shall be affected by exchange rate stability and foreign capital flow mobility. How the monetary policy process influences various economic and financial variables prior to achieving the economic targets can be illustrated as transmission mechanisms, described in Chapter 5.

How does the central bank develop a framework to formulate and implement monetary policy in line with the strategic and operational targets? Theoretically, from a monetary policy perspective, this question is fundamentally linked to three aspects, often cited as dualities, namely (1) What is the best policy target: output stability (economic growth) or prices (inflation)? (2) What operational target or policy instruments should be applied to achieve the final target: quantity-based instruments or price-based instruments? (3) What is the best policy response: rules versus discretion? This chapter reviews various issues relating to the first duality, while the second and third dualities are presented in subsequent chapters of the book.

In other words, the discussion on the monetary policy strategy framework is inextricably linked to the presence of a final target and the relationship with the policy regime applied. This chapter will review the strategies in five sections. After this introduction, the discussion focuses on conceptual dimensions and theoretical models related to the final policy target and monetary policy regime using a nominal anchor. Several issues concerning the Phillips Curve phenomenon are also elaborated in terms of long-term linkages and the socio-economic costs that arise. In addition, the application of monetary policy strategy frameworks in various countries is also presented from a historical perspective along with policy evolution over time. The concluding remarks at the end of the chapter describe the monetary policy regime trends over the past few years.

6.2. Conceptual Dimension and Theoretical Models

6.2.1. *Final Policy Target: Output Versus Prices*

The selection of a policy strategy that prioritizes output (economic growth) or price (inflation) stability is a salient central bank issue when analyzing monetary policy that continues to this day.¹ The history of monetary policy economics shows that many central banks around the world prioritized output stabilization, especially during the 1970s and 1980s. Beforehand, price stability was the prevailing priority during the 1950s and 1960s under the auspices of the Bretton Woods Agreement. The debate over the past decade, however, has centered on the differences in views between central bankers and academics. In general, central bankers favor price (inflation) stability as the main objective, while academics lean toward inflation and real output stability. In fact, since the beginning of the 1990s, nearly all central banks have adopted price stability as the final target of monetary policy (Bofinger, 2001).

The priority between price stability and economic growth as the final targets of monetary policy, similar to general macroeconomic policy, is to improve public welfare (prosperity). Therefore, to evaluate whether a monetary policy is optimal, the main reference point is achieving the desired level of prosperity. A traditional approach to this issue is through the social welfare function. Per the ideal version, the social welfare function is very difficult to evaluate because it depends on the public's welfare preferences, which are very broad, now and in the future, as well as the alternative policies that are required to achieve maximum welfare. Over time, the approaches used have focused on the simpler objective function, in the form of a macrowelfare function, which links directly to several macroeconomic variables, such as output, employment, and prices/inflation.

A macroeconomic welfare function, which plays an important role in monetary economics as a quadratic loss function that must be minimized by the central bank, can be formulated simply based on the trade-off between price-output stabilization as follows:

$$L = \frac{1}{2} [\omega_p (\pi_t - \pi^*)^2 + \omega_y (y - y^*)^2] \quad (1)$$

The argument is, therefore, inflation deviation (π) from its target (M^*) and output deviation (y) from its potential (y^*). Fundamentally, determining the quadratic function implies that positive or negative deviation from the target is

¹The implementation of monetary policy strategy, as an integral part of macroeconomic policy, can be achieved using multiple targets, namely to achieve the final targets of sustainable economic growth, full employment, price stability, and BOP equilibrium. In practice, however, the central bank tends to focus more on efforts to achieve dual targets, namely economic growth (output stability) and price stability. This is achieved with firm confidence in the close linkages between the achievement of economic growth and more employment, as well as between price stability and exchange rate appreciation and the pursuit of external balance.

equally as harmful to the public. In this case, higher deviation receives a larger weight than lower deviation. Meanwhile, ω_p and ω_y , respectively, reflect the preferences or weight of achieving the objective based on the final target sought by policymakers, namely prices or output.

In addition to that loss function, there is another variation of the loss function, which can be expressed as follows:

$$L = \frac{1}{2}[(\pi_t + \Delta y_t) - (\pi + \Delta y)^*]^2 \quad (2)$$

The argument of that function is nominal output growth deviation (the sum of inflation and real output growth) from the nominal output growth target. Conceptually, the first loss function was primarily used in monetary policy analysis where the strategy framework/ regime was based on inflation targeting, while the second loss function was used based on nominal income targeting.

6.2.1.1. Targeting Long-term Policy Goals. Theoretically, the benefits of the respective policy strategies could be observed from the perspective of implementing monetary policy in the near term and long term. In this case, the characteristics of the economy in the near term are not assumed to be influenced by demand-side or supply-side shocks, which is consistent with general prevailing theory, namely the long-term neutrality of money. From the literature, Walsh (2001) proposed some interesting findings on the relationships between money supply, inflation, and output, namely that in the long term, the linkages between money supply growth and inflation are perfect, while the linkages between money growth or inflation and real output growth fade toward zero. The findings demonstrate consensus that, in the long term, monetary policy only affects inflation and not real output. Meanwhile, consensus in the literature regarding the short-term influence of money shows that a monetary policy shock would prompt a hump-shaped distribution in terms of real economic activities, implying that, in the near term, monetary policy affects real economic activity but the influence fades over time. Therefore, the choice of low inflation (price stability) as the single target of monetary policy is fundamentally a feasible policy goal in the long term as well.

The neutrality of money hypothesis is, therefore, less applicable if output growth or nominal income are defined as the long-term targets of monetary policy, considering nominal income is affected by real factors, hence monetary policy cannot be applied to overcome the problem of dwindling growth in the long term. In addition, it would not make sense to compensate the decline in economic growth by increasing inflation in the long run. Consequently, when determining the long-term monetary policy target, a strategy to achieve price stability would be considered more appropriate than a strategy targeting output/nominal income stability.

6.2.1.2. Targeting Short-term Policy Goals. In the near term, the monetary policy target selected, be it price stability or output stability, will be affected by shocks in the economy, irrespective of demand-side or supply-side sources. A simple analysis of the benefits of each respective policy strategy can be performed

using aggregate demand (AD)–aggregate supply (AS) equilibrium analysis, such as that conducted by Bradley and Jansen (1989). For simplicity, the policy target is defined in level form, thus the respective targets (achievement of policy targets) can be expressed as $P = P^*$ (prices) and $PY = PY^*$ (nominal income). Targeting a certain (constant) level of nominal income, for instance $PY = \bar{K}$, therefore $P = \frac{\bar{K}}{Y}$.

Supposing point a represents equilibrium with price stability (P_0) and full-employment output (Y_f) with a zero output gap. In the event of a negative demand-side shock (Fig. 6.1), which reduces investment activity, the investment saving (IS) curve would move to the left, from IS_0 to IS_1 , implying a shift in the AD curve to the left, from AD_0 to AD_1 (the central bank adopts a wait-and-see attitude at the beginning of the period). Such developments produce a new equilibrium at point b and, consistent with the decrease in prices from P_0 to P_1 , total money supply increases, reflected by a shift in the LM curve from LM_0 to LM_1 .

In the case of a strategy framework targeting nominal income, the central bank must undertake monetary expansion, which moves the LM curve to the right, to drive AD to its previous level, point a , which is the intersection between the AS curve and constant nominal income curve. That process ultimately returns to equilibrium under conditions of full employment output (Y_f) and prices P_0 . In the case of a strategy framework targeting prices, the monetary policy response will be the same as the previous policy response, namely monetary expansion to raise AD and prices to the original equilibrium. Therefore, the conflict between achieving price stability and stimulating output growth and employment can be avoided.

Meanwhile, the consequences and policy response to supply-side shocks (Fig. 6.2) would be different. In this case, a positive shock on the supply side, which would move the AS curve to AS_1 , on one hand would raise prices from P_0 to P_1 but on the other hand would also lower output from Y_f to Y_1 . That trade-off forces the central bank to strive toward achieving one of the targets, implying a decline in the other target. For example, expansive monetary policy ($M = M_f$) could strive to return output growth to its original equilibrium, Y_f , at the expense of driving up prices in P_f . If lower prices are sought, however, contractive monetary policy would be required thereafter.

Under such conditions, applying a monetary policy framework targeting prices would undermine the desired result after driving further output declines. Meanwhile, a policy framework targeting nominal income would facilitate a compromise, or flexibility, in terms of achieving the dual targets, namely by striking an optimal balance between higher prices and lower output. In this case, nominal income may be targeted through monetary policy that is not too tight ($M = M_b$), thus the rise in prices would not be too steep ($P = P_b$), while a drastic decline in output would also be prevented ($Y = Y_b$).

The prominent policy implication of the analysis above is that the final target of monetary policy, whether it be price stability or output growth, could be achieved by setting policy target priorities for different horizons. In this regard, price stability is more appropriate as the policy target in the long term, while nominal

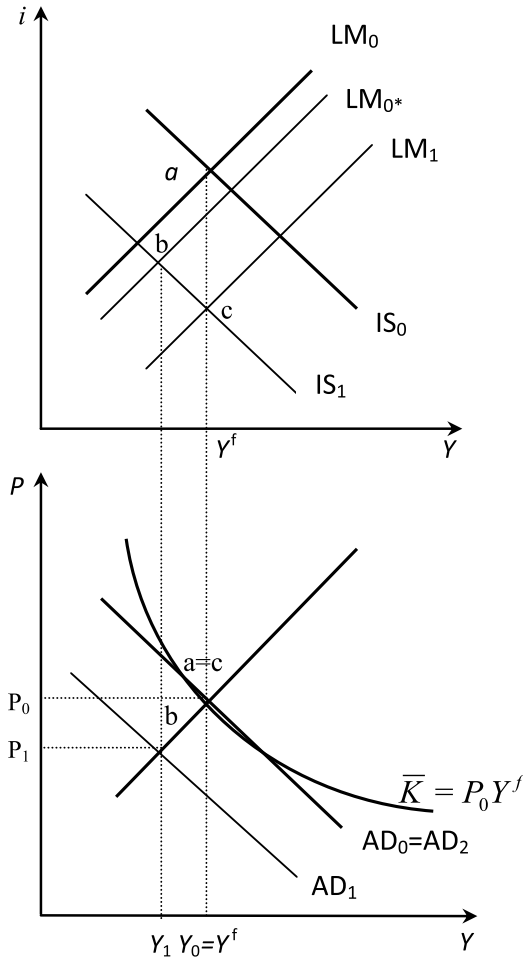


Fig. 6.1: Demand-side Shocks: Targeting Prices and Nominal Income.

income stability could be applied as the near-term policy target. Although, theoretically, the dual targets are a distinct possibility, this is extremely hard to apply in practice, considering the difficulties encountered when identifying the cut-off point between the short and long term as well as the policy response to the prevailing shocks.

If nominal income is targeted (in the near term) the central bank will confront difficulties when gauging the appropriate monetary policy response to extremely short-term shocks, particularly on the supply side. In addition, targeting nominal income would also erode central bank policy effectiveness in terms of anchoring inflation expectations. In general, therefore, central banks have applied a pragmatic solution in the form of achieving the long-term policy target of price

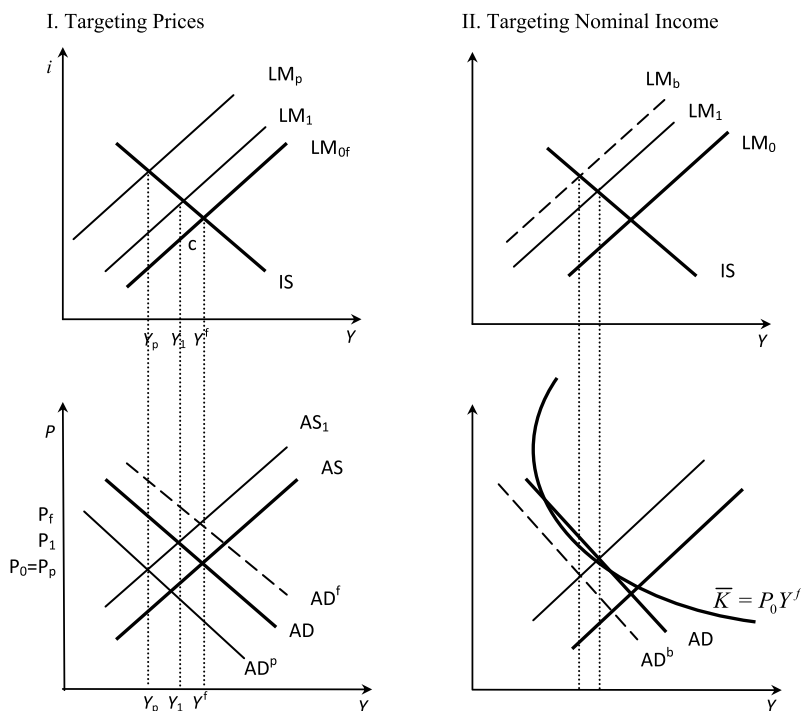


Fig. 6.2: Supply-side Shocks: Targeting Prices and Nominal Income.

stability, defined flexibly considering the possibility of shocks in the near term. Ignoring that problem for a moment, however, there are still advantages and disadvantages to each strategic policy target.

6.2.2. Monetary Policy Regime

In principle, there are several monetary policy implementation strategies available to the central bank to achieve the final policy target. The fundamental difference between the strategies lies in the different indicators used as nominal anchors or intermediate targets to achieve the policy goal.² In the literature, strategy selection depends on the corresponding monetary policy regime applied.

²In the literature of traditional monetary economics, the monetary policy framework is generally translated through the relationships between instruments, operational targets, intermediate targets, and final targets. Over time, congruent with greater use of the interest rate as the operational target, with the direct influence of the final policy target, the presence of intermediate targets has become less relevant. Therefore, the focus has centered more on nominal anchors.

According to Bordo and Schwartz (1999), the monetary policy regime is

a set of monetary institutions and regulations, accompanied by public expectations concerning the policymakers' measures on one hand and the policymakers' expectations regarding the public's reaction to the policy measures taken on the other.

Per Mishkin (1999), however, the main characteristic of all monetary regimes is the use of nominal anchors in various forms. In addition, the success of different monetary policy regimes depends on steering the policy-making measures toward a certain condition in such a way that long-term price stability is the target that must be achieved.

That definition has nearly the same understanding as the strategy or standard targeting rule. Utilizing the targeting strategy is believed to have a number of benefits in the form of commitment to limit discretion in policy decisions and, thus, inflation bias.³ Another benefit is to protect economic uncertainty from the political cycle. Therefore, the sound characteristics of targeting strategy will consider several aspects, including: (1) optimization, namely the best compromise between output and price stability; (2) stability, namely unforeseen contingency proof; and (3) simplicity, namely that the targeting strategy is best achieved transparently and accountably.

Several targeting strategies are practiced at central banks, including: (1) exchange rate targeting, (2) monetary targeting, (3) inflation targeting, and (4) no explicit anchor targeting. In addition to those four targeting strategies, there is also a hypothetical targeting strategy that has not yet been put into practice, namely: (5) nominal income targeting.⁴ The following sections present a brief overview of each monetary policy strategy.

6.2.2.1. Exchange Rate Targeting. There are three alternative approaches to exchange rate targeting as a monetary policy strategy. *First*, by pegging the value of the domestic currency to certain commodities recognized internationally, such as gold (known as the gold standard). *Second*, by pegging the value of the domestic currency to the hard currencies of larger countries with low inflation (known as the currency board system). *Third*, by pegging the value of the domestic currency to the currencies of specific other countries, where the domestic currency is allowed to fluctuate in line with the differences in inflation between the two countries (crawling peg). In Asia, the second alternative is applied by Hong Kong and Brunei, while Singapore applies the third alternative.

³Representing a situation when, because of growth-oriented economic policy, real output does not increase, while inflationary pressures accumulate. In general, inflation bias is linked to time inconsistency in terms of policy-maker behavior. This issue is discussed further in Chapter 10 of this book.

⁴For a more in-depth description of empirical observations concerning monetary policy implementation strategy in several jurisdictions, refer to Mishkin (1999).

Whichever alternative is selected, exchange rate targeting credibility depends on monetary authority discipline in terms of maintaining domestic money supply and credit to remain consistent with optimization of the exchange rate formation process. Similarly, fiscal policy discipline is also required to manage the budget deficit in line with macroeconomic stability. Flexibility in the real economy, in terms of production and labor, is also required to overcome the external shocks faced. Consequently, this regime is typically embraced by small and very open economies in terms of trade and services, such as Singapore and Hong Kong.

Exchange rate targeting also has the following key advantages. *First*, exchange rate targeting may dampen inflation originating from changes in import prices of goods. *Second*, exchange rate targeting may anchor public inflation expectations. *Third*, exchange rate targeting is a rules-based monetary policy strategy that ensures monetary policy implementation discipline. *Fourth*, exchange rate targeting is sufficiently uncomplicated and unambiguous, and thus easy to understand by the public.

In addition to the advantages, exchange rate targeting also contains some inherent weaknesses. *First*, when exchange rate targeting is pursued in a very open economy with foreign capital mobility, monetary policy cannot be implemented independently.⁵ *Second*, exchange rate targeting may cause each structural shock in certain jurisdictions to be transmitted or impact domestic economic stability directly. *Third*, exchange rate targeting is vulnerable to speculative attacks on domestic currency holdings. *Fourth*, in the event of a balance sheet mismatch, and if there is a need to devalue with a high position of external debt, this strategy will trigger fragilities and a possible financial crisis, which could subsequently squeeze fund disbursements in the financial system.

6.2.2.2. Monetary Targeting. In many countries, exchange rate targeting is not the primary choice of monetary policy strategy because of the requirements mentioned above, including monetary policy restrictions or flexibility to meet the domestic economic goals. Consequently, a number of jurisdictions favor monetary targeting, namely by stipulating total money supply growth as an intermediate target, for instance, narrow money (M1), broad money (M2) as well as credit. The main advantage of monetary targeting over exchange rate targeting is the possibility of independent monetary policy, thus allowing the central bank to focus on the targets set, such as low inflation and sustainable economic growth. In addition, as with exchange rate targeting, monetary targeting is completely transparent and allows the public to instantly know the monetary policy stance adopted by the central bank. Such signals are expected to anchor public inflation expectations to future inflation and, hence, alleviate inflationary pressures.

Nonetheless, this strategy has an inherent weakness in the form of dependence on a stable relationship between monetary aggregates and the final policy target (price and output), in addition to exchange rate stability, which may not be

⁵Departing from the concept of institutional independence, independence in this context relates to the extent of monetary freedom or neutrality in terms of shocks originating from the external sector.

maintained. Increasingly developed financial instruments and greater integration between the domestic and global economies has disrupted the stability of the relationship between monetary aggregates and the final policy target, as reflected by income velocity instability. Furthermore, if standard monetary targeting is applied without a feedback mechanism or economic response, the strategy would be unable to achieve output stability despite achieving price stability. Such conditions are the main reasons central banks cite for not applying the standard approach (with a trade-off between credibility and flexibility) or even for abandoning the strategy all together.

6.2.2.3. Inflation Targeting. As the relationship between monetary aggregates and the final target of monetary policy faded, many countries began to adopt inflation targeting, which is implemented by publishing the medium-term inflation target and committing to achieve price stability as the long-term goal of monetary policy. By targeting inflation as the nominal anchor, the central bank is more credible and focused on achieving price stability as the final target.⁶

Although inflation is targeted, this strategy does not neglect the attainment of other monetary policy goals, such as output and employment. In this case, the central bank always considers output stability and employment (at a certain level) in the near term. In addition, to minimize any output decline, the central bank periodically hones the medium-term inflation target toward a lower long-term inflation target. Nonetheless, if there is a conflict between inflation and output, inflation is the priority.

The main advantage of this strategy, in addition to a focus on domestic interests (namely inflation), is policy transparency that reduces uncertainty as well as accountability to prevent inflationary pressures. In contrast, a salient weakness of the strategy is that inflation becomes a policy target rather than an intermediate target, which is difficult to control. Furthermore, there is a lag in the monetary policy transmission mechanism, consequently inflation targeting does not provide an immediate signal concerning the direction of monetary policy. A more in-depth description of this framework is presented specifically in Chapters 8 and 9 of this book.

6.2.2.4. No Explicit Anchor Targeting. Seeking to achieve the desired economic performance, monetary policy strategy can be implemented without an explicit anchor but still paying attention and providing commitment to achieve the final target of monetary policy, including low and stable inflation. This regime provides flexibility to the central bank when determining which target to prioritize in a given period, whether it be inflation, economic growth or creating job opportunities, depending on the prevailing economic dynamics and challenges. Additionally, this strategy is forward-looking, implying that monetary policy is applied by prudently monitoring the signals of indicators for the final policy target, accompanied by anticipatory measures.

⁶A more in-depth description of the ITF is presented in Chapters 8 and 9 of this book.

Although successfully applied in several jurisdictions, this strategy is considered less transparent, hence the public tend to make assumptions regarding the purpose and goal of policy issued by the central bank, which could trigger market uncertainty concerning the output and price outlook. The ambiguity of the strategy could also undermine central bank accountability in the eyes of the public and parliament due to the lack of success criteria, which are typically announced beforehand.

6.2.2.5. Nominal Income Targeting. Fundamentally, targeting nominal (output) income is based on the view that monetary policy can only affect nominal income growth rather than its determinants in the form of inflation and real output growth. In addition, a lack of comprehensive understanding regarding the factors influencing those two determinants implies that the central bank should be oriented toward achieving a certain level of nominal income growth, not controlling inflation and real output separately.

According to McCallum (1988), although this targeting strategy has never been put into practice, counterfactual experiments have shown that this strategy might be superior to the others. Several empirical studies (Hall & Mankiw, 1993; McCallum, 1988) have shown that if this strategy was applied in the past, inflation and output would have hypothetically performed more smoothly than the historical trends. From a theoretical point of view, nominal income targeting is an alternative strategy garnering widespread attention.

6.3. Empirical Studies and Related Issues

The priority between inflation and output as the final target, as well as the most appropriate monetary policy regime for a particular country will depend on the specific economic dynamics prevailing at a given time. In this regard, numerous studies have been performed to investigate the relationship between inflation and economic activity in the short and long term. This is where the debate emerges, not only because inflation does not always impact long-term output adversely as theory would predict, but also because the short-term relationship between the two is not always consistent over time in one country or between jurisdictions at the same time. In addition to the structure and flexibility of the real economy, the sources and magnitude of shocks in the economy, demand-side or supply-side, also have an effect. Furthermore, the trade-off between inflation and output is also influenced by the policies instituted by the government and monetary authority.

Consequently, when formulating the monetary policy strategy framework, the achievement of price stability and output growth are typically sought in tandem, paying due consideration to the degree of flexibility between both in the near term, while prioritizing price stability in the long term. According to Fischer (1996), several empirical facts point to a trend of giving the central bank responsibility for controlling inflation. Nevertheless, when implementing that mandate, the central bank also considers economic growth because monetary policy can indeed influence output, at least in the near term. Furthermore, the understanding of economic players is based more on monetary policy signals and their effect

on economic growth. In this regard, it is important to understand that monetary policy support for economic growth will be best if the central bank consistently maintains low inflation and supports financial system stability.

6.3.1. Short-term Output-prices Trade-off and the Phillips Curve Phenomenon

The empirical study of Phillips (1958) concerning the trade-off between output and price stability represents seminal findings that remain pertinent to the present day, namely the Phillips Curve. The study demonstrated the trade-off between inflation and output, which manifests as deviation from the long-term trend, although this trade-off is comparatively weak in the near term. As explained in Chapter 3, the empirical evidence for the Phillips curve subsequently became the foundation of the synthesis between Classical economics, which opines that monetary policy only influences inflation in the long term, and Keynesian economics, which believes monetary policy can effectively drive economic growth. In several ways, this study underlies monetary policy strategy to actively stabilize output, a regime implemented in several countries during the 1970s and beginning of the 1980s. Nevertheless, in the 1980s, the pendulum swung toward monetary policy with more of a focus on price stability.

The description above shows how important empirical studies of the trade-off between inflation and output are in terms of underlying the monetary policy strategy framework. This is primarily linked to three salient issues underlying the existing research, namely: (1) Whether the relationship or trade-off can be proven empirically and how the relationship or trade-off occurs in the short and long term? (2) Whether the relationship or trade-off is linear (symmetrical) or non-linear (asymmetric)? (3) What are the implications of the relationship or trade-off in terms of implementing economic policy?

6.3.1.1. Presence of the Phillips Curve. The first issue relating to the emergence of empirical facts is stagflation, namely low economic growth coupled with high inflation, as experienced by industrialized nations in the 1970s, who opposed the trade-off shown in the Phillips curve. Stagflation simultaneously demonstrates that, in the long term, the trade-off in the Phillips curve is not always supported by consistent empirical evidence and that high inflation is not necessarily a constraint to robust growth. In fact, research in the past decade (Barro, 1995; Fischer, 1993; Ghosh & Phillips, 1998) concluded a significant inverse correlation or trade-off between output growth and inflation, implying that prevailing economic conditions and the policies implemented will also influence the relationship between inflation and output.

In the context of multivariate relationships, by including demand-side variables, such as money supply and government spending, as well as supply-side variables, such as productivity, the fundamental relationship between output growth and inflation is extremely complex. This is possibly because of the variable interactions and mutual interconnectedness between the determinants of inflation and determinants of output growth. Therefore, it can be concluded that even if the close relationship between inflation and output growth is present, the

correlation is more complex than a linear relationship with a constant parameter (monotonic). In other words, although the Phillips curve could be used as a reference when prioritizing inflation and economic growth as the final targets of monetary policy, the trade-off between the two and the strategy to attain the target is not easily explained theoretically.

6.3.1.2. Linearity in the Phillips Curve. The second issue is based on the findings of Laxton, Meredith, and Rose (1995) concerning the asymmetric effect of economic activity on inflation in seven major OECD countries. The study provided an important conclusion, namely that there is no linear or constant trade-off between economic activity and inflation in the long term. Specifically, strong evidence for the non-linear relationship between inflation and output in the Phillips curve is reflected in the stronger impact of excess demand shocks in terms of driving inflation rather than excess supply shocks in terms of dampening inflation. Several empirical studies have also provided evidence for the non-linearity of the Phillips curve, including the research of Clark, Laxton, and Rose (1996), Debelle and Laxton (1997), and Fisher, Mahadeva, and Whitley (1997).

A number of policy implications may be drawn from the empirical studies mentioned above. *First*, policymakers may align the achievement of output and price stability in the long term more optimally by separating the effect of demand-side shocks and supply-side shocks through supply-management policies, such that achieving robust output growth in the long term would not exacerbate inflationary pressures that could culminate in economic losses. *Second*, the non-linear relationship between economic activity and inflation has an important implication for demand-management policies, namely that in response to demand-side shocks, and to maintain control over inflation, average real economic growth must be maintained below its potential. Consequently, if real output exceeds its potential rate, the central bank must accelerate its monetary policy response to control the early symptoms of inflation, namely a shock in the form of an increase in AD.

6.3.1.3. Formation of Expectations in the Phillips Curve. During development in the 1990s, studies of the Phillips curve tended to focus more on the effect of forming rational expectations on the trade-off between inflation and output per the Neo-Keynesian theoretical framework. Departing from the traditional version of the Phillips curve, the new iteration predicted inflation to have a positive correlation with the output gap when forming forward-looking inflation expectations. This proposition was based on analyzing the heterogeneity of nominal price formation between producers (staggered price setting) inspired by the study conducted by Taylor (1980). The general specification was developed by Calvo (1983), who showed the Phillips curve with forward-looking expectation formation, where the current rate of inflation is linked to future inflation expectations and the marginal cost or output gap. Therefore, inflation in the previous period is not relevant when determining inflation in the current period, implying that while the central bank remains committed to stabilizing the output gap, in other words maintaining economic growth near potential output, the economy could mitigate inflation itself at a low and stable rate. Expressed in a different way, there is no longer a stable trade-off between prices and the output gap.

The problem is that not all economic players act rationally and always form future expectations. The formulation of expectations in the Phillips curve, be they forward-looking or backward-looking, is the subject of another debate. As stressed by Fuhrer and Moore (1995), the dynamic behavior of inflation and output in the forward-looking specification appears awkward if linked to the empirical evidence. That specification implies that inflation anticipates the output gap but the empirical evidence points to the opposite, namely that the output gap tends to drive inflation.⁷ In this regard, the empirical evidence appears consistent with the traditional version of the Phillips curve. Meanwhile, the latest studies from Sbordone (1999), Gali and Gertler (1999), and Gali, Gertler, and López-Salido (2000) provided significant evidence of forward-looking natural inflation behavior as well as linkages between inflation and the real marginal cost or output gap. Regarding both phenomena, Gali and Gertler (1999) and Gali et al. (2000) proposed a hybrid Phillips curve model, namely one that also considered certain fractions of backward-looking expectations in addition to forward-looking expectations in the relationship between inflation and real output.

In Indonesia's case, evaluating and testing the empirical model produced several significant findings that could be concluded as the Phillips curve phenomenon, where the presence and behavior evolved over time in line with changes in the structure of economic fundamentals, particularly due to the economic crisis in 1997 (Solikin, 2004). Specifically, the formation of expectations and linearity of the Phillips curve experienced a significant change between the pre- and post-crisis periods. Pre-crisis, the characteristics of the Phillips curve showed that: (1) the effect of output gap pressures on inflation was moderate (0.2–0.3); (2) forward and backward-looking expectations were relatively well balanced; and (3) the relationship tended to be linear. In contrast, the post-crisis characteristics of the Phillips curve showed that: (1) the effect of output gap pressures on inflation tended to increase (0.4–0.6); (2) forward-looking expectations became dominant; and (3) the relationship tended to be non-linear (convex). The empirical findings showed how crisis conditions have exacerbated the trade-off between inflation and economic growth, which must be considered in the monetary policy strategy framework.

6.3.2. Output-prices Trade-off in the Economic Cycle

Studies into co-movement between output and prices have also become an area of focus when analyzing the economic (business) cycle. One objective of such research is to seek evidence concerning the types of structural shocks that influence comovement. Previous observations have shown that price developments are procyclical in terms of output (Lucas, 1977; Mankiw, 1989), consistent with analysis into the dominance of demand-side shocks in the economic cycle. This is also congruent with the standard Phillips curve findings that provide a rational basis for positive comovement between real output and inflation. On the other hand,

⁷At least if using detrended log GDP as a proxy of the output gap.

however, Cooley and Ohanian (1991) suggested that, fundamentally, the trade-off between inflation and output depends on the relative effect between two shocks, namely demand shocks and supply shocks. In this context, demand shocks cause output and prices to move in the same direction, while supply shocks cause output and prices to move in opposing directions. That price developments explain economic cycle procyclicality was also refuted by Kydland and Prescott (1990) because the correlation between output and prices after World War II was generally inverse, or positive but weak.

Several follow-up studies showed that the trade-off between output and prices could be explained in the standard model assuming sticky prices and only demand shocks (Ball & Mankiw, 1994; Chadha & Prasad, 1993; Judd & Trehan, 1995). Judd and Trehan (1995) showed that the correlation coefficient between inflation and output depended not only on the magnitude of the relative effect between demand and supply shocks but also the dynamic response of each respective variable to the shocks. In addition, Rotemberg (1996) found that the inverse correlation between projected output and prices was consistent with the simple sticky price model using only demand shocks. Nevertheless, the trade-off between inflation and real output is difficult to analyze based on a model assuming flexible prices, considering that that would require different money supply behavior in response to real shocks, forecastable or not.

The intuition of the description above is as follows. Under conditions where the immediate response of prices is slow (sticky prices), nominal (and positive) demand shocks would edge up output in the near term but the trend would not be maintained in the long run. Real output would, therefore, fall back to the steady state value after spiking in the near term because expansive monetary policy would spur demand-side shocks. On the other hand, the response of higher prices to such demand shocks would not appear in the near term after expansive monetary policy because of sticky prices. Nonetheless, the price increase would only become permanent in the long term with output surpassing its long-term trend. Such post-shock economic cycle dynamics explain the trade-off between output and prices in the long run.

Motivated by previous findings, Den Hann (2000) analyzed the comovement of output and prices in post-WWII United States using a different strategy. He argued that the main cause of uncertainty concerning the studies on comovement stemmed from the use of just one correlation coefficient. In his study, Den Hann applied a vector autoregression (VAR) approach to analyze the correlation function of forecast errors over different horizons. At least two important findings were concluded from the study. *First*, using the statistical data proposed, Den Hann found that comovement between output and prices were positive in the near term and negative in the long run. *Second*, Den Hann showed that the results could be explained using a model where demand shocks were dominant in the near term and supply shocks were dominant in the long term. Therefore, this study provided empirical evidence for the relative effect of demand and supply shocks on the correlation between real output and inflation in the economic cycle in the theoretical models developed previously by Chadha and Prasad (1993) and Rotemberg (1996).

6.3.3. *Sacrifice Ratio: Cost of Controlling Inflation*

In general, the main objective of monetary policy is to achieve low and stable inflation, which is expected to contribute to sustainable economic growth in the long term. A related practical question, therefore, is why the central bank does not reduce inflation drastically? The answer is because disinflationary policy may prompt a more significant decline in real output and, therefore, a large sacrifice ratio for the sustainability of the economy in general. For example, disinflationary policy by significantly slowing the growth of money supply would drive up interest rates and trigger a recession.

The central bank dilemma of controlling inflation at the cost of lower output is a controversial topic in economics. The traditional view states that lower inflation would cause a recession, as mentioned above. Opposing that viewpoint, however, Neo-Classical economists believe that the cost is negligible or not even present. According to Sargent (1983), the inflation cost could be negated if anticipated. For example, the central bank could announce its intention to reduce the supply of money supply to subdue inflationary pressures. If the economic players could anticipate that information, the response would be a decline in the rate of price change in the same period, hence real money supply would not change and the recession could be avoided. Even if the decline in money supply growth was not anticipated, if businesses could adjust prices quickly, the effect on output would be comparatively small and quick to dissipate.

Therefore, it can be concluded that the effect of disinflationary policy depends on the fundamental assumptions used, namely: (1) whether the change in monetary policy could be anticipated, meaning whether monetary policy could be announced clearly and whether that announcement is trusted by the markets; and (2) whether the adjustment process to shocks could be implemented quickly enough. Using different assumptions and approaches, several empirical studies have been conducted to observe the inflation cost of disinflationary policy.

Fundamentally, the cost of disinflationary policy can be measured several ways. Using a simple concept, namely the sacrifice ratio, Okun (1978) was the first to measure the sacrifice cost, in the form of output loss, of disinflationary policy based on an assessment of the Phillips curve. Using a back-of-the-envelope approach based on estimates of the linear Phillips curve parameters, Okun (1978) concluded that the sacrifice ratio is 10, implying a 1% decline in inflation would prompt a corresponding 10% decline in economic growth. This concept is illustrated in Fig. 6.3.

Meanwhile, Gordon and King (1982) applied a more complex approach in the modeling structure for inflation, output and the linear Phillips curve, concluding that the sacrifice ratio was much lower than that proposed by Okun at just 3. On the other hand, Ball (1994) criticized potential problems in the assessment based on the linear Phillips curve and estimated the sacrifice ratio using an atheoretical approach. By measuring directly the output loss in periods when inflation decelerated, Ball concluded that the sacrifice ratio was 2.3.

Relevant studies have also tackled the non-linear Phillips curve, including Clark et al. (1995), Laxton et al. (1995), and Filardo (1998). The empirical

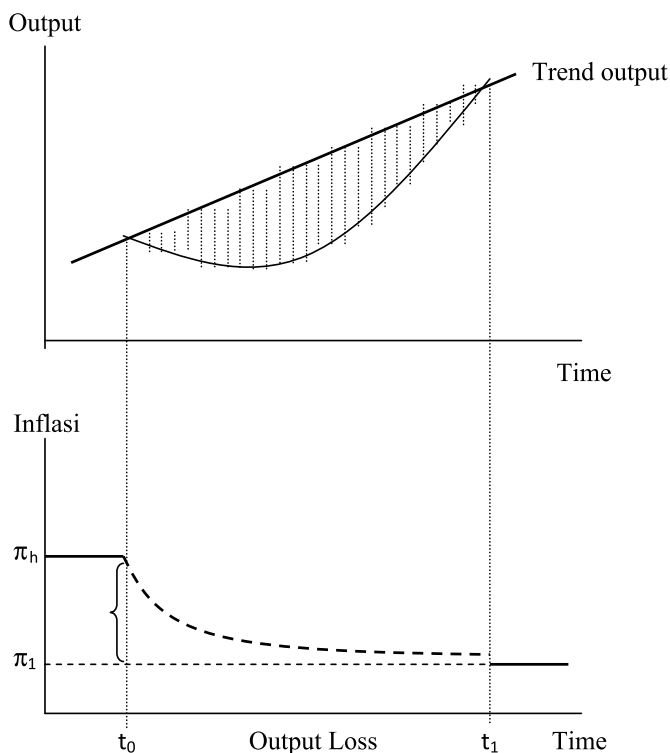


Fig. 6.3: Output Loss Linked to Disinflationary Policy.

facts concerning the non-linear correlation in the Phillips curve created additional complexity when estimating the sacrifice ratio. Christiano, Eichenbaum, and Evans (1994) estimated the sacrifice ratio using a VAR modeling system approach, where the ratio was defined as the ratio of output response relative to the inflation response in the event of policy rate shocks. This involved complex estimations due to dependence on a change in policy regime, whether contractive (output below the trend) or expansive (output above the trend). Using structural equations that took into consideration the piecewise non-linearity of the Phillips curve, Filardo showed that the sacrifice ratio during an economic downswing was 5.0, compared to 2.1 when the economy was overheating. Nevertheless, Filardo's figures differed from the sacrifice ratio estimated using a linear Phillips curve equation, namely 5.7.

6.4. Implementation in Several Jurisdictions

The evolution of the central bank's role and reforms over time was elaborated in Chapter 2, showing how monetary policy is influenced by economic dynamics, the political configuration, and academic thinking. Therefore, the monetary policy

strategy framework applied by central banks has also experienced evolution over the years. The pendulum of the final policy target swung from economic growth to inflation, while the monetary policy regime has also evolved from exchange rate targeting to inflation targeting.

6.4.1. Final Target of Monetary Policy

How is monetary policy practiced around the world to ameliorate public welfare? Is the focus on price stability or also directly linked to stimulating economic growth? The following section describes how political preferences also influence the selection of the final target of monetary policy.

6.4.1.1. A Shift in Priorities from the 1970s to 1980s. Economic dynamics around the world in the 1970s were characterized by the desire to stimulate economic growth and create job opportunities, consistent with the goal of economic development at that time, namely post-WWII recovery in advanced countries and post-independence economic advancement in developing countries. In general, the central bank was still under government control as a member of the cabinet and/or the formation of a monetary board. Consequently, although the legal and institutional frameworks (laws) of the central bank pointed to several monetary policy final targets, many central banks around the world in the 1970s prioritized stimulating economic growth and creating job opportunities rather than controlling inflation and the balance of payments (BOP). In other words, monetary policy was viewed as an instrument of output stabilization policy at that time. This is congruent with Keynes' view developed based on observations of the UK economy after the Great Depression in the 1930s. Output stabilization policy and monetary policy were supported by the correlation between output and inflation per the Phillips curve, which meant the central banks had to determine the desired trade-off in terms of stimulating economic growth on rising inflation that was politically tolerable.

Thereafter, in the 1980s, a fundamental change was witnessed as the pendulum of the central bank's final policy target swung again, this time from economic growth to price stability. Using monetary policy as a pro-growth instrument in the 1970s proved to be a failure because the economy did not grow but inflation soared, leading to stagflation in industrialized nations in 1973–1974, which was followed by soaring inflation in 1979–1980 as the global oil price skyrocketed. Consequently, central banks once again prioritized controlling inflation. Despite remaining under government control, central banks began to consolidate their control over money supply, including for government fiscal financing, which had previously prioritized stimulating economic growth.

Reprioritizing the final target of monetary policy back toward explicitly controlling inflation began in the *UK*. By applying a medium-term policy strategy, monetary policy at the beginning of the 1980s sought to control inflation and support sustainable economic growth and job creation. In that regard, the implementation of extremely tight monetary policy in the 1980s fed through to pound sterling appreciation and triggered an economic recession. The tight monetary policy stance did, however, strengthen monetary policy credibility in terms of

easing wage-push inflationary pressures on the labor market that were previously uncontrolled, while the impact on the financial sector was subdued by government policy. The recession experienced at the beginning of the 1980s ultimately created solid anti-inflation policy credibility in the UK.

In the *United States*, drastic measures to tighten monetary policy in terms of controlling inflation were generally avoided throughout the 1970s because of the huge sacrifice cost involved in the form of declining output.⁸ Consequently, inflation was unacceptably high throughout the 1970s but such conditions were acceptable socio-politically. Radical measures were only implemented in 1979, when the Federal Reserve decided to end accommodative monetary policy. That move triggered a deep recession, with unemployment soaring to around 11% at the end of 1982. Seeking to overcome the issue, the US Administration instituted a number of policies, including expansive fiscal policy and a loose monetary policy stance. Supported by greater wage flexibility, unemployment was eventually reduced gradually. In general, controlled inflation and lower unemployment in the US became a less prominent issue concerning the shift in the final target priority of monetary policy than in the UK and other European countries.

In *Germany*, unemployment was high in the 1980s, increasing two-fold on the level recorded in the 1970s. Despite such conditions, the central bank was not inclined to implement policy to boost demand, although inflation remains very low, approaching zero. Such conditions demonstrated good monetary policy discipline in terms of controlling inflation in Germany. Meanwhile in *France*, although a platform to explicitly reduce unemployment was introduced at the beginning of the 1980s, high unemployment remained, exceeding 6.6% in 1980. Once France decided to join the European Monetary System (EMS) in 1982, the country also committed to controlling inflation to reduce inflation disparity with other EMS members. In fact, stimulating economic growth and creating job opportunities were temporarily set aside in terms of the priority final target of monetary policy. Compared to other advanced countries, in the middle of the 1970s, Japan successfully maintained a relatively low level of unemployment, while inflation simultaneously experienced a sharp decline. At the end of the 1970s and throughout the 1980s, marginal increases in the level of unemployment were observed, albeit moderate, to a level of around 3%.

For *developing countries*, however, the issue of shifting the final target of monetary policy from economic growth or job creation to controlling inflation was not a core issue and therefore did not receive much attention during 1970s–1980s. In general, the policy orientation of developing countries focused on creating a climate conducive to production and employment. Even if controlling inflation was considered, this was achieved jointly in the context of sustainable economic growth and job creation. In this regard, the strategy to determine the final target

⁸From the study by Arthur Okun (1978), the cost of disinflationary policy (sacrifice ratio) was estimated at 10:1, implying a 10% decline in real output would be required to lower inflation by 1%. A full description of the sacrifice ratio is presented in this chapter.

of monetary policy was conducted without explicitly prioritizing a certain target. Such developments were reflected in the relatively high pace of economic growth recorded in developing countries, including *Indonesia, Thailand, and Malaysia* at the end of the 1980s through to the beginning of the 1990s. Meanwhile, although not achieving a particularly low rate, (moderate) single-digit inflation was achieved and maintained.

6.4.1.2. From the 1990s Until the Present Day. Since the 1990s to the present day, nearly all central banks around the world have adopted price stability as the final target of monetary policy. In fact, most central banks have formally applied a monetary policy framework with controlling inflation as the final target. This phenomenon began with the proliferation of the Inflation Targeting Framework (ITF) during the 1990s at many central banks in advanced countries and EME. Fundamentally, ITF emphasizes the desired inflation target, monetary policy consistency to achieve said target as well as stronger central bank independence and transparency to the public. ITF is considered viable to overcome the high inflation experienced in many countries in the near term, while simultaneously providing a solid foundation to achieve sustainable economic growth. A more thorough description of ITF and how it is practiced at various central banks is presented in Chapters 8 and 9.

In practice, a number of central banks focus on inflation, while other central banks select inflation and exchange rate stability as the dual final targets of monetary policy. In many jurisdictions, including Indonesia, there has not only been a shift in priority but also the target focus from multiple objectives, namely economic growth, inflation control, and job creation, to a single objective, namely rupiah currency stability (inflation and the exchange rate).⁹ In many countries, however, the generic target of price stability could be defined more concretely, including the use of relevant price indexes, such as the Consumer Price Index (CPI). For example, the Bank of Canada targets 1–3%, the Bank of England targets $2.5 \pm 1\%$ and the Reserve Bank of Australia targets 2–3%. In general, the inflation target is set by the government or through coordination between the government and central bank. The formal final targets of monetary policy in various jurisdictions are presented in [Table 6.1](#).

It is important to note several exceptions when setting price stability as one of the final targets of monetary policy. One is the case of Germany, where despite not using inflation control as the final target of monetary policy, the Deutsche Bundesbank implicitly targets inflation in the 1.5–2% corridor. Similarly, although many perceive the US Federal Reserve to focus on maintaining low inflation, price stability is not more dominant. In this case, the Federal Reserve Act legally

⁹As contained in Act No. 13 of 1968 concerning the Central Bank, the duty of Bank Indonesia is to assist the government regulate, create, and maintain rupiah stability, while also stimulating production and development as well as creating job opportunities to ameliorate public welfare. Meanwhile, based on the Bank Indonesia Act (No. 23) of 1999, the objective of Bank Indonesia is to create and maintain the value of the rupiah.

Table 6.1: The Central Bank's Legal Monetary Policy Framework.

Final Target of Monetary Policy	Country
Domestic price stability as the final target	Columbia, Czech Republic, Hungary, Iceland, South Korea, México, New Zealand, Norway, Poland, Sweden, Thailand, UK
Currency stability as the final target	Finland, ^a Indonesia, Norway
Currency stability and other targets	Brazil, Australia, Canada, Chile, Israel, South Africa

Source: Carare, Shaechter, Stone, and Zelmer (2002).

^a Also maintains the stability and security of the monetary system and currency in circulation.

stipulates that the goal of monetary policy is to “achieve long-term economic growth to potentially drive production, job opportunities, price stability and moderate long-term interest rates.”

The question, therefore, is how to observe the linkages between setting the final target and monetary policy performance? A simple explanation concerning those linkages based on indicator data for the final monetary policy target (among others), economic growth, and price stability, is presented as follows along with indicator performance over the past 25 years.

From the [Table 6.2](#), at least two salient empirical facts emerge as follows:

- (1) *First*, there is a common trend, namely the phenomenon of declining inflation, both in advanced and developing countries. In the case of Brazil, however, despite experiencing increasing average inflation for more than the past decade in the 5–8% range, this represents a remarkable improvement on conditions at the beginning of the 1990s, when hyperinflation topped 100%. In Indonesia, average inflation remains quite high but has decreased significantly from the position in the 1990s until early 2000s. Sticky inflation in Indonesia stems from potentially intense inflationary pressures on administered prices (AP) in line with rising fuel prices over the past few years.
- (2) *Second*, opposing accelerating and decelerating economic growth trends were observed in advanced and developing countries. In the US, UK, Germany, and México, for example, average economic growth tended to accelerate during the second half of the 1990s, while Japan and Brazil experienced relatively stable average growth throughout the 1990s, contrasting conditions in Thailand and Indonesia, two EME in Southeast Asia that were hit by a severe economic crisis in 1997/98. During the latter half of the 1990s, the economies of both countries, which had previously posted solid average growth (around 8%), suddenly plummeted to just 1%. Meanwhile, during the five years after the GFC, flagging economic growth was observed in advanced countries, especially the United States and United Kingdom.

Table 6.2: Average Economic Growth and Inflation in Several Countries from 1991 to 2014 (%).

Period	Developed Countries											
	United Kingdom		United States		Germany		Japan		Brazil		Mexico	
	y_g	π	y_g	π	y_g	π	y_g	π	y_g	π	y_g	π
1991–1995	1.68	3.02	2.47	2.79	1.34	3.40	1.52	1.01	3.10	> 100	2.57	19.55
1996–2000	3.14	2.71	4.10	2.54	1.69	1.42	1.41	0.31	2.24	5.93	8.35	16.66
2001–2007	2.76	1.83	2.45	2.74	1.43	1.79	1.41	–0.17	3.52	7.67	2.37	4.35
2010–2014	1.94	2.62	2.06	1.70	2.06	1.47	1.47	0.66	3.32	6.11	3.36	3.97
											3.85	2.50
											5.80	6.17

Source: CEIC data.

y_g : annual economic growth; π : annual inflation.

Those two salient facts, namely the downward inflation trend and multispeed economic growth, fundamentally support the previous statement concerning a gradual evolution in the final target of monetary policy, namely toward controlling inflation. Nevertheless, although congruent with the advantages of applying the ITF, it cannot be concluded that the framework is an unmitigated (absolute) success in terms of overcoming the problem of high inflation in many countries, considering that many countries, including the US, Germany and Japan, successfully achieved low inflation without explicitly applying the ITF. In fact, three of the countries did not even mention that domestic price stability was a final monetary policy target in their legal framework.

6.4.2. Monetary Policy Regime

Of the various monetary policy regimes practiced by central banks, exchange rate targeting, monetary targeting, and inflation targeting are the most common. In their evolution, economic, and political dynamics along with academic thinking have shaped the regime change from exchange rate targeting to money supply and inflation targeting.

6.4.2.1. Exchange Rate Targeting. The exchange rate targeting regime was introduced by several countries as part of the efforts to control inflation during the 1980s. With the inherent advantages of the regime, exchange rate targeting successfully lowered inflation in many industrialized countries at that time. For example, the UK and France successfully applied exchange rate targeting to suppress inflation by pegging their currencies to the German mark. In 1987, when France first pegged its currency to the German mark, inflation in the country was 3%, which was 2% above the rate recorded in Germany. In 1992, however, inflation in France had decelerated to 2%, a level not only considered consistent with stability, but also below inflation in Germany. Inflation in France and Germany finally converged in 1996 at slightly below 2%. A similar trend occurred in the UK, where inflation fell from 10% in 1990 to 3% in 1992 after the pound sterling was pegged to the German mark.

The successful implementation of exchange rate targeting to lower inflation was not merely limited to advanced countries but also transpired in several EME. Argentina is an important case study, where the currency board arrangement was initiated in 1991. Accordingly, the central bank applied a fixed peg between the USD and (new) peso at a level of 1:1.¹⁰ At the beginning of implementation in Argentina, the currency board was considered very successful. Inflation fell

¹⁰The currency board was a form of fixed exchange rate regime, with certain criteria. The currency board demanded greater and clearer commitment to the exchange rate target compared to the standard fixed exchange rate regime because the monetary authority was mandated to exchange domestic currency for foreign currency at a certain rate as required by the public. To build credibility, the currency board maintained more than 100% reserve assets as support for the domestic currency, while permitting the monetary authority to deviate from discretion in absolute terms.

dramatically from more than 100% in 1989 and 1990 to below 5% at the end of 1994, while economic growth accelerated with rapidity, averaging 8% from 1991 to 1994.

In practice, however, exchange rate targeting has many inherent weaknesses. During German reunification in 1990, for instance, the attention of policymakers tended to focus on the intense inflationary pressures stemming from expansive fiscal policy to rebuild the former East Germany. Such developments edged up short-term and long-term interest rates in 1991. Subsequently, the shocks that occurred in the German economy, as an anchor country in the exchange rate mechanism (ERM), were transmitted to other countries with a currency pegged to the German mark because interest rates increased in line with interest rates in Germany. Consequently, such conditions culminated in significant economic moderation and increased unemployment in many countries, especially France and UK.

The main problem that emerged in relation to reunification and the exchange rate crisis was speculative attacks on the money market. This is well-founded considering how committed ERM member countries were to maintain the value of their currencies in line with the German mark, while, on the other hand, spiraling unemployment became more questionable. Consequently, a growing trend emerged in other countries, including France, Spain, Sweden, Italy, and UK to depreciate their currencies against the German mark. Therefore, large-scale speculative attacks commenced in September 1992. In reality, only France remained committed to maintain its peg against the German mark. In general, other countries were less inclined to maintain the peg because of the adverse impacts on the real economy, which ultimately led to currency depreciation. In opposition to France, the UK elected to leave the ERM in 1992, replacing the regime with inflation targeting, a framework that subsequently benefitted the implementing economies through increased economic growth, lower unemployment and low inflation.

Meanwhile, the implementation of exchange rate targeting, with a certain degree of flexibility, in EME culminated in spectacular economic crises, such as the 1982 crisis in Chile, 1994 in México, and 1997 in Southeast Asia. The implementation of exchange rate targeting in EME, generally in the form of a fixed or managed exchange rate regime, precipitated a trend of spiraling foreign debt in the corporate and financial sectors. The long-term exchange rate stability associated with this regime was perceived as a guarantee against currency risk by the authorities on their foreign loans. The dominant composition of foreign debt was a common feature of the institutional structure in Latin America and Southeast Asia prior to the crises.

Economic crises are triggered by structural problems in the economy that spur pressures on the domestic currency and, thus, drive strong depreciatory pressures or even devaluation.¹¹ Domestic currency depreciation against foreign currencies

¹¹In several cases, the scale of the economic crisis is exacerbated by a political crisis that prevents the monetary authority or government from taking the most optimal and timely policy measures as required, which occurred in México and Indonesia.

also compounded the burden of the foreign debt held by the corporate and financial sector. Meanwhile, considering most assets are generally in denominated in the domestic currency, the value of corporate assets failed to accrue. Such conditions undermined the balance sheets and net worth of domestic corporations and financial institutions. In the following series of events, aggravated by a loss of public confidence in economic activity, investment, and economic activities nosedived, as occurred in Chile, México, and several Southeast Asian countries, including Indonesia.

6.4.2.2. Monetary Targeting. Two countries that have seriously and successfully implemented monetary targeting are Germany and Switzerland. The success of such a policy on controlling inflation in those two countries is the reason why monetary targeting continues to receive support. This represents a grand departure from the experiences of other countries that have been less successful with monetary targeting, including the US, UK, and Canada, because serious efforts were lacking in its implementation and the relationship between monetary aggregates and the final policy target was unstable.

The key to successful monetary targeting implementation in Germany and Switzerland was simplicity, avoiding overly technical implementation, and relying on a monetary policy communication strategy that focused on the long-term considerations and efforts to control inflation. Therefore, the considerations underlying monetary targeting variations became a public exercise, paying due attention to the following aspects: (1) setting the inflation target as a range or corridor based on a quantity framework; (2) providing regime flexibility, in practice, by considering attaining other targets, such as output and exchange rates; and (3) providing avowed commitment by the monetary authority as well as a clear public communication strategy.

The success of the regime to control inflation in Germany convinced several other countries to follow suit, providing one indication why monetary targeting was more popular than exchange rate targeting in a number of countries. One important success that should be noted was the success of Germany in controlling inflation because of post-reunification economic expansion at the beginning of the 1990s, where, in 1995, the inflation rate had finally dropped below the normative 2% rate of the Bundesbank.¹²

The implementation of monetary targeting in Germany also received serious criticism in relation to the asymmetric Bundesbank reaction when failing to hit the target set (Clarida & Gertler, 1997). In this respect, when overshooting the monetary target, the Bundesbank instituted policy to raise interest rates but when undershooting, efforts to lower interest rates were not taken. Consequently, the policy direction of the Bundesbank in the middle of the 1990s, when inflation was below the normative target of 2%, appears overly tight, which not only exacerbated German unemployment but also in other EMS members, such as France.

¹²With the characteristics of the monetary policy framework adopted and the successful achievement of relatively low and stable inflation, there was a presumption that the Bundesbank was fundamentally an inflation targeter.

Compared to implementation in Germany, monetary targeting in Switzerland was relatively problematic. With a small open economy (SOE), the institutional structure of the Swiss financial system experienced a significant change. In 1977–1978, the Swiss franc experienced hyper-appreciation of 40%, which was intolerable to the Swiss National Bank (SNB). Under such conditions, Switzerland temporarily abandoned monetary targeting (1978–1979) in favor of exchange rate targeting. In 1980, however, monetary targeting was formally reintroduced, with a change in the target indicator used from M1 to broad money. In 1989–1992, monetary targeting in Switzerland encountered more problems, with average inflation accelerating beyond 5%. SNB failure to maintain low inflation was due to currency shocks and a shift in the demand for money function as a consequence of monetary expansion to depreciate the Swiss franc and policy to stimulate financial system development. Consequently, the SNB found it difficult to predict and achieve the inflation target set previously, hence inflation continued to soar beyond an acceptable rate.

The problems encountered when implementing monetary targeting led to ambitious monetary policy easing in Switzerland at the beginning of the 1990s, thus rendering the advantages of the regime, as an instrument to signal the direction of monetary policy, impotent. Seeking to overcome that constraint, the SNB adopted a more flexible framework, by announcing a medium-term (3–5 years) inflation target as well as setting the basis for the expansion path at the end of 1992. Thereafter at the end of 1994, the SNB reiterated its medium-term broad money expansion path for 1995–1999. With a more flexible framework, monetary policy in Switzerland successfully brought inflation back under control at around 1%.

Two invaluable lessons can be taken from the German and Swiss experiences. *First*, monetary targeting can be used to control inflation in the long term, despite the possibility of target inaccuracies. Therefore, rigid rules are considered unnecessary to achieve the inflation target. *Second*, the main reason why monetary targeting was a success in Germany and Switzerland, despite the inaccuracies, was because the final monetary policy target was announced clearly, with the Bundesbank and SNB fully committed to a public communication strategy, which improved central bank transparency and accountability of public perception.

6.4.2.3. Inflation Targeting. As the relationship between monetary aggregates and the final policy target faded, inflation targeting emerged as a new monetary policy framework that appealed to countries previously plagued by inflation control problems in the 1990s. New Zealand was the first country to formally adopt the inflation targeting regime in 1990. Over time, New Zealand proved how successful inflation targeting could be at driving the real economy, while also controlling inflation, which piqued central bank confidence in the new framework. After New Zealand, the following countries adopted inflation targeting: Canada in 1991, UK in 1992, Sweden in 1993, Finland in 1993, Australia in 1994, and Spain in 1994. As a monetary policy framework that has received broad attention from academia and central bankers, the practice of inflation targeting is discussed in more depth in Chapters 8 and 9 of this book.

6.4.2.4. No Explicit Anchor Targeting (Implicit Targeting). Fundamentally, several countries have practiced this strategy, evidenced by impressive macroeconomic performance (including low and stable inflation) without resorting to a nominal anchor, such as exchange rate targeting, monetary targeting or inflation targeting. The real success of this just-do-it strategy is apparent in the US experience, namely when the Fed successfully controlled US inflation from double digits in 1980 to around 3% at the end of 1991. Since then, US inflation has been maintained at around or below 3%.

Strategy implementation was characterized by successful pre-emptive measures, particularly from February 1994 until the beginning of 1995, when the Federal Reserve raised the federal funds rate (FFR) incrementally from 3% to 6% despite stable inflation at the time. Economic expansion followed shortly thereafter, which brought unemployment down to below 5%, an achievement not attained since the 1960s. Nevertheless, despite large economic expansion inflation dropped to below 2%, while robust economic growth in the 1990s endured.

The success of this strategy in the United States was not without incident. Many times, the policy measures were not communicated to the public, therefore the markets regularly second-guessed the policies taken by the Fed, which triggered financial market shocks and uncertainty concerning the economic outlook. Likewise, The Fed's accountability in front of Congress and the public was also called into question, considering the lack of clear criteria to evaluate policy performance, which created time inconsistency problems.

One potential problem with the strategy was that success depends entirely on the preferences, capabilities and individuals at the top of the central banking organization. Regarding the US experience, Alan Greenspan, as Governor of the Federal Reserve, was considered a strong individual on the back of successful policy to control US inflation, thus garnering public confidence in the Fed. Nonetheless, there are no guarantees that subsequent Governors will be as strong as Greenspan, while the situation faced has perhaps become more complex.

In the face of political dynamics, US experience has also shown that central bank leaders play a key role in the success of monetary policy. Experience from the 1970s showed that strong political pressures can spur inflationary policies from the Fed and, using no explicit anchor targeting, the potential problems could return. Political pressures were also observed in 1997, when, after reducing the FFR several times in the previous period, the Fed hiked the interest rate target by 25 bps. Despite the relatively small bump in the interest rate, with robust economic performance at that time, the Fed faced a storm of criticism from Congress and the public in general.

6.5. The Monetary Policy Regime in Indonesia (1968–1998)

Three decades prior to enactment of the BI Act (No. 23) of 1999, which provided the foundation for the ITF, Indonesia experienced monetary policy regime changes from one period to the next. Starting with the period of economic stabilization and rehabilitation in the 1960s and the era of oil-based economic growth in the 1970s, as well as the period of economic deregulation, debureaucratization,

and liberalization in the 1980s to the period before the Asian Financial Crisis of 1997/98. This section describes the monetary policy regimes implemented in Indonesia during those three broad periods, while the ITF is explored specifically in Chapters 8 and 9.

6.5.1. Economic Stabilization and Rehabilitation (1968–1972)

Experience during the period after national independence until the middle of the 1960s provided invaluable lessons on the importance of prudential principles when implementing macroeconomic policy. *First*, fiscal policy must control the budget deficit at normal levels. To that end, budget spending must be selected stringently and prioritized to spending that stimulates real sector activity, thereby avoiding excessive and non-strategic expenditure. *Second*, monetary policy may not be used to finance the budget deficit on the fiscal side. Monetary policy must remain focused on controlling inflation and, therefore, printing money to offset the government deficit would threaten price stability in particular and monetary stability in general. *Third*, fiscal policy and monetary policy must be well coordinated, while maintaining the independence of both institutions to ensure policy synergy in terms of maintaining economic stability for development sustainability.

In the subsequent period, namely from the end of the 1960s, economic and financial development were maintained. Initially, government policy prioritized economic stability, which came under threat in the middle of the 1960s. Budget spending was selected judiciously, the government's budget deficit was under control and financing was sourced from soft foreign loans, which avoided economic instability, especially in terms of controlling inflation. In the monetary sector, no more currency was printed to finance the government's budget deficit and total money supply was brought under control. Such endeavors were accompanied by the provision of much needed goods and services. Through fiscal and monetary discipline, economic stability was quickly restored, with inflation falling rapidly to single digits, which also restored confidence in the economic recovery. After successfully restoring economic stability, the Government subsequently commenced long-, medium-, and short-term national development planning, which gradually brought structure to the national economy and accelerated growth.

The structure of the economy, particularly the monetary and banking sectors, was solidified by the *Central Bank Act (No. 13) of 1968*. Pursuant to the Central Bank Act, BI was tasked with assisting the Government in two areas as follows: (1) regulate, create, and maintain rupiah currency stability; and (2) promote development, while creating job opportunities to ameliorate public welfare. Monetary policy was formulated by the Monetary Board and BI undertook its monetary policy duties in line with the decisions of the Monetary Board.

It is important to note, on one hand, institutional regulation such as this was considered positive because of monetary policy integration and coordination with fiscal policy and other macroeconomic policies. On the other hand, however, such a regulatory framework blurred task focus, discipline and accountability at both institutions when formulating and implementing policy. There was also a lack of checks and balances between monetary policy, fiscal policy, and other

macroeconomic policies. Furthermore, such institutional regulation still facilitated the exploitation of monetary policy to finance the fiscal side, thereby undermining prudential principles and macroeconomic policy discipline.

6.5.2. Era of Oil-based Economic Growth (1973–1982)

National economic growth subsequently accelerated on the back of increasing oil proceeds at the beginning of the 1970s. The discovery of oil fields in Indonesia had advantages and disadvantages. On one hand, oil proceeds provided a windfall in terms of state receipts and could be used to finance routine and development spending on the fiscal side. With an active role and dominated by the government, fiscal policy could stimulate real economic activity. On the other hand, however, increasing oil foreign exchange revenues and government spending also expanded total money supply on the fiscal side, which demanded monetary expansion absorption on the fiscal side to avoid excess liquidity in the economy that could push up inflation.

Against that backdrop, the Government began to implement selective credit policy on the monetary side in 1974. The goal was to control total money supply and, thus mitigate rising inflation, primarily by regulating the magnitude of credit expansion permitted by the banking industry, often known as the net asset expansion ceiling. Each year, therefore, BI prepared a national credit expansion plan, paying due consideration to total money supply in line with predicted inflation and output growth. The banks were subsequently required to submit their credit plans to BI for the central bank to set the credit ceiling for the upcoming year for each respective bank. Ultimately, the ceiling for each individual bank was used as the basis for credit disbursements/liquidity provided by BI in line with the pre-determined sector/program. Banking regulations also restricted deposit rates and lending rates. At the monetary and banking level, credit and interest rate limits became known as financial repression, which was also practiced in various other countries.

Although banking sector conditions were less than favorable due to scarce funding sources after private savings evaporated and restrictions were placed on lending and interest rates during the period of financial repression, investment activity persisted, particularly government investment. Consequently, to provide more wiggle room for banks to utilize funds, especially in terms of disbursing loans to the private sector, BI halved the reserve requirement in 1978 from 30% to 15%.

6.5.3. Period of Economic Deregulation, Debureaucratization, and Liberalization (1983–1997)

At the beginning of the 1980s, the oil price slumped on the international market due to the onset of a global recession, which squeezed state revenues to offset the State Budget. Government dominance to support increasing economic activity was no longer tenable and, consequently, the sustainability of national development was threatened. Therefore, the Government instituted a series of

policy reforms in the economy to overcome the crisis threat due to tumbling oil prices. The goal was to grow, nurture, and expand the role of the private sector in every aspect of economic life to replace the role of the Government in terms of maintaining national development. As a result, at the beginning of the 1980s, the Government introduced deregulation, debureaucratization, and liberalization policy to various economic sectors, including the banking and financial sectors, trade, investment, and so on.

On June 1, 1983, the Government issued bank deregulation policy, signaling an era of liberalization in the banking sector specifically and financial sector in general. Policy focus was to liberalize interest rate setting by the banks as well as limit the net asset ceiling to just the credit disbursement program. This policy prompted rapid banking and financial sector development in Indonesia, by mobilizing private funds not only in the form of demand deposits, saving deposits and term deposits, but also in the form of loans and other types of financing offered by the banking sector to the business community. Likewise, the financial sector experienced rapid development in terms of financial transaction volume as well as the variety of financial products available (stocks, bonds, securities, and derivative products) to trade. Consequently, more funds circulated in the financial sector, which affected the relationship between money, inflation, and output compared to previous periods.

Prevailing economic conditions, particularly the financial sector, had fundamental implications on the implementation of monetary policy by BI. Monetary policy, which had been implemented directly using selective credit policy began to shift to more indirect and market-oriented channels, including open market operations to control liquidity in the economy. Monetary controls tended to follow monetary targeting by setting total money supply (M1 and M2) as the intermediate target and narrow money (M0) as the operational target. Meanwhile, money market operations were conducted through auctions of BI Certificates (SBI), which were first issued in 1984 as the main instrument of monetary policy. Rupiah money market intervention helped to control liquidity by providing short-term lending facilities, from overnight to seven days. Money market operations were directed toward achieving the operational target (M0), which itself was geared toward keeping money supply (M1 and M2) under control in line with current projections.

Thereafter, driving domestic economic activity in the face of global competition, the Government issued a policy package on October 27, 1988, containing regulations to refine financial, monetary and banking policies.¹³ Deregulation focused on the ease of opening a bank office or branch as well as expanding bank operations, including foreign exchange activities. Therefore, deregulation accelerated

¹³Unhealthy economic development occurred in 1987, when the public speculated on foreign currencies after expecting the Government to initiate rupiah devaluation. To overcome that problem, the Government introduced policy measures known as “Gebrakan Sumarlin,” which reduced bank liquidity dramatically and dampened speculative activities.

fund mobilization, lending, product development, and various other banking and financial sector business activities. In addition, a more conducive climate of competition was created in the banking sector through less stringent licensing to establish a new bank or joint venture bank. Concerning the efforts to enhance monetary control effectiveness, the measures introduced included reducing the reserve requirement from 15% to just 2%. In other words, loose deregulation policy precipitated a fundamental change in terms of rapid banking and financial sector development as well as monetary policy implementation in Indonesia.

Congruent with rapid banking and financial sector development in Indonesia, new problems emerged in terms of implementing monetary policy, particularly linked to efforts to control total money supply (M1 and M2). Bank operations and products, to mobilize funds and finance the business community, were not only limited to demand deposits, savings deposits, term deposits, and loans, but also diversified into various forms of money market instrument, such as negotiable certificates of deposit (NCD), commercial papers, promissory notes, automated teller machines (ATMs), and so on. On the other hand, capital market development began to accelerate with rapidity in the form of transaction volume and the types of securities traded. Consequently, decoupling between the financial sector and real sector occurred, creating a more tenuous relationship between money supply and inflation and real output, especially in the near term.

In addition, as a consequence of financial sector liberalization, the inflow of funds to Indonesia, particularly in the form of private foreign loans, surged rapidly. The influx of funds exploited the current economic boom at that time and was supported by the nascent wave of globalization in the financial sector, trade, and investment. On one hand, the deluge of foreign capital flows closed the saving-investment gap, which drove economic growth and national development. On the other hand, however, the influx of non-resident capital spurred a number of problems too. Most foreign capital was in the form of short-term private foreign loans that were not hedged against currency risk and were primarily utilized to fund long-term private projects that did not produce foreign exchange. On the monetary side, the magnitude and mobility of foreign funds also complicated monetary policy implementation by BI.

Striving to prevent the adverse impact of money supply expansion originating from foreign capital flows on higher inflation and rupiah stability, BI absorbed the excess liquidity in the economy, which edged up domestic interest rates. Higher domestic interest rates, however, attracted more foreign capital, primarily in the form of short-term securities. Consequently, total private foreign loans spiraled in various forms and with different maturities. Such conditions were exacerbated further by the implementation of private projects financed by the foreign loans in compliance with good corporate governance, which has been the main cause of crisis since 1997.

6.6. Concluding Remarks

In practice, selection of the monetary policy regime by a central bank is based on the perceived benefits of that regime in terms of achieving the monetary policy targets, adjusted to local economic conditions. Nevertheless, there are two other

aspects that underlie monetary regime selection, namely policy commitment and prudence (discretion). In this regard, the success of monetary policy regime implementation, in terms of achieving price stability and robust growth in the long term, fundamentally depends on striking the optimal balance between policy commitment and prudence (Stone & Bhundia, 2004). According to a previous viewpoint, Mishkin (1999) classified several monetary policy regimes, as practiced, based on the selection of various nominal anchors, namely exchange rate targeting, monetary targeting, inflating targeting, and no explicit anchor targeting.

Of course, the diverse preferences of the various policymakers in different jurisdictions provided a hotbed for other policy regime designs, with different combinations of commitment and prudence. Stone and Bhundia (2004) proposed a new classification of monetary policy regimes, defined as the choice and clarity of the nominal anchor. Departing from the classification proposed by Mishkin, the emphasis of the new classification system, according to Stone and Bhundia, was not only from the aspect of choice, but also clarity of the nominal anchor used. In this case, the different taxonomy of regimes that could be classified based on a combination of commitment and prudence included monetary non-autonomy, exchange rate peg, full-fledged inflation targeting, implicit price stability anchor, inflation targeting lite, weak anchor, and money anchor.

Monetary non-autonomy is a regime where a country does not issue currency independently but is tied to certain regulations, for instance in the form of a currency board arrangement. The exchange rate peg uses a nominal anchor in the form of the domestic exchange rate to other currencies, usually trading partner currencies with low inflation. This regime is the focus of exchange rate targeting classification according to Mishkin. Meanwhile, the two iterations of inflation targeting are full-fledged, where there is clear and formal commitment to the inflation target, as well as inflation targeting lite, where the commitment is not fully toward achievement of the inflation target but also other variables, such as the exchange rate and monetary targets. Fundamentally, the implicit price stability anchor is an eclectic regime with the same characteristics as the implicit nominal price anchor per Mishkin's classification. Likewise, the money anchor regime is the same as monetary targeting, where monetary aggregates are used as the nominal anchor. Finally, the weak anchor regime is aimed at countries without a nominal anchor in practice, where inflation is typically north of 40% year on year.

The gradation of regime classification implies that the empirical analysis performed by Stone and Bhundia stressed a trade-off between commitment and prudence, induced by a shift or refinement in the policy regime, thereby creating space for the emergence of regimes with a new classification. Currently, there is a tendency among countries using the implicit price stability anchor to increase their commitment to achieving the inflation target, thus giving the impression of full-fledged inflation targeting. On the other hand, however, as credibility is more important when applying a full-fledged inflation targeting regime, there is a trend toward softening commitment to the inflation target, thereby increasing the element of prudence in terms of seeking output growth and financial system stability. Therefore, what occurs is a regime that stands somewhere in between implicit price stability and full-fledged inflation targeting.

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Chapter 7

Monetary Policy Operational Framework

7.1. Introduction

In line with the nascent issue of “duality” in terms of achieving the final goal of monetary policy, namely output (economic growth) stability or price (inflation) stability, as explored in Chapter 6, the longer discussion on how monetary policy should be implemented also occurs in theory and practice. When controlling price or output stability, the central bank has two options. It can apply money supply (monetary base) or the interest rate as the operational target. In relation, a critical review by Poole (1970) found that during certain periods, the monetary authority was unaware of the types of shocks or level of output in the economy. Therefore, during such periods, the monetary authority would have the option to target money supply or the interest rate, while accommodating demand for currency.

Furthermore, in terms of the monetary policy operational framework, the monetary authority would also have to consider other equally important factors, including the monetary policy response itself. To that end, intensive discussions since the middle of the twentieth century have highlighted the importance of applying rules or discretion, representing opposite ends of the same spectrum, as a reference for the central bank to determine an optimal monetary policy response.

Naturally, the selection of policy instruments and the most appropriate monetary policy response to prevailing economic dynamics is not a simple task. It is considering that in the monetary policy implementation the monetary authority is not only faced with a range of contextual alternatives, but also factual. In practice, therefore, it is very possible that a central bank may be less assertive concerning the foundation of the policy that must be instituted. In order to explore that issue further, this chapter is organized into six parts. After the introduction, this capture presents the conceptual dimensions and theoretical models concerning the existence and role of operational targets, instruments, and the policy response in the monetary sector. The section will also cover aspects relating to the formulation of an optimal monetary policy framework. The following section will highlight monetary management as practiced in several countries as well as in Indonesia. This chapter will end by presenting hypotheses on state-contingent rules to formulate an optimal monetary policy response, especially for developing economies.

7.2. Conceptual Dimension and Theoretical Models

7.2.1. Policy Instruments: Money Supply Versus Interest Rate

In general, from the various literature, it can be concluded that the selection of optimal monetary policy instruments is determined by, among others, the origin of the shock and how the monetary policy transmission mechanism functions in an economy. An intuitive elaboration of the issue was presented by Poole (1970). According to Poole, when selecting the optimal policy instrument, the monetary authority does not necessarily know the confronted shocks faced or the current level of output, and is thus faced with the decision to target money supply or the interest rate, while also accommodating demand for currency. The standard Poole model fundamentally only accommodates demand-side shocks in the form of monetary shocks and spending shocks. Over time, however, several variations of the Poole model have been developed to include supply-side shocks and the rational expectations theory (Beare, 1978; Walsh, 2001) as follows:

7.2.1.1. Standard Poole Model with Demand-side Shocks. Analyzing the conditions where each respective instrument is appropriate, Poole applied a simple Keynesian approach for goods market (investment savings (IS)) and money market (liquidity preference money supply (LM)) equilibrium as follows:

$$\begin{aligned} Y &= Y_r r + u \\ M &= L_y Y + L_r r + v \end{aligned} \quad (1)$$

The two equations represent the investment savings - liquidity money (IS-LM) equation with respective stochastic residuals of u (spending disturbance) and v (monetary disturbance). Meanwhile, the expected loss function of the monetary authority is targeting output stability is formulated simply as follows¹:

$$E(L) = E[(Y - Y^*)^2] \quad (2)$$

In this case, further analyses were conducted to compare the expected loss function of monetary authorities when applying monetary policy through money supply or the interest rate, such that the level of output or income would arrive at the expected level. In the deterministic model, where $\sigma_u^2 = \sigma_v^2 = 0$, both choices are equivalent but different in the stochastic model.

If the interest rate is determined in r^* and $Y^* = Y_{r^*}$ thus $Y = Y^* + u$, then the expected loss function can be expressed as follows:

$$\begin{aligned} E(L)_{r=r^*} &= E[(Y - Y^*)^2] \\ &= E[(u)^2] = \sigma_u^2 \end{aligned} \quad (3)$$

¹During initial development of the Poole model, the objective function was formulated referring only to the final target, namely output stabilisation, because that was the most popular issue at the time.

If the monetary authority targets money supply, the solution to the IS–LM equation is as follows:

$$\begin{aligned} r &= \frac{M - L_y Y - v}{L_r} \\ Y &= Y^* + \frac{L_r u - Y_r v}{L_r + Y_r L_y} \end{aligned} \quad (4)$$

and the expected loss function is:

$$\begin{aligned} E(L)_{m=m^*} &= E[(Y - Y^*)^2] \\ &= E \left[\left(\frac{L_r u - Y_r v}{L_r + Y_r L_y} \right)^2 \right] = \frac{L_r^2 \sigma_u^2 - 2\rho L_r Y_r \sigma_u \sigma_v + Y_r^2 \sigma_v^2}{(L_r + Y_r L_y)^2} \end{aligned} \quad (5)$$

where ρ is the correlation coefficient between u and v . Therefore, the strategy employed will be the policy that incurs the lowest level of loss, namely by comparing Equations (3) and (5) as follows:

$$\frac{E(L)_{m=m^*}}{E(L)_{r=r^*}} = \frac{L_r^2 - 2\rho L_r Y_r \sigma_v / \sigma_u + Y_r^2 \sigma_v^2 / \sigma_u^2}{(L_r + Y_r L_y)^2} \quad (6)$$

If the ratio is less than 1, the policy selected is $M = M^*$. In contrast, if the ratio is more than 1, the policy selected is $r = r^*$. If $\sigma_v / \sigma_u = L_y$ and $\rho = -1$, then the numerator and denominator values will be the same or have a ratio equivalent to 1. Empirical evidence is available by referring to the behavior of demand for currency as well as demand for goods and services. If the demand function for currency is relatively stable compared to the demand function for goods and services, then σ_v (monetary shocks) will be smaller than σ_u (spending shocks), thus policy strategy using total money supply is superior to interest rate instruments to achieve the final target of monetary policy, and vice versa.

7.2.1.2. Poole Model Variations with Supply-side Shocks and the Role of Market Expectations. Fundamentally, the standard Poole model only calculated demand-side shocks (from the IS–LM equation). Poole model variations, however, also take into consideration supply-side shocks, while alternative strategies price targeting can be formulated through the demand–supply model as follows:

$$\begin{aligned} Y &= -\alpha r + u \\ M - P &= \beta Y - \lambda r + v \\ Y &= \gamma P + w \end{aligned} \quad (7)$$

where the final equation is the aggregate demand equation. The stochastic residuals, u and v , are respective demand-side shocks, while w represents shocks on the supply side. When calculating the influence of supply-side shocks, w than u and v , are assumed to be zero, while the expected loss function of the monetary authority when setting the price stabilization strategy is as follows:

$$E(L) = E[(P - P^*)^2] \quad (8)$$

Therefore, if the monetary policy strategy is to achieve output stability, then through the stages of finding an equilibrium solution as previously done, we get:

$$\begin{aligned} E(L)_{r=r^*} &= E[(Y - Y^*)^2] \\ &= E[(u)^2] = \sigma_u^2 \end{aligned} \quad (9)$$

$$\begin{aligned} E(L)_{m=m^*} &= E[(Y - Y^*)^2] \\ &= E\left[\left(\frac{\alpha w}{\alpha + \gamma(\alpha\beta + \lambda)}\right)^2\right] = \frac{\alpha^2}{[\alpha + \gamma(\alpha\beta + \lambda)]^2} \sigma_w^2 \end{aligned} \quad (10)$$

Meanwhile, if the monetary policy strategy is to achieve price stability, the following equations are used:

$$\begin{aligned} E(L)_{r=r^*} &= E[(P - P^*)^2] \\ &= E[(-w / \gamma)^2] = \sigma_w^2 / \gamma^2 \end{aligned} \quad (11)$$

$$\begin{aligned} E(L)_{m=m^*} &= E[(P - P^*)^2] \\ &= E\left[\left(\frac{(\alpha\beta + \lambda)w}{\alpha + \gamma(\alpha\beta + \lambda)}\right)^2\right] = \frac{(\alpha\beta + \lambda)^2}{[\alpha + \gamma(\alpha\beta + \lambda)]^2} \sigma_w^2 \end{aligned} \quad (12)$$

The optimal monetary policy strategy is selected the policy which resolution the smallest loss, namely by comparing Equations (9) and (10) if the strategy is output stability, and by comparing Equations (11) and (12) if the strategy is price stability. Therefore, the following can be concluded:

- (i) If output stability is the target, targeting the interest rate is superior to targeting the money supply, because:

$$\frac{\alpha^2}{[\alpha + \gamma(\alpha\beta + \lambda)]^2} \sigma_w^2 > 0 \quad (13)$$

- (ii) If price stability is the target, targeting the money supply is superior to targeting the interest rate, because:

$$\frac{E(L)_{m=m^*}}{E(L)_{r=r^*}} = \frac{[\gamma(\alpha\beta + \lambda)]^2}{[\alpha + \gamma(\alpha\beta + \lambda)]^2} < 1 \quad (14)$$

The model was developed further by taking into account rational expectations. Blanchard and Fischer (1989) applied nearly an identical model framework as that used by Sargent and Wallace (1975), a supply-demand model as follows:

$$\begin{aligned} y_t &= -b[i_t - (E[p_{t+1} | t] - p_t)] + z_t \\ m_t - p_t &= y_t - ai_t - v_t \\ y_t &= \beta(p_t - E[p_t | t-1] + u_t) \end{aligned} \quad (15)$$

where the last equation is the aggregate supply equation. It was assumed that the three shocks are white noise, and total money supply approaches zero, thus $E[p_t | t-1]$ and $E[p_{t+1} | t]$ are zero.

If the final target of monetary policy is output stability, the solution for each respective instrument, money supply and the interest rate, is as follows:

$$y_t(m|_{m=0}) = \frac{\{\beta b v_t + b(1+a)\beta u_t + a\beta z_t\}}{[a\beta + b\beta + b + ba]} \quad (16)$$

$$y_t(i|_{i=0}) = \frac{\{\beta b u_t + \beta z_t\}}{(\beta + b)} \quad (17)$$

The two equations generate three solutions.

First, it is conclusive with the Poole analysis, namely that if the policy aims to dampen output fluctuations, and if the primary source of uncertainty is currency demand shocks (v), then an interest rate instrument would be superior because when applying such a strategy, output is not influenced by the shocks.

Second, the output response to demand-side shocks of goods and services (z) is larger when applying an interest rate strategy than the same response using a money supply strategy.

Third, the output response to supply-side shocks is larger in the interest rate strategy if $b > 1$ because such conditions guarantee a flatter aggregate demand curve than aggregate supply.

7.2.2. Policy Response: Rules Versus Discretion

The policy instrument, money supply or the interest rate, to achieve the final target, either output stability or price stability, can be selected in different ways relating to the selection of optimal policy measures in response to economic fluctuations. The conceptual background also implies that the policy variables of macroeconomic stabilization policy can be influenced based on rules that consider permanent feedback in the relationships among the economic variables. That is an alternative to active strategy, or discretion, which is based more on certain considerations and assessments as well as fine-tuning by the policymakers. An alternative way of understanding discretion is as an anti-rules strategy.

Fundamentally, analysis of framing the response is the manifestation of the long-standing debate known as “rules versus discretion.” Interpretation and the extreme differences of “rules versus discretion,” the terminology and approach of which were first developed by Kydland and Prescott (1977) and echoed by Barro and Gordon (1983), have also fundamentally experienced a shift in line with

the ongoing debate over the past few decades.² Analytically, Barro and Gordon (1983) found that a rule-based monetary policy response (or selecting the policy instruments based on rules) is more appropriate in response to current dynamics, which have already been taken into account in the previous formulation of policy instruments. In contrast, a discretion-based monetary policy response is based on evaluations over time of ongoing dynamics, while disregarding past developments and policies. Meanwhile, Taylor (1993), explained that, departing from discretion-based policy, rule-based policy is symmetrical, meaning that it is based on methodologies and planning, not casual and random measures.³

The debate on “rules versus discretion” also takes into consideration the formulation of more rigid rules, where the instrument variables are determined using a constant value, not feedback. This implies the presence of non-active policy, namely policy that is not adjusted in line with the short-term stabilization needs. Per the Keynesian framework, monetary policy is best implemented actively, and is directed by a formal feedback rule or discretion as well as an informal assessment by the policymakers while considering the economic stabilization goals. In contrast, Friedman and other monetarists referred more to the k -percent rule, where money supply is increased based on a constant percentage over a certain period. The standard version of the rule is based on the Quantity Theory of Money (QTM), which implies that the central bank must maintain money supply growth at a minimum of $k\%$, even in the near term; while the soft version allows deviation over a longer period.

²The debate on “rules versus discretion” now refers to a new argument that highlights the time-inconsistency problem when framing policy strategy. The time-inconsistency problem refers to the differences in (optimal) policy measures announced by the central bank to the public – if the central bank’s credibility is sound – with the upcoming policy measures to be implemented by the central bank after the public takes a decision based on their expectations (Kydland & Prescott, 1977). For example, the central bank announces its intention to achieve a given inflation target, and the public executes contracts or work agreements based on that announcement. Under such conditions, the central bank has an incentive not to fulfil its promise by seeking to achieve stronger output growth, which would exacerbate inflationary pressures. Ultimately, however, the public would find out what happened and would raise their inflation expectations, which would constrain real output. If this chain of events was to reoccur, inflationary bias would emerge, a situation when real output does not increase but inflationary pressures heighten. The issue of monetary policy credibility and time inconsistency is elaborated in Chapter 10.

³Although both responses can be developed based on the optimisation method, optimisation for rule-based policy and discretion-based policy are completely different. Woodford (1999) stressed that, departing from discretion-based policy that uses evaluations over time, the characteristics of rule-based policy are grounded in rules with a timeless perspective, namely that the rule is not influenced by current macroeconomic developments. If new information is obtained, however, which causes the central bank to revise its rules, the adjustment can be made without creating a bias in the achievement of the policy target (McCallum, 2000).

The preference for a constant rule is based on the consideration that economic conditions, fundamentally, are unstable, thus making forecasting to frame stabilization policy extremely difficult. Consequently, it is possible that active stabilization policy could destabilize the economy due to imprecise policy timing and uncertainty. Assuming rational expectations, however, uncertainty leads to policy effects that were not anticipated by the real sector. Conversely, a constant rule can steer public perceptions and, thus, the direction of monetary policy, thereby reducing output fluctuations.

7.2.2.1. Feedback Rule and Optimal Monetary Policy. Through development to the present day and consistent with money demand function instability (money multiplier), the use of constant rules, standard or soft, has started to fade, replaced by feedback rules. Furthermore, due to market imperfections and uncertainty, feedback rules when framing the monetary policy response are believed to achieve the policy goal more optimally. An important initial contribution was put forward by Theil (1961), based on the analysis conducted by Tinbergen (1952). The simple version is as follows.⁴ Assuming that the relationship between the final policy goal (policy variable) and policy instrument (instrument variable) can be expressed in a stochastic equation, namely:

$$Y = \alpha + \beta X + u = Y_p + u \quad (18)$$

where Y and Y_p are, respectively, the actual and target values of the final policy target variable (GNP for instance). X is the policy instrument variable, which is under the full control of the monetary authority. α and β are parameters assumed to be accurately known, while u is the stochastic variable with an average value of zero and variance of σ_u^2 .

It is assumed that the loss function of the monetary authority can be expressed as follows:

$$L = \phi_1 (Y - Y^*)^2 + \phi_2 (X - X^*)^2 \quad (19)$$

where ϕ_1 and ϕ_2 are the fixed weights, and Y^* and X^* are the respective desired levels of the final target and policy instruments. Considering that u is a stochastic variable, then Y is also a stochastic variable, thus the loss function is expected as follows:

$$\begin{aligned} E(L) &= E[\phi_1 (Y_p + u - Y^*)^2 + \phi_2 (X - X^*)^2] \\ &= \phi_1 \sigma_u^2 + E[\phi_1 (\alpha + \beta X - Y^*)^2 + \phi_2 (X - X^*)^2] \end{aligned} \quad (20)$$

Optimal policy is determined by minimizing the loss function, with the following first-order condition:

$$X = \frac{\phi_1 \beta}{\phi_1 \beta^2 + \phi_2} (Y^* - \alpha) + \frac{\phi_2}{\phi_1 \beta^2 + \phi_2} X^* \quad (21)$$

⁴For clarification, refer to Beare (1978).

Fundamentally, the equation is an optimal linear decision rule, showing that the optimal value of X is a linear function of the desired level of the final target (Y^*) and policy instruments (X^*).

If no costs are incurred using policy instruments, $\phi_1 = 1$ and $\phi_2 = 0$, then the optimal conditions are as follows:

$$X = \frac{(Y^* - \alpha)}{\beta} \quad (22)$$

From the two equations above, the inclusion of uncertainty through the stochastic variable μ , will influence the desired level of the loss function. On the other hand, however, such conditions do not affect the rule to determine the optimal policy taken, thus the monetary authority implements a strategy of certainty equivalence.

Under extreme circumstances, the feedback rule can be determined using target-based rules, for instance the inflation target and nominal-income target, or through instrument-based rules such as the money growth rule and interest rate rule. Therefore, variations when stipulating an optimal monetary policy rule are fundamentally based on linkages between instruments and the monetary policy target, for instance in the form of correlation between the structural equations derived from the simple monetary policy transmission model, combined with the quadratic objective function that analyses the costs-benefits of achieving the policy target. To fulfill the criteria as an ideal rule, the correlation must be simple, thus the solution of the corresponding optimization problem can easily be understood and resolved.

7.2.2.2. Simple Feedback Rule. One monetary rule, in the form of a simple feedback rule, is the monetary growth rule or *McCallum rule*. Where monetary growth reflects the policy response to the average change in base velocity, as a correction to the ongoing changes due to amended regulations and technological advances, as well as the cyclical development of nominal output (McCallum, 1988). The McCallum rule can be expressed as follows:

$$\Delta b_t = \Delta x^* - \Delta v_t^a + \lambda(\Delta x^* - \Delta x_{t-1}) \quad (23)$$

where b_t is the total monetary base, x_t and x_t^* are the nominal output and desired level respectively, and Δx^* is the constant parameter that reflects the average nominal output growth per period in a year. Δv_t^a is the average growth of base velocity, where v_t^a can be expressed as $(1/k) \sum_{i=1}^{\infty} k(x_{t-i} - b_{t-i})$. Meanwhile, λ is the positive parameter that reflects the stimulative effect of cyclical nominal output on base money growth.

Although the formula is very intuitive, due to the instability problem of linkages between the monetary aggregates and the final policy target (output), the McCallum rule is less desirable. As a replacement, however, another monetary policy rule has gained attention recently, namely the interest rate rule or *Taylor*

rule, where interest rate developments reflect the response of output and inflation (Taylor, 1993). The original formula of the Taylor rule is as follows:

$$i_t = i^* + \Delta p_t^a + \alpha(y_t - y^*) + \beta(\Delta p_t^a - \pi^*) \quad (24)$$

where i_t is the short-term interest rate as a monetary policy instrument, y_t is real output, Δp_t^a or π_t is average inflation, i^* is the long-term interest rate equilibrium, Y^* is the potential output, and π^* is the inflation target. Meanwhile, α and β , respectively, are the positive parameters that reflect the impact of real output and inflation on the interest rate.

According to Ball (1997) and Svensson (1997a), the original version of the Taylor rule is not an ordinary equation but fundamentally reflects feedback that can be resolved through optimization by the central bank, calculating the Phillips curve (backward-looking) on the supply side and IS curve on the demand side. Meanwhile, the general objective function of monetary policy is to minimize the loss function, namely the weighted expectation value of output deviation from its potential and inflation from its target. Nonetheless, the Taylor rule equation is not, fundamentally, forward looking. The interest rate response to output deviation from its potential (output gap) and inflation deviation from its target, are contemporaneous. The response to contemporaneous shocks aims to stabilize current output and inflation, which is not relevant considering the time lag of monetary policy (Svensson, 1997a). Meanwhile, in the context of an open economy, the Taylor rule equation can also be modified with other variables, particularly changes in the exchange rate due to the influence of certain exogenous variables (Ball, 1997).

The design of monetary policy rules varies greatly in line with the theoretical background it is based on. Clarida, Gali, and Gertler (2000), for instance, included interest rate smoothing and exogenous shocks in the system. Meanwhile, in their study of the inflation targeting framework (ITF) for Canada's economy, Macklem-Srou (2000) used a forward-looking orientation for an explicit horizon of six to seven quarters, referring to the monetary policy lag and efforts to mitigate excessive output shocks. In general, the above rules have failed to include possible non-linearity or asymmetry in the Philips curve. Schaling (1999) developed the inflation-forecast-based rule proposed by Svensson (1997a), by calculating a convex Philips curve. Using the optimal control technique, the derived asymmetric monetary policy rules are expected to correct interest rate determination in the inflation-forecast-based rule that has a tendency to underestimate.

7.2.3. Several Aspects of Optimal Monetary Policy Formulation

In terms of utilizing monetary policy instruments, money supply and the interest rate, in general, are not targeted rigidly. Similarly, the policy response, using rules or discretion, is more flexible in its implementation. Irrespective of whether there are inconsistencies between the theoretical conclusions and empirical practices, it is generally understood that such inconsistencies are due to two factors,

namely inconsistent theoretical assumptions and the unique fundamentals of an economy that underlie the behavior and patterns of linkages between the economic variables.

In relation to the use of rules, the benefits or advantages of applying a pure (unmodified) policy rule can be viewed from at least three aspects, namely transparency, consistency, and predictability as follows:

First, the application of a rule provides a mechanism to analyze how committed the monetary authority is to that rule because, technically, the rule can be verified by economic players.

Second, the application of a rule guarantees policy consistency, thus, if the monetary authority is credible, the application of a mechanistic rule will reduce uncertainty surrounding the measures available to the monetary authority.

Third, affects predictability. Although comparatively simple, the application of a rule when analyzing monetary modeling, in general, generates sound recommendations for the monetary authority.

On the other hand, the disadvantages of applying a pure policy rule are as follows:

First, the results generated may not be optimal because the design of a rule only responds to a small part of the information on economic dynamics. Such conditions are reflected by the Taylor rule, which, in general, explains no more than two-thirds of empirical interest rate developments (Judd & Rudebusch, 1998).

Second, there are many situations of deviation from the rule that necessitate judgment. In fact, Taylor (1993) himself conceded discretionary departures from the rule. Consequently, a rule is considered incomplete because it does not accommodate the misaligned rule.

Third, although empirical studies support the need to apply a policy rule, for instance the Taylor rule, no benchmarks are available to test the success of the rule considering that no central banks are fully committed to the rule (Svensson, 2002).

Consensus was reached after a long debate between economists in relation to the selection of policy response, rule versus discretion, stating that the central bank cannot fully implement monetary policy based on discretion. Conversely, rules are a prerequisite for the implementation of sound monetary policy, thus the application of monetary policy without using a certain rule could possibly lead to the opposite.

As alluded to previously, when using rules, economists now traditionally focus their observations on two types of rule, namely money growth rules (McCallum, 1988) and interest rate rules (Taylor, 1993). The best type of rule is a question that remains unanswered. Nonetheless, it has generally been agreed that rule-based policy can be applied with a certain level of discretion. The opposite is also true,

although ideal conditions have been met, discretion is still recommended, while paying due consideration to the rules.

In the design of policy rules, namely growth rules and interest rate rules, two distinct ways of thinking are prevalent. *First*, policy rule design fundamentally reflects linkages between the final policy goal (output and prices) and the operational target or policy instrument (base money and short-term interest rates). In general, if output growth is selected as the final policy target according to GDP nominal targeting, then the monetary growth rule will be the first choice. Conversely, if price stability is selected as the final policy target per the ITF, then the interest rate rule will be the first choice. Nevertheless, McCallum and Nelson (1999) stressed that the role of the final target indicator in the designs of both rules are interchangeable, thus allowing the possibility, for instance, of designing a rule with the monetary base as the operational target/instrument, where its performance is a response to output and inflation. Similarly, it is also possible to set the interest rate rule based on feedback from output developments.

Second, the analysis of monetary policy rules does not have to reflect optimal central bank behavior. If the objective of the analysis is to seek an optimal policy, then the design of the policy rule should be produced from optimization measures that refer to the function of the central bank's goal, which is based on public utility or behavior; where, in practice, no central bank explicitly states the objective function. Therefore, not all analyses recommended for use must assume optimal measures from the central bank. In this case, it is reasonable to expect positive analysis, examining the impact of hypothetical rules that reflect various alternative approaches, is more beneficial (McCallum, 2001).

Irrespective of the diverse views described above, understanding the relationships between output and price stability play an extremely important role in the analysis of monetary policy, at least from the perspective of two concepts, namely to explain the influencing factors (determinants), while simultaneously forecasting inflation, as well as to formulate an optimal monetary policy framework with the single target of price stability. Technically, the essence of the first concept relates directly to elements of uncertainty and forming market expectations to influence the dynamic behavior of inflation. Therefore, careful observation of both elements through an appropriate econometric approach is crucial. Meanwhile, the essence of the second context relates more to which monetary policy is required and how to implement it. The policy framework selection strategy is based on various considerations or choices, for instance the choice between credibility and flexibility as well as between forward-looking or backward-looking, and the need to consider a cost-benefit analysis of disinflationary policy by the central bank, especially in relation to the ever-present problem of output variability when pursuing price stability.

Therefore, the monetary policy framework selection strategy should fundamentally be directed toward an optimal monetary policy framework, namely policy that is based directly on the assumption of optimal behavior among the economic agents. To conclude that a policy is optimal, an evaluation is typically required of changes in the welfare of each individual economic agent because of applying the alternative policy, which can be derived through a simple modeling system or

general equilibrium analysis. Theoretically, several studies, including Theil (1961), Ball (1997), Svensson (1997a), Rudebusch and Svensson (1998) as well as Blake, Weale, and Young (1998), showed that optimal policy is generally formulated by considering (at least) the basic elements, namely a cost-benefit analysis of applying the policy, for instance output or price stabilization, based on optimization measures in the decision-making process of the central bank. In this regard, the analysis could be described with reference to the loss function that must be minimized by the central bank, with the determination of policy instruments as the subject and several constraint functions relating to intervariable structural linkages in the economic system. Based on such conditions, optimal monetary policy can be mathematically resolved as an optimal reaction function by the central bank, otherwise known as the optimal decision rule or optimal policy rule, which stipulates the presence of policy instruments as a function of all the information that is available.

Independent of the basic elements above, to achieve the best policy application framework and to facilitate monitoring, policymakers specifically need to consider two important aspects:

First, forward-looking policy while paying due consideration to the time lag of monetary policy (Svensson, 1997a). In this respect, Svensson submitted a proposal on implementing monetary policy based on inflation forecasts rather than actual values. The advantage of this theoretical approach, in addition to producing a response equivalent to optimal conditions, is to increase monetary policy credibility by focusing on inflation expectations.

Second, informational requirements in policy formulation while paying due consideration to the short-term behavioral adjustment mechanism (Blake, 2000). In this regard, central banks were urged to adopt more generic policy rules, considering the factors used to make the forecasts. With the possibility of using a more detailed model specification, per Blake, model representation with an error correction mechanism could produce the best policy control framework. In the study, Blake et al. (1998) showed that a simple rule, for instance the Taylor rule, is inferior to other, more optimal and more complex rules, estimated paying due consideration to deviation from the simple rule in the event of an economic shock.

The fundamental issue is how to implement an explicit monetary policy target in the utilization of instruments and choice of policy response. This is linked to the empirical condition that the actual economic behavior observed may appear consistent with the behavior of policymakers, based on optimizing the simple or relatively more complex policy response. With the conditions outlined above, studies into how monetary policy should be applied optimally in a realistic economic modeling framework, congruent with the economic characteristics or fluctuations, are a strategic choice.

From the perspective of monetary policy directed toward inflation targeting, where the principles of policy transparency and credibility are imperative, the selection of rule-based policy is considered more relevant than discretion-based policy (Hubbard, 2002; McCallum, 2000). To that end, an alternative approach to study the application of optimal monetary policy is to look at alternative ways to determine the optimal monetary policy rule, namely by considering the aforementioned elements and forward-looking policy procedures.

7.3 Application in Several Countries

7.3.1. Operational Targets and Monetary Policy Instruments

In terms of monetary policy implementation, the central bank must choose an operational target along with monetary policy instruments to achieve the policy target. In line with the common operational framework, the policy instruments must be under the control of the central bank, while the transmission of changes in policy instruments to changes in the operational target, and the attainment of the final policy target must be managed well. In this context, the choice of the operational target and policy instruments depends on the financial market structure in each respective country.

7.3.1.1. Operational Target: Monetary Aggregates Versus Interest Rates. In practice, the central bank may choose alternative operational targets, including short-term interest rates, monetary aggregates, and exchange rates. Departing from trends in advanced countries (Australia, Canada, UK, and Japan), where short-term interest rates are the preferred operational target, but where monetary aggregates were previously used, developing countries, at least until the early 2000s (including México, India, Thailand, and Indonesia) continued to apply monetary aggregates. It is important to note that in line with commitment to apply Inflation Targeting, South Korea and Thailand also take short-term interest rates into consideration, along with bank reserves, as the operational target. In Indonesia, during the transition to inflation targeting through to July 2005, the operational target was still monetary aggregates, while short-term interest rates were used as a supporting operational target. Exchange rates are generally only used as a supporting operational target. Nonetheless, only countries with very open economies, such as Singapore and Hong Kong, still apply exchange rates as the operational target.

Monetary policy, as applied in several developing countries using bank reserves or broader indicators, such as the monetary base, is starting to garner attention. This possibly reflects the strong perception that bank reserves or base money can be used to predict broader monetary aggregate trends, such as M1, M2, and credit. Another interpretation is possibly that signals from price indicators are not as sufficiently strong as in advanced countries, where the financial systems are more established and stable. Particularly in countries that have experienced an economic crisis, including México, Thailand, and Indonesia, the drastic switch to interest rates when vulnerable to inflationary pressures would trigger shocks or distortions in terms of monetary policy implementation.

Congruent with the domestic structural economic changes and significant financial sector deregulation/liberalization policy, many central banks have finally realized that the monetary base is quite fluctuative and does not always accurately mirror M1 or M2 developments or economic dynamics in general. In addition, financial sector deregulation/liberalization tends to expand the role of the interest rate in the monetary policy transmission mechanism. Therefore, nearly all central banks ultimately selected short-term interest rates as the operational target, or at least as a supporting operational target, when applying monetary policy day-to-day. Among the few countries using monetary aggregates, interest rate fluctuations are still carefully taken into account to monitor daily liquidity.

Experience in México is the exception. In general, interest rates do not play a role in the monetary policy framework of México, even during the currency crisis in 1994/95. Strengthening the value of the peso, the Bank of México targeted the settlement balances of commercial banks at the central bank (positive or negative). Therefore, interest rates and exchange rates were left to fluctuate in line with market forces. Experience of this operational framework through to the end of 1998 showed that changes in the cumulative reserves target can drive faster and more significant price adjustments on the money and foreign exchange markets. Nonetheless, such conditions were disrupted when the financial sector in México faced disruptions at the end of 1998. The use of operational targets and supporting operational targets in several countries is presented in [Table 7.1](#).

In line with the growing trend of using short-term interest rates as the operational target, the question of which interest rate is best to use has emerged. In general, advanced countries and several developing countries apply the overnight (O/N) interbank rate as their first choice because the O/N rate represents the shortest term or maturity in the term structure of interest rates, which can be controlled relatively easily by the central bank. In addition, the O/N interbank money market is very liquid, with a dominant transaction volume in the economy. Theoretically, in its role as supplier or through its ability to influence demand for bank reserves through various policy instruments, the central bank can influence the O/N rate relatively accurately as well. According to the expectations hypothesis

Table 7.1: Monetary Policy Operational Targets (2015).

	Main Target	Supporting Targets
Advanced Countries		
United Kingdom	Overnight interest rate	—
United States	Overnight federal funds rate	—
Germany	Monetary Base	Money market rate
Japan	Overnight interbank rate	—
Australia	Overnight interbank rate (cash rate)	—
Canada	Overnight interbank rate	—
Emerging Market Economies		
México	Overnight interbank rate	—
Brazil	Overnight repo rate	—
India	Overnight repo rate	Repo/reverse repo
South Korea	7-day repo rate	Exchange rate
Thailand	1-day repo rate	—
Indonesia ^a	Overnight interbank rate	—

Source: Several IMF and BIS publications.

^aInterest rate used as the operational target since July 2005.

of the term structure of interest rates, changes in the O/N rate are transmitted to longer tenors, including deposit and lending rates. Ultimately, policy transmission through the interest rate channel is expected to effectively influence real sector activity.

Many central banks, however, tend not to focus exclusively on the O/N rate because it might not play a dominant role in the monetary policy transmission mechanism based on the structure and characteristics of the respective financial system. In this case, a central bank exercising control over the O/N rate may not produce the expected results for all maturities and, similarly, in terms of economic performance. Consequently, longer tenors are considered more relevant.⁵

The Bank of England in the 1990s, for instance, utilized short-term interest rates with a maturity of two weeks because the base rates of commercial banks correlated closely with relatively longer money market interest rate maturities. Nevertheless, the Bank of England currently uses the O/N rate. Considering the emergence of the FFR as *the* money market benchmark, it has naturally become the focus of the Federal Reserve when implementing monetary policy in the United States. Similar practices are also applied in several emerging market economies (EME), reflected in the tactical selection of each respective central bank, where the O/N interbank rate is typically favored.

Another interesting phenomenon is the proclivity of all central banks that apply full-fledged inflation targeting, with the exception of México, to use short-term interest rates as the operational target, with maturities ranging from O/N to 3 months. This trend implies that such central banks prefer to accommodate fluctuations in bank reserves or settlement balances. In this regard, a number of central banks adjust the short-term interest rate by considering the response to deviation in terms of inflation or inflation expectations to the target and output gap.⁶ In practice, however, the central bank complements that approach with judgment based on all information that supports the inflation forecasts. Meanwhile, central banks in EME rely more on qualitative information because model development in emerging markets tends to contain a comparatively larger amount of uncertainty than models developed in industrialized nations.

7.3.1.2. Monetary Control Instruments. In terms of the monetary policy instruments used, in practice, there are two types of instruments employed by central banks around the world, namely direct and indirect market-based instruments. The choice of policy instrument by each central bank depends on

⁵Conceptually, using interest rates with longer maturities (than overnight) has its own disadvantages. *First*, it is possible that the central bank wishes to dominate the supply side and demand has dwindled, therefore control over bank reserves would also decline significantly. *Second*, because the central bank uses information on interest rates to gauge market expectations, the central bank pays less attention to the actual formation of interest rates. *Third*, a concentration on longer maturities tends to exacerbate overnight interest rate shocks. Refer to Van 't Dack (1999).

⁶Interest rate determination based on the response is fundamentally similar to the Taylor rule.

respective financial system development. At the end of the 1970s, central banks in advanced/industrialized countries began to move away from direct instruments for monetary control in favor of indirect instruments. This trend was followed in subsequent years by developing countries and transition economies⁷ due, primarily, to increasingly open global economic dynamics, which drove financial market development and subsequently made direct instruments ineffectual because of inefficiencies and the trend of disintermediation.

Naturally, the transition from direct to indirect policy instruments required in-depth preparations. The IMF, in 1995, published the results of a study regarding the experiences of 19 developing and transition countries that had changed from using direct to indirect instruments. The transition period was assumed to begin when central bank or government securities were sold (auctioned) for the first time and ended when interest rate policy controls were terminated and the credit program accounted for less than 25% of total credit in the economy. The study concluded that the average transition period was 3.7 years from direct to indirect policy instruments, substantiating the fact that a protracted period of time is required to refine the expertise and institutional arrangements to use indirect instruments, as well as the behavioral change required in terms of interest rate determination and competition on financial markets. In addition, other supporting policies are required, including payment system development, enhanced bank supervision quality, overcoming the barriers to competition in the banking sector, and technical development of monetary analysis.

In practice, the common direct instruments include credit ceilings, controlling interest rates, direct credit, and statutory liquidity ratios. On the other hand, the most common indirect policy instruments include primary reserves, reserve requirements, discount facility and open market operations (OMO). Besides, rediscount facility, credit auctions, and foreign exchange operations (swaps) are also commonly used.⁸ Nonetheless, the main instruments used as the backbone of monetary policy implementation are somewhat different. Most central banks apply OMO as the primary policy instrument, while some jurisdictions have used other instruments, such as the discount facility applied in Brazil.

The widespread use of OMO by most central banks in advanced as well as developing countries is understandable because OMO are very flexible and the involvement of banks and brokers (second parties) is not binding. Furthermore, central banks can control the frequency of OMO and the number of auctions desired, which is very beneficial in terms of stabilizing the monetary base or short-term interest rates. In addition, of no less importance is that OMO do not place a tax burden on the banks. The main instrument of OMO are government debt obligations, such as T-Bills, complemented with central bank securities. The

⁷For more information, refer to "The Adoption of Indirect Instruments of Monetary Policy," IMF Occasional Paper No. 126, 1995.

⁸A detailed review of monetary policy instruments is available in "Monetary Control Instruments," by Ascarya, Centre for Central Banking Studies and Education, Bank Indonesia, 2002.

use of government securities will considerably alleviate the cost of OMO borne by the central bank.

Table 7.2 presents the general monetary policy instruments used in various countries. From the table, developing countries tend to follow advanced countries in terms of selecting monetary policy instruments consistent with the respective level of financial market development. Nevertheless, developing countries generally also use other supporting instruments in relation to the transition from direct to indirect instruments per the level of development in the respective jurisdiction. For example, India used a Switching Facility instrument until 1992. Through the switching facility, the Reserve Bank of India (RBI) purchased notes at a low coupon rate and sold/replaced them with notes at a higher coupon rate in order to improve the portfolios of banks and other financial institutions. In addition, more specific policy instruments are still applied to attain certain targets. Thailand, for instance, uses the credit ceiling to allocate financing to priority economic sectors.

Concerning the application of reserve requirements, however, developing countries tend to use lower ratios than advanced countries. Meanwhile, since the beginning of the 1990s, advanced and developing countries have been inclined to lower their respective reserve requirement. For example, the reserve requirements applied in the United States, Germany, South Korea, and India have decreased significantly over the past two decades, while primary reserves in the UK are approaching zero. Therefore, primary reserves are not actually part of monetary policy implementation, merely a form of tax revenue from the banking sector. In fact, several countries, including New Zealand, Canada, and Sweden, have abolished primary reserves as the effectiveness of the instrument faded, in terms of monetary policy, in line with financial sector innovation. Experience has shown that the position of bank reserves determined by the central bank tends to decrease toward zero when the central bank applies short-term interest rates as the operational target of monetary policy.

Conceptually, lower reserve requirements have two implications that must be addressed by the central bank to apply effective monetary policy. *First*, in an environment of non-binding reserve requirements, there are fundamental linkages between the structure of the payment system and the implementation procedures of monetary policy. In this regard, if strong demand for central bank balances is driven to meet payment needs rather than the reserve requirement, a change in the payment system structure becomes important in the design of monetary policy implementation. *Second*, a system in which the reserve requirement is low or even zero will exacerbate short-term interest rate shocks because of the relative difficulty in predicting demand for reserves or the lower demand sensitivity of reserves to interest rates.⁹ From the experience of New Zealand, Canada, and Sweden, it can be concluded that there are tight linkages between changes in payment system structure and monetary policy operating procedures. Nonetheless, there is also a weak correlation between reserve requirements and the degree of short-term interest rate volatility. In this case, interest rate volatility occurs when

⁹For more comprehensive analysis, refer to Sellon and Weiner (1997).

Table 7.2: Monetary Policy Instruments in Several Countries (Until 2004).

Country	Reserve Requirement	Discount Facility	Open Market Operations	Other Instruments
<i>Advanced Countries</i>				
United Kingdom	Not Applicable	LOLR	Government securities, private securities	Not applicable
United States	3% and 10%	Short-term, as required	Government securities, bonds, notes	Not applicable
Germany	1%	3 Months, medium-term, long-term	Government securities	Credit extension facilities
Japan	0.75%	Per bank, with discretion	Government securities, bonds, CDs	As required
Australia	Not applicable	Overnight	Government securities, private securities	Foreign Exchange Swaps
Canada	Not applicable	Short-term, as required	Government securities, private securities	LVTS

Developing Countries

Brazil	25–45%	Short-term, as required	Government securities, Central Bank securities	Overdraft facilities, credit extension facilities, credit auctions
México	10.5%	Emergency	CETES, BONOS	CDs, emergency funding facilities
South Korea	0–7%	Emergency, to meet the RR	Government Bonds, MSBs	Credit auctions
Thailand	6%	7 Days, LOLR	Government securities, Central Bank securities	SLR, FXW
Indonesia	6.5% SR: 1–3%	Emergency, short term <3 months	Central Bank securities, money market securities	Government deposits, credit liquidity, FXO
India	4%	Short term, 4–8 weeks	Government securities	SLR, switching facility

Source: Several IMF and BIS publications.

Notes: LOLR – lender of last resort, namely a central bank function that offers loans to banks that are experiencing financial difficulties; Government Securities – government securities (T-Bills); Central Bank Securities – central bank securities; Money Market Securities – money market securities; Private Securities – private securities; CDs – certificates of deposit; LVTS – large-value transfer system, a direct method to adjust the cost of money market operations through special purchase and resale agreements as well as sale and repurchase agreements; CETES – government securities, such as T-Bills, released by the Ministry of Finance of México; BONOS – bonds issued by the Government of México; MSBs – monetary stabilization bonds issued by the Bank of Korea (similar to central bank securities); SR – secondary reserves, supplemental reserves to the mandatory primary reserves, on which the banks receive remuneration (interest); SLR – statutory liquidity ratio is a direct instrument used by the central bank to require other banks to maintain primary reserves and certain securities or currencies at a specified percentage (typically government debt); FXW – foreign exchange window. Here, the BANK of Thailand administered an exchange equalization fund (EEF) to buy and sell USD/baht at a previously announced rate; FXO – foreign exchange operations or rupiah intervention.

the country is influenced more by institutional arrangements in terms of providing and absorbing liquidity rather than the lack of reserve requirements.¹⁰

7.3.2. *Monetary Policy Response: Policy Rules*

As mentioned previously, academics and central bankers believe that there are numerous benefits to using a rule when determining the optimal monetary policy response. In practice, however, no central bank is fully committed to following a particular rule (Svensson, 2002). In general, a rule is used as an implicit guide underlying policymaking. Several studies on the application of monetary policy rules have been conducted intensively (Clarida et al., 2000; McCallum, 2000; Taylor, 1993), showing that a rule can at least provide a rough direction in the policy-making process in advanced countries, including UK, US, Germany, and Japan.

Counterfactual analysis of historical data to implicitly observe the application of policy rules by central banks was conducted by comparing the behavior of determining a variable or instrument with the value specified by the rule in response to prevailing conditions. In this regard, whether erroneous policymaking, through an *ex post* assessment, could be detected by a rule was tested. The Taylor rule and McCallum rule were found, as candidate rules, to play a role in policy formulation.

The original application of the Taylor rule for the US economy was as follows:

$$i_t = r^* + \Delta p_t^a + 0.5(y_t - y^*) + 0.5(\Delta p_t^a - \pi^*) \quad (25)$$

where $r^* = 2$ and $\pi^* = 2$. This shows confidence that: (1) the average real interest rate value in the short term is predicted at around 2%; and (2) the annual inflation target set by the Federal Reserve is around 2%. Although the target varies according to prevailing dynamics, 2% is considered an acceptable target. Meanwhile, the original application of the McCallum rule in the US was as follows:

$$\Delta b_t = \Delta x^* - \Delta v_t^a + 0.5(\Delta x^* - \Delta x_{t-1}) \quad (26)$$

The nominal GDP growth target, Δx^* , is determined by $\pi^* + \Delta y^*$, namely the sum of the annual inflation target plus average annual real GDP growth in the long term.

7.3.2.1. Implicit Interest Rate Policy Rule in Several Countries. A study by Clarida et al. (2000) provided a complete analysis framework for monetary policy implementation in the United States during the period from 1990 to 1996. The backward-looking Taylor rule used was specified as follows:

$$i_t = r^* + \beta(\pi_{t-1} - \pi^*) + \gamma((y_{t-1} - y^*) / y^*) \quad (27)$$

¹⁰Sellon and Weiner (1997).

If the interest rate target is formulated in real terms, it implies that the nominal interest rate, i_t , will be deflated with inflation, thus:

$$r_t = R^* + (\beta - 1)(\pi_{t-1} - \pi^*) + \gamma((y_{t-1} - y^*)/y^*) \quad (28)$$

Interest rate smoothing is also assumed to occur, where the Fed calculates the interest rate adjustment, i_t , to its target after several periods.

$$i_t = \rho(L)i_{t-1} + (1 - \rho)i^*, \quad \text{where } r = r(1) \quad (29)$$

Therefore, the following reaction function is produced:

$$i_t = (1 - \rho)[R^* - (\beta - 1)\pi^* + \beta\pi_{t-1} + \gamma((y_{t-1} - y^*)/y^*)] + \rho(L)i_{t-1} + \varepsilon_t \quad (30)$$

Solving the equation produced the following results:

The results provided a particularly instructive description of monetary policy implementation by the Federal Reserve from the 1960s until the 1990s. In this case, significant disparities were found between monetary policy implementation before and after 1979. *First*, the inflation target during the first period (1960–1979) was higher than during the second period (1979–1996). *Second*, the value of β during the second period (<1) was significantly larger than the β value during the first period (>1), signifying the importance of how monetary policy responds to inflationary shocks.

With $\beta < 1$, the central bank reacts to higher inflationary pressures by raising the nominal interest rate slightly, thus lowering the real interest rate. Such conditions show that the destabilization process continues. The reverse occurs during the second period, however, when a hike to the nominal interest rate is of relatively greater magnitude than the increase in inflation, therefore the real interest rate tends to increase. Such results clarify the general question about why, since the leadership of Paul Volker, the monetary policy focus has been on more organized inflation control.

Table 7.3: US Monetary Policy Reaction Function.

	π^*	β	γ	ρ
Pre-Volker	5.95	0.86	0.39	0.68
(1960–1979)	(1.92)	(0.07)	(0.08)	(0.05)
Volker–Greenspan	4.08	1.72	0.34	0.71
(1979–1996)	(0.56)	(0.28)	(0.19)	(0.05)
Post-1982	2.96	2.55	-0.15	0.89
	(0.27)	(0.56)	(0.28)	(0.03)

Source: Clarida et al. (2000, Table IV).

In terms of output stabilization as the focus of attention, the use of a more active approach ($\gamma > 0$) was apparent in the pre-Volker era. Meanwhile, after 1982, the influence of the output gap had a negative sign and was not significant. The high ρ value shows that interest rate smoothing plays an important role in the decision-making process of the Federal Reserve. This strategy became even more important in the subsequent periods, except from 1979 to 1982, when the Fed operationally practiced monetary targeting (monetary base), which ultimately spurred larger interest rate shocks. Furthermore, observations failed to pick up clear indications that the output gap played an important role in terms of monetary policy implementation by the Federal Reserve during the period after 1979. It is interesting to note that large deviation was observed in the interest rate compared to the normative channel based on the Taylor rule, particularly during several episodes prior to 1987 and in 1992, when John B. Taylor revised his rule.

Despite monetary targeting, operationally, monetary policy in Germany was not directed by a simple rule linked to the monetary base as the operational target, for instance the McCallum rule. Nonetheless, certain trends were believed to steer decisionmaking at the Bundesbank. In that respect, as occurred in other industrialized nations, the Taylor rule was considered to play an important role in the operational framework of monetary policy in Germany, where the short-term interest rate was used as a supporting target.

Analysis conducted by the Bundesbank (Monthly Report, 51(4), 1999) also showed that during the period after 1987, interest rates in Germany were more aligned with the normative channel determined by the Taylor rule, assuming an average real interest rate of 3.4%. Meanwhile, using the original Taylor rule, Bofinger (2001) showed that during the 1970s and first half of the 1980s, interest rate behavior, as determined by the Bundesbank, was relatively aligned with that determined by the Taylor rule.

In terms of the UK economy, as experienced by several other central banks as well, the decision-making process at the Bank of England was also directed by an iteration of the Taylor rule. McCallum (2000) analyzed the behavior of UK interest rates based on the Taylor rule, using an average interest rate value, r^* , of 2.25% and inflation target, π^* , of 2%. The analysis showed that monetary policy in the 1970s was considered too loose, contrasting monetary policy from 1983 to 1987, which was deemed excessively tight. Furthermore, during the period after 1987, with 1994 as the only exception, the monetary policy response instituted by the Bank of England was considered relatively optimal. Bofinger (2001) also showed that short-term interest rate trends in the UK after the application of inflation targeting could be replicated using the original version of the Taylor rule.

The Taylor rule also explains the salient periods of monetary policy application in Japan (Bofinger, 2001). After the collapse of the Bretton Woods system (1973–1978), nominal interest rates were too low and, therefore, real interest rates became negative. Consequently, inflation soared, primarily during the years after the first oil price shock (1973–1975). During the 1980s, however, monetary policy

was comparatively restrictive, where the nominal interest rate was higher than the normative interest rate per the Taylor rule.

From 1985 to 1995, the Taylor rule could also explain the interest rate policy of the Bank of Japan (BOJ). Since 1995, Japan has applied a zero interest-rate policy (ZIRP), close to the normative interest rate per the Taylor rule. This demonstrates the BOJ's efforts to provide strong monetary stimuli in an economy experiencing deflation. In that context, relatively small deviation of the actual interest rate to the normative interest rate per the Taylor rule from 1989 to 1995 also showed that persistent deflation in Japan during the 1990s was not caused by erroneous monetary policymaking.

7.3.2.2. Role of Monetary Aggregates When Applying an Implicit Rule.

Congruent with the shift in the preferred operational target from monetary aggregates (money base) to short-term interest rates, particularly in advanced countries, use of the interest rate-based Taylor rule has outstripped use of the McCallum rule based on the money base. Nonetheless, this does not imply that the McCallum rule has no place in monetary policy application. To test that role, in a counterfactual study, McCallum (2001) compared the historical performance of the Taylor rule and McCallum rule in three advanced countries, namely the United States, United Kingdom, and Japan.

In the US case, both rules indicated excessively tight monetary policy during the 1970s, with the McCallum rule providing stronger indications than the Taylor rule. During the 1980s and 1990s, disparities emerged between the two rules, while during the period after 1987, both rules indicated undesirable monetary policy. In the UK, both rules showed excessive inflationary pressures during the 1970s but differences appeared in the 1980s, when the McCallum rule indicated excessively loose monetary policy during the middle and toward the end of the decade, while the Taylor rule did not. In Japan, the focus of attention concentrated on the period since the 1990s. In this case, both rules indicated overly tight monetary policy in 1998, but the McCallum rule indicated excessively tight monetary policy from 1990 to 1998, while the Taylor rule did not.

Operationally, the role of monetary aggregates can also be seen in the context of applying monetary policy in situations that require the central bank to support financial system stability. Interestingly, McCallum (1994) found that central banks possibly apply a macro-oriented monetary policy rule together with the role as lender of last resort (LOLR) in the financial system. In this respect, McCallum proposed a procedure using both interest-based rules and monetary based rules simultaneously in the case of the United States. Technically, the system consisted of: (1) a policy rule for the FFR designed to achieve the monetary base target; and (2) a policy rule that highlights the response of monetary growth to changes in the FFR. From the observations, McCallum concluded that the procedures explored could accommodate the macroeconomic goals together with the central bank acting as LOLR during the periods from 1974 to 1979 (September) and 1988 to 1991.

7.4. Monetary Operations in Indonesia

7.4.1. *Quantity-based Approach*

Monetary policy implementation prior to the crisis referred to Act No. 13 of 1968 concerning the Central Bank, which stipulated the final targets (plural) of monetary policy at BI. In addition to low and stable inflation, BI was also mandated to achieve robust economic growth, low unemployment and a zero balance of payments (BOP). Operationally, however, BI relies heavily on base money (M0) as the operational target, with the nominal exchange rate target used as the policy anchor. In this case, the exchange rate is tightly controlled within a narrow band and depreciated at a relatively constant rate (managed exchange rate system). According to that monetary policy strategy framework, a controlled monetary base is expected to translate into controlled money supply (M1 and M2) and, therefore, economic activities (economic growth and prices).

The quantity-based approach may be adopted with some considerations. *First*, when economic and financial conditions are undergoing structural change, as in the case of Indonesia, BI needs an indicator that can be controlled, namely the money base. *Second*, base money developments also influence money supply, output and inflation.

Operationally, the process of monetary control begins with monetary programming. Fundamentally, monetary programming is policy planning to control total money supply in order to achieve the final target of monetary policy. Monetary programming includes determining the operational target of monetary policy.¹¹ Thereafter, BI determines the measures required and the instruments to be used to influence the operational target.

The monetary instruments available to influence the operational target are OMO, the discount facility, reserve requirements and appeals. OMO are conducted through auctions of securities, which aim to increase or decrease liquidity on the money market. Meanwhile, the discount facility is a credit facility offered to financial institutions with the discount determined by BI. The reserve requirement is the minimum amount of liquid assets that must be maintained by the banks held at BI. Finally, appeals are made by BI so that all banks are on the same page and can follow the monetary policy measures expected by BI.¹²

¹¹When preparing monetary programming, the operational target is determined using the following assumptions:

- (a) Policy and developments in other sectors (fiscal, trade, investment, etc.) will progress as expected.
- (b) A stable relationship between money supply (as the intermediate target) and the real economy (as the final target). Such conditions require stability in terms of income velocity and demand for money.
- (c) A stable relationship between the monetary base (as the operational target) and money supply (as the intermediate target). Such conditions require money multiplier stability.

¹²For an in-depth explanation, refer to the Central Banking Book Series, No. 3, Monetary Control instruments, by Ascarya, PPSK-Bank Indonesia, 2002.

It is understood that the relationship between monetary policy aggregates, output, and inflation in the near term may not have been stable during past episodes of financial liberalization and the recent crisis periods.¹³ Congruently, a new way of thinking emerged concerning an alternative approach considered more appropriate to the monetary policy system in Indonesia, namely the price-based approach.¹⁴ This approach emphasizes the role of prices in terms of monetary aggregates, namely the interest rate as a salient variable in transmitting the impact of monetary policy to economic activity.¹⁵

A new way of thinking emerged with the development of the financial sector. With the emergence of the financial sector came features such as advanced financial product innovation securitization, and the loss of linkages between the monetary sector and real sector, which had adverse implications for monetary policy implementation. Several supporting research papers, focused on Indonesia, showed that rapid change in the financial sector of Indonesia caused a twofold increase in money creation by the financial system beyond that created by the central bank. Such developments created an unstable money multiplier.¹⁶ Furthermore, instability not only affected the money multiplier but also income velocity and the demand for money function.¹⁷

7.4.2. The Shift from the Quantity-based Approach to the Price-based Approach

Understanding that the ultimate targets are diverse, focused monetary policy is difficult to implement. Moreover, operationally, monetary policy that relies on the monetary base also has several inherent problems. Although the quantity-based approach is considered effective over a longer time horizon, the approach has begun to face stiffer challenges, especially since the beginning of the 1990s. Extremely rapid money market development due to deregulation and greater economic integration internationally has destabilized the relationship between monetary aggregates and output and inflation. Consequently, quantity-based monetary policy has become less effective. Addressing that challenge, BI subsequently adopted a more pragmatic (eclectic) policy framework. Without abandoning the quantity-based approach, more attention was put on interest rate developments, while the extent of BI intervention bands on the foreign exchange

¹³Such phenomena have long been found in advanced countries and are the primary reason the quantity-based approach was abandoned in the monetary policy framework.

¹⁴This idea was delivered by Boediono at the seminar on monetary policy mechanisms in Indonesia, held in May 1998. Refer to the Monetary and Banking Economic Bulletin (1998).

¹⁵The rationale to use the interest rate as the operational target emerged prior to the crisis in 1997.

¹⁶Sarwono (1996).

¹⁷Solikin (1998).

market under a managed exchange rate regime was broadened to reduce the cost of monetary policy. Ultimately, in August 1997, BI abandoned the intervention bands and inevitably applied a floating/flexible exchange rate system.

In 1998, however, BI, which had lost control of exchange rates when forced to disburse large-scale liquidity facilities to the banking industry to unwind the Asian financial crisis of 1997/98, began to apply a tight monetary policy stance. The monetary policy stance was tightened to prevent an escalation of inflationary pressures, by attempting to stop all forms of monetary expansion to prevent excess liquidity in the economy. In this case, monetary base targeting was reinforced through a series of policy innovations and improvements. A stiff penalty system for negative bank balances at BI was also introduced to dampen bank demand for the liquidity facilities provided by BI. Limits were placed on deposit rates and interbank lending guaranteed by the Government, which eased the risk of liquidity flows from Bank Indonesia. Meanwhile, the auction system for Bank Indonesia Certificates (SBI) through OMO was changed from targeting the interest rate to targeting the quantity. Thereafter, the “rupiah intervention” instrument was created (subsequently named Bank Indonesia Facilities – FASBI) as an additional contractionary and expansionary instrument to alleviate interest rate shocks on the money market.

Under the floating exchange rate system, rupiah exchange rate fluctuations on the market became particularly vulnerable to market sentiment. To reduce excessive currency shocks, BI conducted intervention policy, at that time, by focusing on the lower bound of Net International Reserves (NIR). The actual NIR value is announced publicly on a weekly basis and has never dropped below the lower bound. In terms of the relationship with the management of monetary aggregates, foreign exchange sterilization has a contractionary impact on the money base and, therefore, can be considered an instrument to bolster OMO.

Crisis experience provided several invaluable lessons for BI. In terms of monetary management, after losing control of the exchange rate in August 1997, monetary policy lost its exchange rate anchor during the crisis. Consequently, BI applied monetary base targeting as the concurrent operational target and nominal anchor of monetary policy. The effectiveness of the monetary base as the sole indicator and, therefore, anchor of monetary policy in Indonesia was also called into question. As was alluded to previously, the quantity-based approach was used as the monetary policy framework during the crisis period because of monetary leakage that had to be absorbed due to excess liquidity in the banking industry as a result of the liquidity facilities provided through BI's function as LOLR, not because of the more fundamental consideration of the stable relationship between inflation and monetary aggregates. In fact, several research papers have debunked the assumptions underlying monetary base targeting, for instance that the money multiplier and velocity of money are inadequately stable in the case of Indonesia. The monetary policy framework using this program was subsequently formalized as part of the IMF program, where achievement of the monetary base target was considered a viable performance criteria indicator.

Monetary base targeting could not guarantee sound attainment of the inflation target however. The low performance of monetary base targeting was primarily

due to difficulties predicting public behavior in terms of holding currency. Post-crisis public demand for currency experienced a structural change that was difficult to explain from the motive perspectives of transactions or for a rainy day. Control over the monetary base was also considered more difficult when the bank intermediation function was not normal. Under such conditions, raising the SBI interest rate monetary instrument (policy rate) to absorb currency back into the banking system was oftentimes constrained by the limited deposit rate response, which meant that coercing currency back into the banking system required a hike to the policy rate of higher magnitude that would normally be expected.

This reality caused BI to regularly face a monetary policy dilemma. On one hand, to reduce demand for base money, BI had to adopt a tight policy stance, which raised interest rates. On the other hand, however, the ineffective transmission of monetary policy implied that the tight monetary policy stance would require a much larger hike to the interest rate for a positive impact on the economic recovery. When banking and corporate sector conditions were comparatively weak, however, the adverse impact of higher interest rates on the economy was much more pronounced than under normal conditions (Warjiyo & Agung, 2002). The dilemma made it difficult for BI to fulfill its monetary policy commitments to achieve the monetary base target set.

Difficulties controlling the monetary base were also found when the money base fell well short of its target, as transpired in 2002. Efforts to grow the monetary base were ineffective when the banking industry was still weak and real sector risks were considered high. While the bank intermediation function was not considered normal, additional economic liquidity through the banking sector would flow back to the central bank. Therefore, the monetary base is said to be more demand determined than supply determined.

The problems that emerged in relation to monetary base targeting as the operational target of monetary policy corroborated the implications presented in preliminary studies, which recommended a price-based approach to monetary policy, using the interest rate as the operational target (Solikin, 2005; Warjiyo & Agung, 2002). This is also supported by the empirical fact that, in the case of an emerging economy, the characteristics of base money are more endogenous and, therefore, the indicator is less flexible as the operational target. In contrast, however, considering that the monetary base and interest rate mutually respond to one another, simultaneously targeting the money base and interest rate, as has occurred recently, tends to trigger problems relating to policy transparency and consistency. Fundamentally, the implications stress the importance of BI focusing the orientation of the operational target on just one indicator, namely the interest rate. Congruent with ITF application since July 2005, BI officially applies the interest rate as the operational target of monetary policy in Indonesia.

Consequently, two questions emerge, namely which interest rate indicator should be used as the operational target? And which interest rate is representative of the policy rate and can therefore reflect the monetary policy stance? Studies specifically addressing the choice of an operational target for monetary policy under an inflation targeting regime found that the O/N interbank rate was a feasible operational target of monetary policy based on controlling the interest

rate under an inflation targeting regime. In addition to the findings of previous research that concluded several advantages of using the O/N interbank rate per international best practices, such as measurability, ability to affect the ultimate target and easily understandable by the public, the study also found strong indications that BI could control the O/N interbank rate (controllability) through the signaling and liquidity adjustment mechanisms. Other important empirical evidence includes low O/N interbank rate volatility during the observation period, despite not targeting the interest rate in the operational framework at that time (BI, 2004b).

In terms of selecting a representative interest rate for the policy rate, there are several schools of thought, specifically concerning the criteria and essence of the policy rate itself. There are at least three salient criteria for interest rate indicators that could be used as the policy rate, namely that the interest rate indicator selected must be controllable by BI, can influence the cost of funds in the banking industry, and can effectively be used as a reference rate by economic players in their economic decision-making process. Those criteria are congruous with the essence of the policy rate, namely that the policy rate is part of the (transparent) communication strategy, is a variable that reflects the monetary policy direction (as the simplest form of central bank communication), can be steered to influence the (pricing) expectations behavior of market players, indicates the level of short-term interest rates desired by the central bank, reflects the central bank's response to various macroeconomic indicators, and is not the actual interest rate on the money market. Based on those considerations, BI chose the BI rate as the policy rate. In terms of OMO, the BI rate was used, until the middle of 2008, as the reference rate for auctions of 1-month SBI. Since 2008, however, the BI rate has been used as a reference rate for the O/N interbank rate.

7.5. Concluding Remarks

Academically, analysis of and support for rules-based policy strategy using the Taylor rule (Taylor, 1993), at least in the context of advanced countries, has increasingly reverberated considering the main problems associated with fine tuning a discretion-based policy strategy. The issues include difficulties formulating stabilization policy in terms of policy timing and elements of uncertainty in terms of the policy influence, as well as the tendency for time-inconsistency and inflationary bias problems to emerge. Meanwhile, in the case of emerging economies, Taylor (2000) stressed that rules are also very useful but require modification to the unique characteristics of the economy involved. In fact, Taylor (1996) also emphasized the flexibility by referring to conditions where central banks have abandoned rules, in addition to the possibility of applying discretion combined with rules.

For the Indonesian economy, there remains broad research scope concerning the possible application of monetary policy strategy using rules, or even

state-contingent policy rules as mentioned by Keynes three decades ago.¹⁸ What is meant by state-contingent rules? King (1997) explained that

The choice of monetary strategy is often described as a choice between rules and discretion ... The optimal strategy is a state-contingent rule, which allows flexibility in the response of policy to shocks while retaining a credible commitment to price stability.

Meanwhile, Woodford (2003) stressed that “The fully optimal policy commitment would lead not only to the optimal long-run average values of target variables, but also to the optimal responses to disturbances.” Therefore, state-contingent rules are considered an optimal policy strategy that allows the central bank to respond to shocks without neglecting the target of price stability.

The choice of policy strategy will depend on the characteristics of the respective economy and the monetary policy response. For example, Indonesia’s economy is unique and has experienced economic gyrations over the past three decades.¹⁹ Paying due consideration to large macroeconomic fluctuations as a consequence of structural changes in Indonesia, particularly during the past decade, along with the reorientation of monetary policy strategy with the final target of price stability, monetary policy strategy formulation using state-contingent rules, which can accommodate rules and discretion, is incredibly relevant. Therefore, one salient issue for further study is how, technically and operationally, to formulate the state-contingent rules.

In his pioneering frontier study on formulating state-contingent rules for the Indonesian economy, Solikin (2004b) showed that policy response formulation based on state-contingent rules successfully represented an optimal design of policy rule for the Indonesian economy. The policy rule design accommodated the long and short-term policy responses with an error correction mechanism, as well as the presence of specific shocks that surfaced due to sudden structural changes and dynamic changes because of elements of uncertainty that affected monetary policy response volatility.

¹⁸Boughton.

¹⁹Several studies on the (possible) use of rules (Taylor rule) in Indonesia have been conducted, including Darsono et al. (2001) and Hutabarat (2002) of the Directorate of Economic Research and Monetary Policy – BI. the studies focused on the relevance and use of rules with the possible implementation of the ITF.

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Chapter 8

Inflation Targeting Framework: Concept and Implementation at Central Banks

8.1. Introduction

Since the beginning of the 1980s, a growing number of central banks in various jurisdictions have applied a monetary policy regime or framework based on inflation targeting. In advanced countries, the Inflation Targeting Framework (ITF) has been implemented, among others, in New Zealand (1990), Canada (1991), Israel (1991), United Kingdom (1992), Sweden (1993), Finland (1993), Australia (1993), and Spain (1994). On the other hand, in emerging market economies (EMEs), ITF has been applied in several Latin American countries, including Chile and Brazil, in Eastern European countries, such as Poland and the Czech Republic, as well as in a number of Asian countries, like South Korea, Thailand, the Philippines, and Indonesia. Furthermore, ITF has also become a hot topic in numerous discussions and research papers authored by economists and as educational material to be explored at various universities.

In the context of Indonesia, ITF was implicitly introduced in the year 2000 upon enactment of the BI Act (No. 23) of 1999 on May 17, 1999. Pursuant to that law, BI was mandated with achieving rupiah currency stability, meaning price (inflation) stability, and rupiah exchange rate stability. Seeking to achieve that goal, monetary policy is consistently directed toward attaining the published inflation target. In addition, Bank Indonesia also independently formulates and implements monetary policy as well as the other duties stipulated in the Bank Indonesia Act. Notwithstanding, BI cannot extend loans to the Government. Those provisions represent the salient characteristics and form the legal basis of ITF implementation by BI in Indonesia. Indirectly, the legislation contained within the BI Act mandates that monetary policy be based on the framework known as ITF in theory and as practiced by other central banks. Nevertheless, congruent with the protracted economic recovery process after the Asian Financial Crisis of 1997/1998 that persisted into the 2000s, ITF was implemented gradually through a transition period and officially applied in full on July 1, 2005.

This chapter will discuss the ITF-based monetary policy framework along with its application at various central banks, while implementation in

Indonesia will be explored in the subsequent chapter. The discussion is divided into seven sections. After this introduction, the second section will present a detailed overview of the ITF, including the rationale and characteristics as well as advantages and disadvantages. The third section will discuss the institutional arrangements, focusing on provisions in the central bank statutes, inflation target formulation as well as accountability and transparency. The fourth section will describe the operational arrangements of ITF, specifically inflation projection techniques, monetary policy transmission mechanisms and monetary operations. The fifth section will discuss the most common ITF regimes applied around the world, namely full-fledged inflation targeting (FFIT), eclectic inflation targeting (EIT), and inflation targeting lite (ITL). The sixth section will explore the contribution ITF implementation has had on economic performance in various countries, particularly inflation, macro-economic performance, and in terms of the central banks' proclivities to monetary policy. To conclude, the seventh section will present several important takeaways from ITF application.

8.2. Conceptual Dimension and Theoretical Model

8.2.1. Rationale, Characteristics, Advantages, and Disadvantages

8.2.1.1. Rationale. Fundamentally, ITF is a framework where monetary policy can be directed toward achieving a future inflation target that is published transparently as a tangible form of central bank commitment and accountability. This definition contains several implications. *First*, there must be an official statement, supported by central banking laws, that the final goal mandated by the public in terms of monetary policy is to achieve and maintain price stability/low inflation. *Second*, a medium–long term inflation target must be determined and announced as an anchor of monetary policy. *Third*, there must be commitment and discipline from the central bank to steer monetary policy toward the inflation target set, while enjoying full independence to formulate and implement monetary policy, without interference from the government or a third party. And *fourth*, there must be monetary policy transparency to the public and accountability mechanisms that hold the central bank accountable for its monetary policy to the public in pursuance of the price stability mandate given.

In theory and in practice at various central banks, there is different rationale underlying ITF implementation. *First*, a growing body of empirical evidence has shown that monetary policy only influences prices in the long term, although monetary policy can affect economic growth (real output) in the near term. *Second*, there is an increasing requirement to base monetary policy formulation and implementation on a specific framework to enhance central bank discipline, consistency, and credibility. *Third*, transparency and accountability of public policies, including monetary policy, have become increasingly important as the embodiment of good governance and democratic accountability in the government's institutional reforms, including the central bank. The following section will explain the rationale behind ITF implementation.

In Chapter 3, the effect of money on the economy was discussed, specifically the impact on prices and real output. Various empirical evidences have shown that money only influences prices in the long term, thus proving the neutrality of money on real output congruent with the quantity theory of money. Meanwhile, in the near term, money can influence real output to differing degrees depending on the various definitions of money supply, the analysis period, and cross-border analysis. The studies have provided empirical evidence that in the medium–long term, monetary policy must be directed toward price stability. Nonetheless, there is a trade-off in the near term between inflation and real output when price stability is set as the final target of monetary policy.

Over the subsequent few chapters, the need for consistent and transparent monetary policy implementation to enhance effectiveness and credibility in terms of influencing inflation and real output shall be discussed. The time-inconsistency theory proposed by Kydland and Prescott (1977) showed that monetary policy directed toward a medium–long term inflation target is more optimal than policy oriented toward a short-term inflation target because a short-term policy target would trigger time inconsistency and not take into consideration the policy implications on long-term public welfare. Meanwhile, Barro-Gordon (1983) proposed several requirements for monetary policy credibility as follows: (a) the central bank must publish its inflation target and the monetary policy direction taken to achieve said target; (b) the public must have confidence that the central bank will consistently orient monetary policy toward achieving the inflation target; and (c) the central bank can successfully achieve the inflation target set.

Monetary policy transparency could be contained in rational expectation hypothesis theory, indicating that only anticipated monetary policy has an influence on real output,¹ considering that under conditions of increasing uncertainty, economic players will tend to act rationally and take into account the direction of monetary policy implemented by the central bank when taking business decisions. Therefore, policy transparency will facilitate decisionmaking among economic players and avoid an unnecessary build-up of uncertainty that could arise if the direction and target of monetary policy are unknown. In other words, transparency can enhance monetary policy effectiveness in terms of influencing economic and financial activities to achieve the inflation target set.

¹This theory is discussed in more detail in Chapter 12 along with other theories underlying central bank monetary policy transparency. Fundamentally, the theories are linked to the problem of asymmetric information in monetary policy, primarily in the context of the focus and clarity of the central bank's propensity towards the goal of monetary policy, the relevance and magnitude of market expectations in response to and in terms of transmitting monetary policy as well as the debate between rules versus discretion in monetary management. The various theories have been motivated by the phenomenon of increasing monetary policy transparency at various central banks in line with the proliferation of central bank independence under democratic government rule, the application of a monetary policy framework based on inflation targeting, as well as the importance of managing expectations in increasingly advanced and complex financial markets.

Meanwhile, the importance of central bank independence and monetary policy accountability is based on agency theory in various public policies as a form of good governance and democratic accountability in the government system. In this respect, the public, as the principal, gives a mandate to the central bank, as the agent, to achieve and maintain price stability as the goal of monetary policy. On the other hand, to implement its public mandate, the central bank must be given full independence to formulate and execute its monetary policy. Per agent theory, the principal–agent relationship must also be accompanied by accountability mechanisms for the independence given to the agent, namely the central bank, in monetary policy to implement the mandate given by the agent, namely the public, toward the final goal of price stability. Furthermore, because public policy is involved, the principal–agent relationship in terms of mandating the goal, monetary policy independence as well as the accountability mechanisms are stipulated in central banking laws formulated and enacted through parliament.²

8.2.1.2. Characteristics. The rationale detailed above underlies the ITF-based monetary policy framework. In this regard, ITF is a monetary policy framework with a number of salient characteristics, namely an official statement from the central bank that the final goal of monetary policy is to achieve and maintain low inflation in the medium–long term, as well as to publically announce the inflation target. The announcement implies that the central bank is avowedly committed and assures the public that every monetary policy issued shall consistently strive to achieve the inflation target, while ensuring that the central bank is accountable for its monetary policy if the inflation target is not achieved. In other words, ITF application satisfies the principles of monetary policy oriented toward a medium–long term inflation target, with central bank commitment to focus its monetary policy on achieving the inflation target as well as giving independence together with increased transparency and accountability in the monetary policy implementation process by the central bank.

In more depth, the characteristics of inflation targeting per Bernanke, Laubach, Mishkin, Posen (1999) are presented in [Table 8.1](#).

[Table 8.1](#) presents the general characteristics associated with ITF. *First*, there is an official statement that price stability is the final goal of monetary policy and the inflation target is announced to the public. Regarding who sets the inflation target, whether it be the government or central bank itself, remains the subject of debate in theory and in practice. In addition, the inflation target should be determined in the medium term. *Second*, the central bank is given independence to formulate and implement monetary policy (instrument independence) to achieve the inflation target set. In this case, the monetary policy response will be based on inflation and output projections and could apply rules in its application. *Third*, there is monetary policy transparency with intensive public communication, not only concerning the inflation target but also explaining how the central bank

²A more detailed description of the theories and practices associated with independence and accountability at several central banks and at BI will be discussed in Chapter 11 of this book.

Table 8.1: Characteristics of Inflation Targeting.

Criteria	Bernanke et al. (1999)	Svensson (2000)	King (2004)
1 Price stability as the final goal of monetary policy	Yes	Yes	Yes
2 Inflation target announcements	Yes	Yes	Yes
3 Medium-term inflation target	Unclear	Yes	Yes
4 Intensive communication with the public	Yes	Yes	Yes
5 Use of a specific monetary policy rule	Unclear	Inflation projection targeting	Inflation targeting + supply-side response
6 Published inflation and output projections	Not required	Yes	Unclear
7 Government-set target (goal dependent)	Yes	Yes	Not required
8 Instrument independent	Yes	Yes	Yes

Sources: Bofinger (2001) and King (2004).

conducts monetary policy along with performance in terms of achieving previous inflation targets.

Mishkin (2004) suggested five core elements that reflect ITF application as follows: (1) the public announcement of medium-term numerical targets for inflation; (2) an institutional commitment to price stability as the primary goal of monetary policy, to which other goals are subordinated; (3) an information inclusive strategy in which many variables, and not just monetary aggregates or the exchange rate, are used for deciding the setting of policy instruments; (4) increased transparency of the monetary policy strategy through communication with the public and the markets about the plans, objectives and decisions of the monetary authorities; and (5) increased accountability of the central bank for attaining its inflation objectives. Those five aspects explain the key elements of ITF, which cover more than just announcing the future inflation target. It is important to understand, however, especially for EMEs, most of which routinely report the inflation target or other economic projections as part of the government's economic plan for the upcoming year, that such monetary policy cannot be categorized as ITF because it does not meet the four other requirements.

As emphasized by Bernanke et al. (1999), ITF is not a hard and fast rule but more of a framework for the central bank to formulate and implement its monetary policy. Furthermore, ITF application does not necessarily imply that

monetary policy only considers inflation, per the “inflation nutters” moniker suggested by King (1996).³ As explained in Chapter 7, the term “rule” in the monetary policy implementation strategy is linked to the theoretical thinking behind applying one operational target based on one rule or instrument. For example, the money rule à la Milton Friedman established that growth of the monetary base at a certain rate is based on the predicted requirement for money supply in the economy as the operational target of monetary policy. In addition, the interest rate rule à la Taylor linked a specific short-term interest rate as the operational target based on future projections of inflation and real output.

As a monetary policy framework, ITF represents a comprehensive regulatory framework to formulate and implement central bank monetary policy concerning institutional arrangements, monetary policy implementation, and organizational implications for the central bank (Schaechter, Stone, & Zelmer, 2000). As an institutional framework, ITF encompasses central banking legislation that stipulates price stability as the main target of monetary policy, central bank independence as well as restrictions on the central bank financing the Government’s budget deficit. In addition, determining the inflation target as well as the accountability and transparency mechanisms are also regulated.

As an operational framework, ITF covers monetary policy formulation and implementation strategies to achieve the inflation target set, including regulations on modeling inflation projections, monetary policy transmission mechanisms and the monetary operations available. In principle, based on an assessment of inflation projections (as well as other economic and financial variables) and the expected transmission mechanisms, monetary policy is formulated in such a way to guarantee that future economic and financial fluctuations remain within a certain corridor to achieve price stability. In its implementation, monetary operations are oriented toward achieving the given operational target, usually short-term interest rates, based on a specific rule, such as the Taylor rule. Nevertheless, the use of a rule is only one consideration underlying monetary policy formulation by the central bank in addition to various other economic and financial indicators. In other words, rules are only used as a guideline for consideration when determining the operational target of monetary policy, otherwise known as rule-constrained discretion in ITF implementation.

ITF also covers the organizational/institutional arrangements at a central bank required to support effective and successful implementation. This includes regulations concerning the monetary policy formulation process, which can guarantee that the policies to be instituted have been through a rigorous analysis process considering all the relevant data and information. In addition, committees are required with specific decision-making regulations to enhance the quality of

³In theory and in practice, when the inflation target is set, its impact on economic growth is always considered. When setting the inflation target in the context of the trade-off between inflation and output per the Phillips curve, refer to Chapter 7. The empirical evidence described in the sixth section of this chapter also shows how real output is considered when applying ITF in various countries.

monetary policy. Various adjustments to the internal organization of the central bank are also required, including data collection, inflation and economic projection modeling techniques as well as policy analysis, in addition to enhancing the competence of human resources.

8.2.1.3. Advantages and Disadvantages. The growing number of countries applying ITF demonstrates that the framework offers several advantages over monetary policy regimes using money supply, the exchange rate or other anchors (Bernanke et al., 1999; Leiderman & Svensson, 1995; Svensson, 1997) as follows:

- *First*, ITF helps divert public attention away from short-term interventionist policies, where monetary policy plays a smaller role, to the achievement of low and stable inflation in the medium–long term to support macroeconomic stability and economic growth.
- *Second*, ITF fundamentally increases monetary policy and fiscal policy accountability and discipline.
- *Third*, established ITF implementation provides space for the central bank to respond to short-term developments with a smaller risk to inflation credibility in the long term.
- *Fourth*, ITF helps to motivate institutional reforms at the central bank, while also encouraging more general structural reforms, especially in the context of lowering inflation (disinflation).

Of course, ITF implementation also has several potential weaknesses, including instrument instability if the framework is applied too strictly, the need for a relatively strong fiscal position and the possibility of higher short-term exchange rate volatility, especially in a small open economy (Schaechter et al., 2000).

8.2.2. Theoretical Model

As described in previous sections, there are at least three salient aspects in the implementation of monetary policy. *First*, expectations play a fundamental role in the workings of monetary policy. This is evidenced when determining the policy rate, for example. Expectations concerning how monetary policy will manifest in upcoming periods as well as the current interest rate are given the same degree of importance by policymakers. *Second*, there is understanding that economic dynamics will constantly evolve. Therefore, there are no hard and fast rules to support optimal monetary policy strategy formulation. *Third*, the complexity of optimal economic agent behavior allows the central bank to consider the possibility that the public applies simple and practical rules.

Congruent with that perspective, macroeconomic theory, particularly monetary theory, has developed with rapidity over the past three decades. Evidence that theoretical advancements have had a significant effect on monetary policy practices is relatively simple to find with a foothold in historical evolution and tied to the long-term future perspective. The growing trend of central bank independence, ITF implementation and the use of policy rules to guide monetary policy implementation represent tangible examples of the impact macroeconomic

theory has had on monetary policy practices. Although the substance or rationale behind ITF, especially in relation to building monetary policy credibility, can be explained by theories existing prior to the emergence of ITF, for instance, the theory of credibility and time consistency by Kydland and Prescott (1977), forming credibility and reputation by Barro-Gordon (1983), credibility and conservative/independent central banks by Rogoff (1985), the general view states that the growing number of central banks applying ITF is part of the “practices ahead of theory” phenomenon (King, 2005).

The practice ahead of theory phenomenon is based on guiding the achievement of today’s monetary policy goal on efforts to maintain price stability, a rationale that did not develop during the three decades prior to ITF proliferation. From the end of WWII until the mid-to-late 1970s, most economists, academics and policymakers held the same view, that monetary policy was less related to inflation and an ineffective instrument of monetary management. The underlying theoretical basis of that opinion is not particularly clear, but empirical experience has shown that price instability causes fluctuations in real output and expensive job opportunities. Meanwhile, monetary policy oriented toward price stability has been shown to be a key to success for aggregate demand management. Congruent with the empirical facts, theoretical support for monetary policy practices through evolution has been a significant acknowledgment during the past two decades, reflected by the accelerating development of theoretical views based on ITF implementation in various countries.

ITF, which was initially applied as an “art of central banking” with a grounding in strengthening the institutional framework of monetary policy to support policy credibility and manage expectations (Bernanke et al., 1999), gradually began to be considered through the foundational development of macroeconomic theory. In this case, there are two main schools of thought providing a foundation for ITF macroeconomic theory, namely New Keynesian (Clarida, Gali, & Gertler, 1999; Rotemberg & Woodford, 1997; Svensson, 1997) and New Neoclassical Synthesis (Goodfriend & King, 1997). New Keynesian represents supply-side equilibrium, via the theory of inflation expectations in the New Keynesian Phillips Curve (NKPC) and demand side (IS Curve), as well as policy rule behavior. Meanwhile, New Neoclassical Synthesis offers a formal opinion on the achievement of price stability based on achieving prosperity.

8.2.2.1. Baseline ITF Macroeconomic Model. Consensus from both sides, New Keynesian and New Neoclassical Synthesis, has produced a viewpoint concerning the substance of ITF theory, namely that “inflation dynamics depend on expected (future) inflation and output gap” (Clarida et al., 1999) and “the central bank should seek to keep output at its potential by targeting the minimum mark-up consistent with near-zero core inflation” (Goodfriend & King, 1997). In general, such conditions are reflected in a system of three fundamental equations (baseline model) considered as a form of modern macroeconomic thinking, namely (Walsh, 2003):

$$\text{Supply equation (NKPC)} \quad \pi_t = E_t \pi_{t+1} + \alpha x_t + \varepsilon_t \quad (1)$$

$$\text{Demand curve (IS)} \quad x_t = E_t x_{t+1} - \delta(i_t - E_t \pi_{t+1}) + u_t \quad (2)$$

$$\text{Interest-based policy rule} \quad i_t = \phi_\pi \pi_t + \phi_x x_t + \phi_i i_{t-1} + v_t \quad (3)$$

By emphasizing the optimal policy rule design, an additional two equations were created in the system as follows:

$$\text{Central bank loss function} \quad L_t = E_t \sum_{\tau=t}^{\infty} \delta^\tau \left[(\pi_{t+\tau} - \bar{\pi})^2 + \lambda x_{t+\tau}^2 \right] \quad (4)$$

$$\text{Targeting rule} \quad \pi_t + \frac{\lambda}{\alpha} x_t = 0 \quad (5)$$

where π is inflation, x is the output gap, π is the inflation target, and I is the nominal interest rate.

Several fundamental points emerged when developing the equation system above as follows:

- *First*, the modeling system above is based on a microfoundation that fundamentally and explicitly can be derived from the optimal behavior of economic agents (representative agents), namely the central bank as the policymaker, producers as prices setters, and the public or households as the end users of goods and services (consumers). In this case, the optimal policy taken aims to maximize public welfare.
- *Second*, the equations in the system are still in basic form (generic), where NKPC and the IS Curve are fully forward looking. In the empirical literature, however, further development facilitated a hybrid NKPC with a mixture of forward- and backward-looking expectations. Meanwhile, the policy rule can be stated explicitly in the form of the Taylor rule.
- *Third*, NKPC and IS Curve behavior are fundamentally already in equilibrium, thus interaction between the two is theoretically in equilibrium. Nonetheless, a system with forward-looking expectations triggers multiple equilibria as a form of behavior or expectations, which directly or indirectly become self-fulfilling expectations identified in the rational expectations model. Such conditions create indeterminacy. From a general perspective, the problem of indeterminacy can be overcome by proposing an active policy rule design, namely with a reaction parameter of the policy rate to changes in inflation of more than 1. Therefore, the presence of a policy rule that emerges as the third equation is the optimal policy rule design, which is typically oriented toward efforts to achieve system stability in line with prevailing policy commitment.
- *Fourth*, the central bank's preference, be it inflation or output stability, will influence the impact of the monetary policy taken. Theoretically, strict ITF application, where $\lambda = 0$, will provide a larger short-term impact on inflation, output and interest rates compared to flexible ITF, where $\lambda \neq 0$.

8.2.2.2. ITF in the Inflation Forecast Targeting Format. According to Svensson (1997), in line with the forward-looking policy perspective, the substance of ITF could connote inflation forecast targeting. This has implications on the specification of the inflation target and how monetary policy must operate to achieve that target. The view interpreted per optimal policy rule can be presented with the following basic considerations:

- (1) Bearing in mind the time lag inherent with monetary policy, the target of monetary policy must be based on (expected) inflation forecasts, not current or past inflation.
- (2) Inflation targeting implies inflation forecast targeting, where the weight allocated to output stabilization determines the speed at which the inflation forecast adjusts to the inflation target.

With those basic considerations, adapting the optimal policy rule based on modifications to the ITF equation system as the baseline model can be expressed as follows:

Economic system:

$$E_t[\pi_{t+1}] = \pi_t + \alpha y_t + \varepsilon_{t+1} \quad (6)$$

$$E_t[y_{t+1}] = \beta_1 y_t - \beta_2 (i_t - \pi_t) + \eta_{t+1} \quad (7)$$

Loss function (a normalized natural rate of output at zero):

$$L(\pi_t, y_t) = \frac{1}{2} [(\pi_t - \pi^*)^2 + \lambda y_t^2]$$

Intertemporal loss function:

$$E_t \sum_{\tau=t}^{\infty} \delta^{\tau-t} L(\pi_{\tau}, y_{\tau})$$

With the given lag of monetary policy (1 year for output and 2 years for inflation), then:

$$E_t[\pi_{t+2}] = a_1 \pi_t + a_2 y_t - a_3 i_t + (\varepsilon_{t+1} + \varepsilon_{t+2} + \alpha_1 \eta_{t+1}) \quad (8)$$

where

$$a_1 = 1 + \alpha_1 \beta_2$$

$$a_2 = \alpha_1 (1 + \beta_1)$$

$$a_3 = \alpha_1 \beta_2$$

$$\text{and } \frac{\partial \pi_{t+2}}{\partial i_t} = -a_3$$

Case 1: There is only one policy target (inflation target), $\lambda = 0$

$$\text{Min}_t E_t \delta^2 L(\pi_{t+2})$$

FOC:

$$\frac{\partial E_t \delta^2 L(\pi_{t+2})}{\partial i_t} = E_t \left[\delta^2 (\pi_{t+2} - \pi^*) \frac{\partial \pi_{t+2}}{\partial i_t} \right] \quad (9)$$

$$= -\delta^2 a_3 (E_t [\pi_{t+2}] - \pi^*) = 0$$

$$\Rightarrow E_t [\pi_{t+2}] = \pi^*$$

$$\Rightarrow \pi^* = a_1 \pi_t + a_2 y_t - a_3 i_t$$

The optimal policy rule can be expressed as follows:

$$i_t = \pi_t + b_1 (\pi_t - \pi^*) + b_2 y_t \quad (10)$$

where

$$b_1 = \frac{1}{\alpha_1 \beta_2}$$

$$b_2 = \frac{1 + \beta_1}{\beta_2}$$

Case 2: Considering other policy targets (output stability), $\lambda \neq 0$

$$\text{Min}_t E_t [\delta^2 L(\pi_{t+2}) + \delta L(y_{t+1})]$$

$$\text{FOC: } \Rightarrow E_t [\pi_{t+2}] - \pi^* = -\frac{\lambda}{\delta \alpha_1 k} E_t [y_{t+1}] \quad (11)$$

$$\text{where } k = \frac{1}{2} \left(1 - \frac{\lambda(1-\delta)}{\delta \alpha_1^2} + \sqrt{\left(1 - \frac{\lambda(1-\delta)}{\delta \alpha_1^2} \right)^2 + \frac{4\lambda}{a_1^2}} \right)$$

From (6), $E_t [y_{t+1}] = \frac{1}{\alpha_1} (E_t [\pi_{t+2}] - E_t [\pi_{t+1}])$ is obtained.

The FOC can be written as follows:

$$E_t [\pi_{t+2}] = c \pi^* + (1-c) E_t [\pi_{t+1}] \quad (11')$$

where

$$c = \frac{\delta\alpha_1^2 k}{\lambda + \delta\alpha_1^2 k}$$

$$\text{From (7): } \Rightarrow E_t[\pi_{t+2}] = \pi_t + \alpha_1(1 + \beta_1)y_t - \alpha_1(i_t - \pi_t) \quad (12)$$

The optimal policy rule is derived from (11) and (12) as follows:

$$\Rightarrow i_t = \pi_t + \frac{\delta\alpha_1 k}{\beta_2 \lambda} (E_t[\pi_{t+2}] - \pi^*) + \frac{\beta_1}{\beta_2} y_t \quad (13)$$

$$= \pi_t + \hat{b}_1(\pi_t - \pi^*) + \hat{b}_2 y_t$$

where

$$\hat{b}_1 = \frac{c}{\alpha_1 \beta_2}$$

$$\hat{b}_2 = \frac{c + \beta_1}{\beta_2}$$

After solving the optimal policy rule above, fundamentally the Taylor rule could be modified into a forward-looking, inflation-forecast-based rule. A forward-looking policy orientation that considers the monetary policy lag supports the relevance of implementing monetary policy based on the inflation forecast, not actual inflation. The theoretical advantage of this approach, in addition to producing a response equivalent to optimal conditions, is to enhance monetary policy credibility by focusing on inflation expectations.

8.3. Institutional and Operational Framework

An institutional framework that supports ITF application encompasses provisions in central banking laws, formulation of the inflation target as well as accountability and transparency mechanisms. As mentioned previously, ITF has been implemented in the following advanced countries: New Zealand, Canada, Israel, United Kingdom, Sweden, Finland, Australia, Spain, and South Africa. On the other hand, ITF has been adopted by EMEs in Latin America, such as Chile and Brazil, in Eastern Europe, including Poland and the Czech Republic, as well as Asian countries, like South Korea, Thailand, the Philippines, and Indonesia. In this case, there are several iterations of ITF practiced in full at different central banks around the world. In general, EMEs tend to apply comparatively more institutional regulations than advanced countries.

8.3.1. Institutional Framework

8.3.1.1. Central Banking Laws. ITF implementation requires regulations in central bank laws stipulating price stability as the target of monetary policy and providing full authority to the central bank to achieve that target (instrument independence). In this case, some variation arises concerning the clarity of the monetary policy target and the weight given to the attainment of price stability in the regulations of various central banks (Table 8.2). In many advanced countries that have applied ITF, including New Zealand, United Kingdom, Spain, Sweden, and Finland, price (currency) stability as the target of monetary policy is stipulated in the statutes of the central bank law. In other advanced countries, however, including Australia and Canada, although the central bank's mandate covers broader economic goals, in practice, price stability has become the main monetary policy target of central banks.

In EMEs, the goal of monetary policy is legislated in central banking laws, which facilitate the internal and external dimensions of currency stability. Therefore, laws typically stipulate currency stability as the goal of monetary policy, encompassing price (inflation) stability and exchange rate stability, which reflects the role of a managed or floating exchange rate system in several EMEs. In practice, however, the focus of the monetary policy target has shifted away from domestic price stability in line with the proliferation of jurisdictions adopting a flexible exchange rate system.

Table 8.2: Provisions in Central Bank Laws.

Legal Provisions	Country
Monetary policy goal	
• Currency stability as the primary goal • Price stability as the primary goal • Currency stability and other goals	• Finland, Czech Republic, and Indonesia • New Zealand, Poland, Spain, Sweden, and United Kingdom • Australia, Canada, Israel, Brazil, Chile, and South Africa
Instrument independence	• New Zealand, Australia, Canada, and United Kingdom, Poland, Spain, Sweden, and Israel, Finland, Czech Republic, Brazil, Chile, South Africa, and Indonesia
Financing the fiscal deficit	
• Restricted • Proscribed	• Canada, Czech Republic, and South Africa • Brazil, Chile, Finland, Israel, Poland, Spain, Sweden, and Indonesia

Source: Schaechter et al. (2000, p. 7).

Instrument independence in monetary policy is generally safeguarded in central banking laws. All EMEs stipulated instrument independence in prevailing laws prior to ITF implementation. In advanced countries, the United Kingdom and Sweden amended their central banking laws to accommodate instrument independence. In Australia, Canada, and New Zealand, however, although prevailing central banking laws permit a parliamentary amendment to the monetary policy decisions of the central bank, none of the respective governments have taken such action in over 30 years (Schaechter et al. 2000).

Central banking laws in EMEs also explicitly regulate restrictions on the central bank financing the government's budget deficit. This includes the purchase of government debt securities by the central bank on the primary market due to previous experience of hyperinflation stemming from the overprinting of money to finance the deficit. Government debt is relatively large in EMEs compared to their advanced counterparts, which demonstrates lower inflation volatility to monetary expansion to offset the deficit (Kumhoff, 2000). Nevertheless, it is important to note that restrictions on financing the government's budget deficit do not extend to buying government debt securities on the secondary market for monetary operations.

In many countries, ITF implementation has been reinforced by amendments to central banking laws. When to amend and the magnitude of the change to central banking laws vary from country to country, reflecting a trade-off between the efforts required to garner public support for ITF adoption compared to the increase of credibility brought about by the new legislation (Walsh, 1995). In advanced countries, including New Zealand, Spain, and Sweden, legal amendments were implemented in conjunction with ITF implementation. Conversely, in United Kingdom, Canada, and Israel, public support for a change in legislation was obtained after ITF credibility was proven.

On the other hand, EMEs have been more inclined to amend prevailing central banking regulations prior to ITF application. In other words, the legal amendments were considered a way to enhance monetary policy credibility in terms of stabilizing inflation, which, in general, has historically been comparatively higher in EMEs than advanced countries. Of course, the contribution of central banking laws to improving monetary policy credibility must be taken in perspective because monetary policy credibility needs to be built through the ability of the central bank to achieve and maintain low and stable inflation (Blinder, 1998).

8.3.1.2. Formulation of the Inflation Target. Formulation of the inflation target also varies among ITF countries in terms of the price index used, size of the target, target horizon, and publication method. Inflation target formulation primarily depends on several factors specific to the country involved, including the level of central bank credibility, economic vulnerability to shocks as well as data availability. In several cases, a number of countries have adjusted how the inflation target is formulated consistent with changes in those specific factors. What is most important, however, is that when formulating and specifying the inflation target, the central bank also considers the relationship and impact on various other macroeconomic variables, especially economic growth.

Selection of the price index. Fundamentally, the choice of which price index to use as the inflation target is affected by the influence monetary policy has on

controlling prices (controllability) and the level of public trust in the price index selected (credibility). In this case, the CPI, also known as headline inflation, is a sound measure of inflation that provides a better level of credibility than the other price indexes.⁴ CPI is the earliest price indicator available periodically, the focus of expectations, easy to understand by the public and not thought to be subject to manipulation by the central bank. On the other hand, however, CPI often contains components that are beyond the control of the central bank (administered prices (APs), terms of trade (ToT), and indirect taxes), which could, therefore, undermine the influence of monetary policy. Consequently, the use of CPI could spur excessively tight monetary policy if the price components that are beyond the control of the central bank are comparatively significant.

Another available price index is core inflation, which fundamentally shows the components of CPI that are affected by monetary policy. Various methods have been developed to measure core inflation (Roger, 1998). The exclusion method omits the price components beyond the control of the central bank from the CPI. The trimmed method statistically excludes price components within certain percentiles of the CPI distribution. Although this price index is more indicative of inflation that can be controlled through monetary policy and is, therefore, the responsibility of the central bank, this method is hampered because it is difficult to understand by the public and the measure could reduce the credibility of the inflation measure used.

In general, EMEs applying ITF typically use the CPI to formulate the inflation target because it is easy to understand by the public and supports credibility (Table 8.3). Nonetheless, core inflation is also generally measured and monitored, not only because it is a measure that underlies monetary policymaking but also because core inflation indicates future CPI trends. Furthermore, core inflation could fundamentally be used in terms of central bank accountability by showing which components of inflation are influenced by monetary policy and which are

Table 8.3: Price Indexes Used.

Price Index	Country
Consumer price index (CPI)	Australia, New Zealand, Brazil, Chile, Israel, Poland, Spain, Sweden, and Indonesia
Core inflation	Canada, United Kingdom, Finland, Czech Republic, and South Africa

Source: Schaechter et al. (2000, p. 10).

⁴An alternative to CPI is the GDP deflator, which is a measure of the level of prices of all new, domestically produced, final goods and services in an economy. A weakness of the GDP deflator in terms of ITF is the availability of monthly data, which could impair its transparency and credibility.

outside the control of the central bank. In this case, to enhance credibility, core inflation should be measured by a third party, independent of the central bank, usually an institution dealing with economic statistics, based on a method jointly agreed and announced to the public. Subsequently, CPI could be replaced with core inflation as the inflation target after core inflation and the application of ITF have been widely accepted by the public, as has occurred in Australia (1998) and New Zealand (1999).

Size of the inflation target. The inflation target can be set at a certain point or within a specific range. Using an explicit tolerance band provides greater flexibility to the central bank in response to shocks in the economy and allows discretion concerning the other policy targets in addition to inflation. Furthermore, the breadth of the inflation target corridor indicates how tolerant the central bank is to inflation fluctuations around the midpoint. On the other hand, however, a target range undermines central bank accountability and can complicate the formation of public inflation expectations. A broader range also reduces the focus on the formation of public inflation expectations and the central bank's commitment to its monetary policy.

Several countries prefer an inflation target anchored to a specific point, such as in United Kingdom, Spain and Finland or to a point with a stipulated margin, such as in Brazil and Sweden, because both cases are easier to understand by the public (Table 8.4). Conversely, a number of other countries prefer an inflation target specified as a certain range. In this regard, the breadth of the range will depend on the frequency and magnitude of the shocks in the economy, in addition to central bank credibility. Many central banks set the inflation target range at 2% or less, excluding Brazil and South Africa, where a larger range is preferred to cover price changes that are beyond the control of the central bank.

Target horizon. The target horizon indicates the period during which the central bank is committed to achieve the inflation target set. In this context, as inflation represents the long-term target, the target horizon illustrates the lag of monetary policy transmission in terms of influencing inflation as well as the desire to avoid excessive changes in the use of monetary instruments and the impact on output. In Australia, New Zealand, United Kingdom, Sweden, and Chile, where the inflation target is set at a level approaching long-term inflation, the target horizon is set at more than one year (Table 8.5). A longer target horizon allows the central

Table 8.4: Inflation Target: Point or Range.

Point or Range	Country
Target point	• United Kingdom, Spain, and Finland
Target point and margin	• Brazil, Sweden, and Indonesia
Target range of 2% or less	• Australia, Canada, Chile, Czech Republic, Israel, and Poland
Target range of more than 2%	• New Zealand and South Africa

Source: Modified from Schaechter et al. (2000, p. 11).

Table 8.5: Inflation Target Horizon.

Target Horizon	Country
Annual target	• Israel, Poland, and South Africa
Annual target announced several years in advance	• Brazil, Czech Republic, Spain, and Indonesia
Medium-long target horizon or not announced	• Australia, Canada, Chile, Finland, New Zealand, Sweden, and United Kingdom

Source: Modified from Schaechter et al. (2000, p. 8).

bank to respond to economic shocks and catalyze relatively more stable economic growth, in addition to anchoring public inflation expectations. On the other hand, a shorter target horizon could trigger instability, primarily if shorter than the lagged effect of monetary policy on inflation.

In countries where inflation is above its long-term trend, a shorter target horizon could accelerate disinflation. Monetary policy can respond to shocks by adjusting the inflation target rather than through a change in policy stance. Many EMEs pursue a disinflation path in conjunction with a long-term inflation target. Some countries announce the disinflation path years in advance, such as in Czech Republic and Indonesia, while others announce the path annually, as in Poland. In general, setting an annual inflation target makes it easier to assess the success of efforts to reduce inflation, especially if disinflation is achieved more quickly than expected. Furthermore, an annual target allows monetary policy to absorb short-term inflationary pressures and output fluctuations, in addition to preventing an overly ambitious inflation target that would be impossible for the central bank to attain.

Periodic monitoring of the inflation target is required, usually through the publication of an inflation report or monetary policy report, monthly or quarterly for instance. Inside, the reports describe annual inflation developments (year on year) compared to the inflation target, the determinants of inflation and current monetary policy. Furthermore, the reports contain future inflation projections (for instance with an inflation forecast fan chart) and the macroeconomic outlook in order to explain the future monetary policy direction and targets. In addition to monetary policy transparency in the eyes of the public, periodic monitoring, and the publication of an inflation report or monetary policy report will simplify the central bank's accountability mechanism in terms of achieving the inflation target set.

Announcing the inflation target. The central bank may announce the inflation target itself, or the government, or the government in conjunction with the central bank, depending on the mechanism used to determine the inflation target (Table 8.6). Announcing the inflation target by the government or together with the central bank has several advantages, including enhancing the credibility of the inflation target because the government also displays its commitment to ITF implementation by the central bank as well as fiscal policy implementation to

Table 8.6: Announcing the Inflation Target.

Institution	Country
Government	• Brazil, Israel, and United Kingdom
Central Bank	• Chile, Finland, Poland, Spain, Sweden, and Indonesia
Government and Central Bank	• Australia, Canada, Czech Republic, New Zealand, and South Africa

Source: Modified from Schaechter et al. (2000, p. 8).

support attainment of the inflation target. Theoretically, Debelle and Fischer (1994) suggested that the government must instruct the central bank of the target because the government represents the interests of the people, while the central bank, depending on the level of independence enjoyed, could set an inflation target considered too strict by the government. On the other hand, the government could prioritize short-term interests in terms of stimulating economic growth, thus setting an inflation target well above its long-term trend. Both of these issues can be overcome if the inflation target is set by the government after consultation with the central bank and announced jointly to the public.

The inflation target is typically announced by the central bank alone in countries where the law explicitly mandates price stability as the final goal of monetary policy and provides jurisdiction to the central bank to set the inflation target, such as in Poland, Spain, and Sweden. Meanwhile, the central banks in Chile and Finland, where the goal of monetary policy is stated more implicitly, namely exchange rate stability, announce their inflation targets to show the ability of the central bank to interpret the legal mandate as domestic price stability. By announcing the inflation target, the central bank affirms its accountability to the public and is expected to enhance monetary policy credibility, considering that announcing the inflation target represents a symbol of delegating authority to set the goal of monetary policy from the government to the central bank.

Escape clauses. Formulating the inflation target may also encompass certain requirements that clearly state the level of tolerance to inflation fluctuations (escape clauses). Inflation target formulation must also consider the trade-off between flexibility and credibility. For example, Canada, Czech Republic, New Zealand, and South Africa stipulate escape clauses in ITF to increase flexibility. In this context, the escape clauses state under which conditions the inflation target may not be achieved, while also requiring the central bank to indicate the horizon needed to achieve the inflation target again or communicate the new direction of the inflation target. One weakness of the escape clause is that it could be interpreted as a policy based merely on judgment and unable to anticipate all potential shocks. Therefore, the use of an escape clause in an ambiguous situation could undermine ITF accountability and credibility. Bernanke et al. (1999) showed that, thus far, escape clauses have only been triggered by New Zealand in 1990–1991 and 1993–1994, when the prices of oil and wood were omitted from the inflation basket.

8.3.1.3. Accountability and Transparency. All central banks applying ITF operate transparently because public accountability is an integral element of successful monetary policy. The importance of the accountability mechanism is clear because of the lag between monetary policy and inflation, which complicates the monitoring of policy commitment in the near term. This differs from a regime targeting money supply or the exchange rate, where attainment of the inflation target can be monitored in the short term. In addition, accountability may also help safeguard monetary policy from outside political pressures.

To improve accountability and transparency, central banks applying ITF periodically clarify their stance on achieving the inflation target and existing monetary policy through regular publications, monthly or quarterly, of the inflation report or monetary policy statement, submitted to the government, parliament, and public at large. The reports detail the latest economic and financial developments and their impact on inflation compared to the inflation target, clarify the rationale underlying monetary policy and explain the recent developments as well as discuss inflation projections for the upcoming 12–24 months. Such measures are also accompanied by an elucidation from the central bank to parliament concerning various important aspects as required. In several cases, including Czech Republic and Sweden, the central bank publishes the minutes of monetary policy discussions.

Transparency is also enhanced through the immediate communication of the monetary policy measures decided by the central bank. In general, the announcement is made through a press release, stating the monetary policy decisions made and a summary of the underlying factors. High-ranking officials from the central bank also give presentations on the latest economic and financial developments, attainment of the inflation target as well as the monetary policy stance adopted to various parties, including the mass media. Such measures are required to improve broad and comprehensive understanding of monetary policy objectives and central bank performance. In Brazil, officials from the central bank give lectures and presentations to the private sector and media regarding the goal and implementation of monetary policy, while in Canada, in addition to media briefings and presentations through various forums, meetings are also held with various private sector groups to discuss the latest economic and financial dynamics (Schaechter et al., 2000).

Various other forms of communication are also available to improve accountability. Central banks in advanced countries, and in Brazil, Chile, Israel, and Poland, also publish their research to enhance understanding of the various economic and financial dynamics underlying inflation projections. In addition to posting online, central banks also sponsor seminars and workshops where their researchers can discuss compelling issues with external experts. Hitherto, only central banks from advanced countries have published detailed models used to generate inflation projections. Meanwhile, not many central banks in EMEs have published their models because they are still under development (Schaechter et al., 2000).

8.3.2. Operational Framework

ITF implementation requires the central bank to hone its monetary policy strategy and operational framework. In general, monetary policy at ITF central banks is implemented as follows (Schaechter et al., 2000):

- Regular inflation projections are prepared and updated based on the latest economic data, market sentiment indicators, modeling results, and judgment.
- The direction of monetary policy, in the form of an operational target required to minimize deviations between inflation projections and the inflation target paying due regards to the output gap and other factors, is determined based on modeling and judgment.
- Other factors, like international developments and political events, which could influence the timing and magnitude of changes in the operational target, must also be taken into consideration.
- Changes in policy direction are announced and the operational target is adjusted according to the desired level.

Fundamentally, ITF-based monetary policy requires three elements, namely a model or method to project future inflation, a review of the policy transmission mechanism, and effective monetary control when setting the operational target and monetary instruments to be used. In general, central banks in advanced countries and EMEs implement a similar monetary policy strategy. In all cases, the central bank must rely on judgment when formulating monetary policy. Countries that have already developed statistical models can base their monetary policymaking decisions on such models. In other countries, however, particularly EMEs, many factors must be considered because of underdeveloped statistical models as well as limited data availability and ongoing structural changes in the economy. Furthermore, emerging economies also face far more economic and financial shocks, in addition to underdeveloped financial markets.

8.3.2.1. Inflation Projections. In ITF, inflation projections play a key role in terms of monetary policy formulation, primarily considering the lag between monetary policy and its impact on inflation. Therefore, the ability to predict inflation will determine how well monetary policy can steer the economic and financial outlook in line with the inflation target set. In many cases, inflation projections can be viewed as an intermediate target to achieve the inflation target as understood in the monetary policy strategy of monetary economics (Svensson, 1997).

In practice, the ability to project inflation is based on a combination of three factors: namely inflation indicators, economic modeling, and qualitative judgment (Schaechter et al., 2000). In this case, central banks applying ITF typically assess the accuracy of projections from economic modeling as well as the various inflation and economic indicators selected. Subsequently, the central bank makes a judgment based on the qualitative information obtained through contact with various economic sectors to decide the most appropriate monetary policy direction.

Inflation indicators. Monetary inflation indicators, encompassing an observable data and information set that can signal the direction of future changes in inflation, are used to formulate monetary policy (Schaechter et al., 2000).⁵ Numerous inflation indicators are available in countries that apply ITF. Fundamentally, inflation indicators cover aggregate demand and supply variables, measures of interest rates and exchange rates, inflation, price measures, and expectations.

Many EMEs tend to focus on indicators of aggregate demand and supply, considering the large economic shocks encountered and the lack of alternative indicators to reflect market conditions. Furthermore, new inflation indicators need to be created and old ones improved. Many central banks in EMEs work in conjunction with existing statistical institutions to measure core inflation, which can signal inflation trends. Central banks also identify or create new inflation indicators based on market conditions.

Surveys of inflation expectations conducted by third parties or the central bank itself are also often utilized as inflation indicators. In this case, such surveys are designed to reveal inflation expectations at the corporate and consumer levels (New Zealand). Central banks also summarize the results of existing surveys to indicate inflation expectations (United Kingdom). Private sector qualitative surveys are also used in Canada and New Zealand. Furthermore, the central bank could also consider informal information gleaned from discussions or meetings with various private parties.

Government bond yields are also an indicator of inflation. Shifts in spread between various bond yields have also been used in Australia, Canada, Israel, New Zealand, and United Kingdom to assess the impact of monetary policy and various other factors on inflation expectations and the inflation premium contained in the bond yield spread. Inflation expectations can also be measured using the spread between nominal and real forward interest rates. Nevertheless, this kind of data must be used with caution considering that spread between bond yields or forward interest rates contain the markets' perception of default risk.

Economic models. Structural economic modeling is also often used by central banks to calculate inflation projections. In general, such models cover the aggregate demand curve for open economies, international asset market equilibrium and a monetary policy reaction function (Bogdanski, Springer, Goldfjan, & Tombini, 2000). These elements have been applied in various empirical research on central bank behavior, monetary policy rules, and the sacrifice ratio (Taylor, 1999). In addition, the equations are oftentimes incorporated in structural macroeconomic models at central banks, including the Quarterly Projection Model at the Bank of Canada and the FRB model at the US Federal Reserve Board.

⁵Per the monetary policy framework, there are fundamentally two types of monetary policy indicators (Bofinger, 2001; Handa, 2000). The first are economic and financial indicators that show the direction of future inflation fluctuations, also known as leading indicators. The second are the economic and financial indicators selected as policy indicators. In terms of ITF, inflation indicators are considered leading indicators.

The use of large structural macroeconomic models is common in advanced countries. Such models have helped guide monetary policy formulation for a long time and maintain a relatively stable relationship. Canada and New Zealand use dynamic models to produce future inflation and economic projections (Drew & Hunt, 1998). The models are calibrated to generate simulations for theoretical assessment as the models are based on historical data, which is less useful for policy simulations. The development of large-scale models requires a protracted amount of time, while experience of ITF implementation remains comparatively brief. Many advanced countries also utilize small structural models to analyze specific sectors or key variables (Drew & Hunt, 1998).

Meanwhile, many EMEs use judgment combined with economic models. Relatively brief experience of ITF implementation impairs the comparatively stable relationship between economic variables when projecting inflation, on top of the structural changes that often occur in the economy (Leiderman & Baror, 2000). Nevertheless, an increasing number of central banks in EMEs are benefitting from using such models to formulate monetary policy and have, therefore, allocated more resources to the development of economic models. Brazil, Czech Republic, and Israel have developed models with three or four key equations that are often used in advanced countries (Bodganski et al., 2000; Clinton, 2000).

In addition to producing inflation projections, economic models can facilitate monetary policy in many significant ways (Issard & Laxton, 2000). Small-scale models can help the central bank rationalize policy transmission through the various channels considered dominant. More importantly, the models can help as frameworks to discuss policy and even facilitate the presentation of inflation projections to increase transparency. Fan charts, which show the distribution of inflation projections covering various scenarios, are the latest upgrade in terms of representing inflation projections based on economic models (Allen, 1999).

Time series models are thought to place fewer restrictions on economic structure but make short-term projections, as well as cross-checking the inflation projections produced by large-scale models. Univariate models for an aggregate inflation index, the components of inflation or other key indicators are comparatively easier to build and facilitate more cross-checking with the direction of inflation. In New Zealand, the results of such models are used to generate projections for the last two quarters after the historical data have ended. Multiple time series models, such as vector autoregressive (VAR) models, also provide simultaneous interactions between key variables together with fewer structural restrictions. Such models are used by the central banks in Brazil, Chile, and Israel.

8.3.2.2. Policy Transmission. As elucidated in the previous chapter, deep understanding of the monetary policy transmission mechanism is an integral element of implementing monetary policy. Such understanding becomes even more important in ITF because it will determine the effectiveness of monetary policy in terms of steering the future direction of inflation in line with the target set. In other words, considering that the impact of monetary policy on the economy and inflation occurs after a lag, inflation targeting necessitates pre-emptive monetary policy.

Issues arise during implementation. Monetary policy has an imperfect influence on inflation. Not only because sources of inflation originate from non-monetary factors, such as APs as well as disruptions to the supply and distribution of goods and services, but also, in the words of Milton Friedman, monetary policy works with a relatively long and varied lag. As elaborated in Chapter 5, the channels through which monetary instruments influence inflation include money supply, interest rates, exchange rates, and expectations. Central bank understanding of the monetary policy transmission mechanisms is a prerequisite to formulate monetary policy.

Studies on monetary policy transmission mechanisms are also required to comprehend the relative effect of each channel in an economy. Such studies can be used to identify and create important indicators for monetary policy formulation because they show the magnitude and lag of monetary policy's impact on inflation and real output through each respective channel. Considering that the use of intermediate targets (other than inflation projections), as proffered by Svensson (1997), is not in keeping with the ITF, most central banks underscore several economic and financial indicators as the basis of monetary policy formulation. Such indicators cover a variety of economic and financial variables that contain a wealth of information on the future direction of inflation.⁶ Although not stressing which indicators should be used, many central banks depend on various information sources when setting monetary policy.⁷

In an open economy, the transmission channels between monetary policy actions and inflation work directly and indirectly (Ball, 1999; Svensson, 1998). In the near term, the direct impact of monetary policy occurs through changes in the exchange rate, where a hike to the interest rate directly reduces inflationary pressures through a stronger exchange rate, which feeds through to import prices. Over the longer term, however, a hike to the interest rate indirectly lowers inflation, first by lowering inflation expectations and, therefore, corporate investment, and secondly by reducing consumption and, thus, prompting exchange rate appreciation that diverts spending from traded to non-traded goods.

Of course, a central bank cannot rely solely on the exchange rate to transmit monetary policy because economic players will gauge the extent to which monetary policy influences inflation through the interest rate or exchange rate. Besides, monetary policy can also affect inflation indirectly through changes in wealth and aggregate demand. Furthermore, in the case of underdeveloped and unestablished financial markets, monetary policy transmission channels can be compromised by corporate balance sheets and bank lending behavior. In the Czech Republic, for example, monetary policy transmission was impaired by the inelasticity of bank lending policy to the interest rate due to high non-performing

⁶Examples include the monetary conditions index developed by the central banks of New Zealand and Canada, and the broad money (M2) indicator used by the central bank of Spain.

⁷In other words, the most common monetary regime practiced in ITF countries is constrained discretion.

loans (NPL) and lending to unproductive businesses. In Poland, the credit channel was undermined by underdeveloped financial markets and a deluge foreign funds.

The monetary policy transmission channels in EMEs, where inflation is typically high, are characterized by sticky prices (price rigidity) and the rapid impact of the exchange rate on inflation. Brazil, Chile, and Israel, which had a long history of high inflation, downward price inertia and an almost immediate exchange rate impact on inflation prior to ITF, have become more credible (Bogdanski et al., 2000). Exchange rate pass-through to inflation has also faded in advanced countries, to around six to eight quarters (Menon, 1995). In New Zealand, the short-term direct exchange rate channel was observed to decline with the advent of hedging and the longer duration impact of monetary policy on inflation. In the case of Brazil, the indirect channels of monetary policy through demand have a lag of two to three quarters on inflation.

8.3.2.3. Policy Implementation. The operational target of monetary policy normally used by ITF central banks are short-term interest rates, usually overnight to 1 month. In addition to closely reflecting the actions of the central bank, short-term interest rates are used because they can be transmitted to the entire interest rate spectrum and, thereafter, to the economy and inflation. Furthermore, the growing popularity of interest rates as the operational target is a corollary of the general central banking trend to move away from targeting money supply in monetary policy that might not be consistent with ITF (Borio, 1997).

Several central banks in countries with an open economy utilize a combination of interest rates and exchange rates, contained in an index known as the monetary condition index (MCI), as the operational target or to signal monetary policy to the markets. MCI is used because of the complexity involved with differentiating monetary policy transmission through the interest rate channel and exchange rate channel in terms of the influence on inflation in a small open economy. In practice, however, the use of MCI also entails complications due to market expectations that all exchange rate fluctuations, even daily changes, will elicit a response from the central bank through changes in interest rates to maintain MCI and the fact that exchange rate fluctuations can impact MCI differently (Freedman 1994; Hunt, 1999). Canada first used such an index at the end of the 1980s but subsequently reduced its role as a signaling tool in 1998. New Zealand used to announce its desired MCI direction and range but shifted to an interest rate operational target in the middle of 1998 to avoid short-term interest rate fluctuations.

In the implementation of monetary policy, all ITF central banks use instruments based on market mechanisms to maintain the interest rate operational target at the desired level. Open market operations, outright and repo, are the main instruments used by central banks to maintain market liquidity consistent with the published interest rate target. OMO involve buying and selling government securities or securities issued by the central bank itself. Meanwhile, direct monetary instruments, such as the reserve requirement or lending requirements, are rarely used because they are considered less effective in terms of overcoming inflationary pressures or as a signal of monetary policy.

The interest rate was set as the operational target in reaction to deviations of inflation projections from the target and the output gap. In this case, the interest rate response in ITF is more forward looking compared to other monetary policy frameworks. In general, interest rates are set based on a mechanism often referred to in monetary economics as the Taylor rule. In practice, however, central banks bolster this approach with judgment based on various information that support future inflation projections. Judgment is more commonly used in EMEs because, among others, economic projection models remain under development or contain unstable inter-parameter relationships, while the corresponding economies are often blighted by relatively large shocks and data availability is limited.

8.4. Inflation Targeting Regimes

Currently, around 40 countries, including advanced countries and EMEs, have applied ITF as their central bank's monetary policy framework.⁸ In general, the 40 countries apply a more flexible exchange rate system and explicitly acknowledge their commitment to controlling inflation in their monetary policy. Furthermore, the countries avoid implementing a fixed exchange rate system in order to limit economic vulnerability to speculative exchange rate attacks, while also increasing their monetary policy independence from external influences. Simultaneously, monetary policy based on targeting money supply is becoming less effective because of unstable demand for money, among others.

8.4.1. Regime and Rationale

In practice, there are variations in the extent to which ITF is applied from country to country depending upon the specific conditions of the country involved. In this case, Carare and Stone (2003) classified three ITF regimes, namely FFIT, EIT and ITL. The classification was based on the clarity and credibility of the central bank's commitment to the inflation target. Clarity was reflected in the announcement of the inflation target to the public and with institutional arrangements that support accountability to achieve the target. On the other hand, credibility was measured using the achievement of actual inflation and the domestic government's debt rating.

FFIT is the most prominent ITF regime. FFIT countries have a medium-high level of credibility, clear commitment to the inflation target and institutional commitment in the form of a transparent monetary policy framework to support central bank accountability to attainment of the inflation target. For such countries,

⁸In the selection of the 42 countries, Carare and Stone (2003) used the following assumptions: (1) that a country applying a flexible exchange rate regime is committed in some way to an inflation target; and (2) a total of 89 small countries with GDP below USD 4 billion and GDP per capita below USD 720 were omitted from the 185 IMF member states.

applying FFIT empowers the central bank to maintain monetary policy consistency in terms of controlling inflation at a certain pace of economic growth.

An EIT regime also enjoys a relatively high level of monetary policy credibility to maintain low and stable inflation without the need to fully emphasize transparency and accountability through a specified inflation target. Low and stable inflation, coupled with financial system stability, allows the central bank to pursue output stabilization and price stabilization.

Countries adopting ITL announce the inflation target but due to low credibility are unable to maintain inflation at the announced target. Relatively low credibility reveals vulnerability to economic shocks and financial instability as well as a weak institutional framework. ITL can be viewed as a transitional regime, where the authorities are implementing the structural reforms required to credibly apply a single inflation target.

ITF regimes can be viewed as a combination of policy objectives to maximize social welfare, each based respectively on the level of credibility enjoyed by the country involved. The final goal of monetary policy is to maximize social welfare through robust and stable economic growth in the long run. Monetary policy can support long-term economic growth through a combination of single-digit inflation, financial stability, and output stability. The combination of those three policy objectives to maximize social welfare in the formulation of monetary policy depends on the level of credibility garnered in a country.

Therefore, the three ITF regimes mentioned above can be explained by the weight given to the three policy objectives and clarity of commitment to the inflation target. FFIT has a medium-high level of credibility and solid financial stability but cannot maintain low and stable inflation without commitment to the inflation target. The clarity and transparency of such commitment incurs a cost in the form of less flexibility to stabilize output. EIT countries have already maintained low inflation and a stable financial system and, therefore, can credibly achieve price stability and enjoy flexibility to stabilize output. The dual objectives mean that EIT countries cannot operate as transparently as FFIT countries. On the other hand, ITL countries do not have enough credibility to maintain low inflation and are vulnerable to financial and cyclical shocks. Consequently, EIT countries are unable to steer their monetary policy toward achieving an inflation target, but still require the flexibility to confront shocks.

8.4.2. Clarity of Commitment to the Inflation Target

The clarity of commitment to the inflation target demonstrates a fundamental difference between ITF regimes and other monetary regimes based on exchange rates or monetary aggregates. In terms of exchange rate targeting, the central bank can easily be held accountable if monetary policy fails to stabilize or achieve a certain exchange rate because exchange rates are easy to monitor in real time. Similarly, in a regime targeting money supply, it is easy to measure whether the target has been achieved because money supply statistics are available with a lag of only one or two months. On the other hand, however, achievement of the inflation target per ITF can only be observed after a lengthy and variable lag

because of the time required from when the policy action is taken until its effect on inflation can be observed. Therefore, inflation target clarity manifests in stages determined by application of the ITF monetary policy framework.

Inflation target clarity contains two elements. The first is clarity and public communication from the central bank concerning the inflation target to be achieved. In general, central banks that apply ITF announce their inflation target. The difference is in the clarity of the level or range of the inflation target, the time horizon, the type of price index used and the monetary policy that will be and has been pursued to achieve the inflation target. Ex-ante and ex-post monetary policy clarity, communicated effectively and in detail, will facilitate central bank accountability in terms of achieving the target.

The second is transparency concerning the institutional arrangements required so that the public and markets can monitor monetary policy and hold the central bank accountable for the target. Institutional transparency is reflected by the preferred communications media used by the central bank, including the publications, frequency, and depth of the analysis contained in the inflation report or similar, the announcement of changes to the direction and target of monetary policy contained in a press release or other form, an assessment of inflation target attainment and the monetary policy implemented, whether the inflation projection model has been published, as well as the various other forms of communication media and publications used.⁹

8.4.3. Inflation Targeting Regime Credibility

Despite clear commitment to inflation, a country is not always considered a serious proponent of ITF if the inflation target set is comparatively high and negative market sentiment exists toward the central bank. In this case, the credibility of commitment to achieving the inflation target is the main differentiator of ITF.

Carare and Stone (2003) used two indicators to measure the credibility of inflation targeting regimes, the first being inflation performance using average inflation over several years or a comparison between inflation realization and the target or the level of public inflation expectations. Low and stable inflation supports robust and stable economic growth in the long run (Sarel, 1996) and, therefore, monetary policy that supports long-term economic growth is deemed more credible. Consequently, countries that maintain a relatively low rate of inflation are considered to have more credible monetary policy.

The second indicator used was the rating of government securities denominated in the domestic currency, for instance the rating affirmed by Standard and Poor's or another international rating agency. This indicator negated the weaknesses inherent to the inflation achievement indicator, for which the data were relatively short term in the few countries new to ITF application, because the rating directly captures market perception regarding confidence in long-term financial

⁹The issues of central bank transparency and communication strategy will be discussed in Chapter 12.

stability, which is ultimately the responsibility of the central bank. Furthermore, this indicator also illustrates the strength of the government's fiscal position, which also determines the credibility of commitment to the inflation target. In countries suffering from a comparatively large fiscal deficit, offset by government issuances of debt securities, any fiscal considerations will influence the central bank's monetary policy and, therefore, undermine the achievement of no fiscal dominance as a prerequisite of ITF implementation.

8.4.4. Inflation Targeting Regime Classification

Based on the analysis of commitment clarity and credibility to inflation targeting, using inflation data from 1999 to 2002 and government debt ratings from 2001, Carare and Stone (2003) classified 43 countries into three inflation targeting regimes (Table 8.7).

The first regime consists of 18 countries with explicit commitment to an inflation target and applying a transparent monetary policy framework to ensure central bank accountability to achievement of said target. These countries were classified as FFIT because they are fully committed to achieving the inflation target set. It is interesting to note that the 18 FFIT countries include small and medium sized advanced countries as well as medium and large EMEs.

The second group consists of five central banks with sound credibility, which allows for more implicit commitment to the inflation target. These countries have been very successful in maintaining low inflation but implement different monetary policy frameworks, particularly in terms of price stability and monetary operations. They appear to enjoy more flexibility to orient monetary policy toward goals other than price stability, such as economic growth or exchange rate stability and are, therefore, known as EIT. All members of this group are advanced countries.

Countries in the third group announce an inflation target, but due to low credibility cannot offer full commitment to the target. They are also heterogeneous in terms of formulating the goals or monetary operations. This regime is known as ITL because the practicing countries cannot explicitly give credible commitment to the inflation target. A total of 19 countries were classified per this regime, all of which are EMEs.

8.4.5. The Importance of Regime Classification

In addition to explaining the practical variations of ITF implementation at different central banks, inflation targeting regime classification also reinforces the analysis of countries moving from one regime to another. From discussions on the criteria of each respective regime, the ideal conditions to be considered include which requirements are necessary for an ITL country to enhance commitment credibility to the inflation target, thus raising its status to FFIT. In this case, a comparative study of FFIT and ITL EMEs would provide a clearer picture.

In terms of the macroeconomic conditions, Carare and Stone (2003) showed that the level of economic and financial development in FFIT EMEs (EMFFIT)

Table 8.7: Inflation Targeting Regime Classification.

FFIT	EIT	ITL
Advanced countries	Advanced countries	EMEs
1. Australia	1. US	1. Peru
2. Canada	2. ECB	2. Venezuela
3. New Zealand	3. Japan	3. Uruguay
4. United Kingdom	4. Switzerland	4. Guatemala
5. Iceland	5. Singapore	5. Jamaica
6. Norway		6. Dominican Republic
7. Sweden		7. Honduras
		8. Russia
EMEs		9. Croatia
8. Brazil		10. Slovenia
9. Chile		11. Slovakia
10. Colombia		12. Albania
11. México		13. Kazakhstan
12. Czech Republic		14. Rumania
13. Hungary		15. Philippines
14. Poland		16. Indonesia
15. Israel		17. Sri Lanka
16. South Africa		18. Mauritius
17. South Korea		19. Algeria
18. Thailand		

Source: Excerpted from Carare and Stone (2003).

was generally higher and more advanced than in ITL EMEs (EMITL). *First*, the average GDP in EMFFIT was six times that of EMITL and GDP per capita was three times that of EMITL. *Second*, monetization and financial markets are more advanced in EMFFIT compared to EMITL. The ratio of money supply to GDP was also around 40% larger in EMFFIT. Capital market capitalization was also much higher and real interest rates lower in EMFFIT. More advanced financial sector conditions explain why EMFFIT can more effectively implement monetary policy through indirect and interest rate instruments as the operational target. *Third*, EMFFIT countries maintain a stronger fiscal position with clear restrictions on central bank financing of government operations. Although the average fiscal surplus/deficit is not particularly disparate between the two regimes, EMITL suffer from a larger position of government debt, which is important as a reflection of the no-fiscal-dominance prerequisite of ITF application.

A comparison can also be made of the operational targets and monetary policy instruments used by the central bank. In general, despite using the interest rate as the operational target and open market operations as the instrument, the clarity of the operational framework is different between EMFFIT and EMITL regimes. In EMFFIT countries, the elements of the monetary policy operation framework are explicit, namely short-term interest rates as the operational target and open market operations as the instrument. Conversely, EMITL countries generally apply a mix of operational targets and monetary instruments. In addition to short-term interest rates, other operational targets are also used, such as the exchange rate and the monetary base. Some EMITL countries also still use direct instruments, including the reserve requirement and lending requirements. On top of comparatively underdeveloped financial sectors, the use of several operational targets and monetary instruments is possibly linked to the prevailing dual goals of monetary policy.

The analysis above shows which preparations are required by EMITL countries to upgrade their status to the EMFFIT regime. Specifically, EMITL countries must introduce structural and policy changes prior to FFIT implementation. Carare and Stone (2003) suggested a number of interesting observations. *First*, EMEs generally apply FFIT during periods of disinflation and to build credibility. *Second*, the ratio of government debt to GDP is typically tracking a downward trend thanks to sounder debt management and disinflation. And *third*, the ratio of money supply to GDP is increasing, as reflected by the growing role of the financial sector. These observations support the preconditions proposed by Bernanke et al. (1999) and other economists, who showed that prior to FFIT implementation, EMEs strived to lower inflation, consolidate government fiscal conditions, and deepen the financial sector. In other words, such conditions illustrate that EMEs need to try harder to build credibility in terms of ITF implementation.

8.5. Inflation Targeting and Economic Performance

8.5.1. Inflation Targeting Successes and Alternative Opinions

With conceptual understanding, institutional and operational frameworks as well as an inflation targeting regime as described in previous sections, the question arises about whether ITF implementation, as a monetary policy framework, contributes positively and significantly to economic performance and the central bank. Specifically, does ITF implementation generate lower inflation and inflation expectations? If yes, how much disinflation must be achieved with lower real output and volatility, particularly economic growth and unemployment, to interest rates in the country involved? Does ITF implementation trigger changes in the way the central bank implements its monetary policy? Is economic performance better in countries applying ITF than in other countries?

Several studies have researched ITF application in different countries and its impact on the central bank and economic performance. Some of the studies analyzed the basic features of ITF, including Haldane and Simon (1995), Masson,

Savastano, and Sharma (1997), Agenor (2000), Blejer et al. (2000), Fry, Julius, Mahadeva, Roger, and Sterne et al. (2000), Schaechter et al. (2000), Carare, Shaechter, Stone, and Zelmer (2002), Sterne (2002), Amato and Gerlach (2002), and Mishkin (2004). Other cross-border studies focused on a comparison of macroeconomic performance and inflation stabilization using a sample of ITF countries, including Schmidt-Hebbel and Werner (2002) for Brazil, Chile, and México, as well as Kahn and Parrish (1998), Mishkin and Posen (1997), Kuttner and Posen (1999), and Siklos (1999) for industrialized nations. Other studies have compared macroeconomic performance in ITF countries with countries practicing other monetary policy regimes, including studies by Bernanke et al. (1999), Cecchetti and Ehrmann (2000), Corbo, Landerretche, and Schmidt-Hebbel (2000), Mishkin and Schmidt-Hebbel (2001), Landerretche, Corbo, and Schmidt-Hebbel (2001), Neumann and Von-Hagen (2002) as well as Ball and Sheridan (2003).

This section will review a number of seminal empirical studies, particularly those containing cross-border comparisons. The discussion will emphasize two main points to provide a more objective view concerning the pros and cons of ITF implementation. *First*, opinions that sum up success stories of ITF implementation as summarized from various empirical studies of ITF countries, such as those suggested by Mishkin and Schmidt-Hebbel (2001). *Second*, opinions that criticize the methodologies and empirical findings of previous studies, such as the latest study by Ball and Sheridan (2003). Reviews of other relatively recent studies are available in Landerretche et al. (2001) as well as Neumann and Von-Hagen (2002).

8.5.1.1. Inflation Targeting Successes. Mishkin and Schmidt-Hebbel (2001) are perhaps the best reference to draw several success stories during the first decade of ITF implementation in the 1990s from various previous empirical studies. Nevertheless, it is important to note that because of differences in the methodologies used, countries sampled, and periods analyzed, the conclusions made by the two economists tended to emphasize a number of salient and relevant empirical findings that provide valuable lessons. In other words, the review was not intended as an incontrovertible conclusion or a substitute for existing empirical studies or those still required.

In general, Mishkin and Schmidt-Hebbel (2001) concluded that ITF implementation was proven as a very successful new monetary policy framework according to the experiences of the countries involved and a viable alternative to the monetary regimes applied in advanced countries during the 1990s. In summary, the two economists put forward eight success stories as follows:

- (1) *ITF has helped implementing countries to lower inflation, albeit not below the levels attained in non-ITF advanced countries.* The empirical evidence shows that ITF countries lowered long-term inflation beyond what would have been possible without ITF implementation. Nevertheless, the introduction of ITF did not create inflation lower than that achieved in advanced countries applying other monetary policy regimes, as evidenced by Bernanke et al. (1999) and Mishkin (1999). The empirical evidence shows that ITF implementation can be interpreted as an investment in strengthening monetary policy

to initially produce significant disinflation, equivalent to the performance of other advanced countries that have successfully controlled inflation.

- (2) *ITF has been proven to overcome economic shocks.* The 1990s, excluding the period when a devastating financial crisis befell EMEs from 1997 to 1999, was a golden age for the global economy, with relatively low inflation and the US economy driving expansion. Such conditions galvanized the opinions of several economists that ITF implementation was untested in terms of overcoming economic shocks that undermined the achievement of low and stable inflation in the affected countries.

Mishkin and Schmidt-Hebbel (2001) were of another opinion, however. Many small open economies applying ITF faced significant shocks in the wake of the Asian Financial Crisis of 1997/1998 compared to advanced countries that were largely unaffected. A combination of financial and ToT shocks experienced by Australia, Chile, Israel, and New Zealand, in addition to other ITF countries, led to significant exchange rate devaluation and served as the perfect proving ground for ITF implementation. Those countries successfully overcame the relatively small external pass-through shocks from exchange rates to inflation in their respective jurisdictions.

Furthermore, the oil price shocks of 1999–2000 were another rigorous test for ITF implementation, especially for large importers of oil, such as Brazil, Czech Republic, Poland, and others. A significant spike in imported inflation, through rising energy prices and exchange rate devaluation, should have threatened the inflation targets set. Nonetheless, evidence from ITF countries showed that the impact of the oil price shocks on core inflation was comparatively small, while the impact on higher headline inflation was small and temporary.¹⁰

- (3) *ITF has helped reduce the sacrifice ratio and output volatility in practicing countries to levels approaching the performance of non-ITF advanced countries.* The cost of disinflation can be measured using the sacrifice ratio, namely the percentage loss of output per the percentage decline of inflation. Technically, the ratio is calculated using the deviation of potential output from the actuals during a given period divided by CPI disinflation during the same period. In other words, the sacrifice ratio is a measure of the costs associated with slowing down economic output to bring inflation toward the target set.

Bernanke et al. (1999) found that ITF implementation did not lower the cost of disinflation per the sacrifice ratio (and Philipps curve) in ITF countries, which was not observed to change dramatically after ITF implementation nor was it different to the levels found in non-ITF advanced countries. Notwithstanding, based on the latest evidence from a larger sample of ITF countries, Corbo et al. (2000) concluded that the sacrifice ratio experienced a decline, especially in EMEs after ITF implementation. They also found that output volatility decreased in EMEs and advanced countries after

¹⁰Similar findings were also put forward by Neumann and von Hagen (2003).

ITF implementation to a level on par with (and sometimes lower than) that observed in non-ITF advanced countries.

- (4) *ITF has helped anchor and influence inflation expectations and overcome inflation shocks.* According to Almeida and Goodhart (1998) as well as Bernanke et al. (1999), ITF application does not immediately lower inflation expectations, only gradually. Corbo et al. (2000) reported that erroneous inflation projections based on VAR models in ITF countries were reduced consistently by ITF implementation to the low levels found in non-ITF advanced countries. They also found that inflation persistence had declined significantly among ITF countries during the 1990s, showing that by setting and announcing an inflation target, ITF had successfully anchored inflation expectations compared to formulation based on previous inflation.
- (5) *ITF-based monetary policy is adequately flexible in its symmetrical response to inflation shocks and in terms of accommodating temporary inflation shocks, which do not affect attainment of the medium-term target.* ITF countries are not considered “inflation nutters” per King (1996), because they generally react symmetrically to positive and negative shocks in the economy, strive to gradually lower inflation and react to temporary output shocks as well. The evidence discovered by Cecchetti and Ehrmann (2000) showed that output deviation has a positive weight in all objective functions of a central bank applying ITF.
- (6) *Monetary policy has a clearer focus on inflation and can be more tightly applied through ITF implementation.* The mandate of the central bank to maintain price stability was reinforced by ITF implementation (Bernanke et al., 1999). Cecchetti and Ehrmann (2000) showed that central bank aversion to inflation shocks (relative to output shocks) was strengthened by ITF implementation, a conclusion that was corroborated by Corbo et al. (2000).
- (7) *Central bank independence could be strengthened by ITF implementation.* Experience from several countries in the 1990s showed that greater independence was afforded to the central bank in support of ITF implementation. In several countries, ITF was introduced after formal independence and instruments were given to the central bank, such as in the cases of New Zealand and Chile. In other countries, including United Kingdom, instrument independence was given after ITF implementation. That positive correlation was further confirmed by the empirical findings concerning formal independence and instruments, but not for target independence.
- (8) *Communication, transparency, and accountability are strengthened through ITF implementation.* In general, ITF implementation is accompanied (or sometimes preceded) by improvements underlying central bank communications with the public and markets to enhance monetary policy transparency. Most central banks applying ITF have begun publishing inflation reports, monetary policy statements, minutes of board of governors’ meetings, central bank models, and inflation projections. Such underlying communication efforts employed by central banks appear to be strengthened by ITF application compared to other monetary policy regimes, primarily because of the crucial role played by policy credibility and the management of inflation expectations to achieve the inflation target per ITF (Bernanke et al., 1999).

8.5.1.2. Alternative Opinions. The conclusions made by Mishkin and Schmidt-Hebbel (2001) generally reflect the views of economists as well as central bankers applying ITF. The ITF success stories underlie the economists' volition to propose ITF as the ideal framework for a new monetary policy regime. Proponents of this monetary policy framework refer to the various advantages associated with ITF implementation, such as those described above. With such a range of advantages, ITF appears to be a suitable new monetary policy paradigm for implementation in other countries, including the US and ECB.

In their latest study, however, Ball and Sheridan (2003) revealed some empirical findings that could change or, at least, soften this view. In general, the two economists concluded that, if the focus is merely on ITF countries, macroeconomic performance has indeed improved with the implementation of the ITF. Nevertheless, macroeconomic performance in non-ITF countries also improved during the 1990s. Therefore, the question arises: were the macroeconomic gains reported in ITF countries due to ITF implementation or other more general reasons that were also found in other countries.

Seeking to investigate this issue more rigorously, Ball and Sheridan (2003) researched the impact of ITF implementation on macroeconomic performance in 20 OECD member countries, seven of which applied ITF prior to 1999 (New Zealand, Australia, Canada, United Kingdom, Finland, Sweden, and Spain) and 13 others that did not. The study analyzed whether ITF affected several dimensions of macroeconomic performance, such as inflation, output growth, and interest rates, which differed between ITF and non-ITF countries. Departing from previous studies based on VAR models or structural macroeconomic models, Ball and Sheridan (2003) based their methodology on the "differences in differences" approach. To that end, the observation sample was divided into several subsamples of before and after ITF implementation. By separating the subsamples, the impact of ITF on inflation, output growth, and interest rates could be analyzed, while also controlling the specific influence of countries, observation periods, and ITF implementation itself.

The results of this study not only confirmed the previous empirical findings but also provided some different conclusions. In general, the salient empirical results concerning the impact of ITF implementation on inflation, economic growth, and interest rates can be summarized as follows.

First, ITF's impact on inflation. Similar to the findings of several other studies, ITF implementation can lower the rate and variability of inflation to levels comparable in non-ITF advanced countries. Likewise, ITF implementation can reduce inflation persistence toward the target set rather than previous inflation (inertia). Nonetheless, when compared to non-ITF advanced countries, Ball and Sheridan (2003) found no irrefutable evidence that ITF implementation provided additional advantages. In terms of disinflation, for instance, if the influence of initial inflation conditions is omitted, ITF does not provide any different advantages than in the other countries. Furthermore, ITF implementation was also not proven to decrease inflation variability any more than in non-ITF countries. Similarly, the decline of inflation persistence found in ITF countries was also observed in non-ITF countries.

Second, ITF's impact on output growth. Ball and Sheridan (2003) found that ITF implementation was not unequivocally shown to stimulate real economic growth, as effused by ITF supporters (e.g., Mishkin, 1999). Average economic growth was indeed observed to accelerate after ITF implementation, rather than decline as in the case of non-ITF countries. Nevertheless, the difference between the two groups was not significant. Similarly, Ball and Sheridan (2003) found no empirical evidence that the implementation of flexible ITF stabilized inflation and output, or that output variability increased if ITF was applied too strictly. During the period when ITF was first implemented, output variability was observed to increase in ITF countries. Nonetheless, output stabilized after ITF implementation and was not significantly different to output variability reported in non-ITF countries.

Third, ITF's impact on interest rates. ITF implementation was said to reduce inflation permanently toward the target and, therefore, lower long-term interest rates. Ball and Sheridan (2003), however, found no evidence for this. Although long-term real interest rates decreased in ITF countries, the improvements were not particularly different from those enjoyed in non-ITF countries. Similarly, ITF implementation has not been shown to significantly change central bank behavior in terms of applying interest rate monetary instruments. Despite higher short-term interest rate volatility in ITF countries, which subsequently declined, similar observations were also found in non-ITF countries.

With their empirical findings, Ball and Sheridan (2003) concluded that ITF-based monetary policy did not significantly improve economic performance in the sample countries. Both economists offered several reasons why ITF-based monetary policy did not significantly improve economic performance. One possibility was that ITF and non-ITF countries implemented similar interest rate policies, previously identified by several studies as using a rule, such as the Taylor rule. If this is the case, ITF implementation, fundamentally, does not change the behavior of monetary policy instruments and, therefore, it is not surprising that economic performance would also not experience a significant change. These findings, however, are not supported by the institutional arrangements or formal ITF regulations, such as inflation target announcements, inflation reports, and central bank independence, which have also been implemented to some extent in non-ITF advanced countries.

8.5.2. Inflation Targeting after the GFC of 2008/2009

During the past decade, especially after the GFC, the global economy has confronted various difficult challenges. Furthermore, the global economy has not yet fully recovered from the crisis, therefore, the challenges to ITF implementation have also become more complex, particularly in terms of confidence in credible ITF implementation on one hand and determining the priority between price stability and output growth to maintain economic recovery momentum on the other. Another pertinent issue concerns how far existing ITF implementation has overcome the structural and specific problems of the respective economies.

Fundamentally, the GFC also provided valuable lessons for monetary policy. *First*, in an open economy, monetary policy faces increasingly complex challenges, thus the instruments used must be diverse and part of an instrument mix. A policy mix of monetary policy, capital flow, and exchange rate management as well as macroprudential policy allows the central bank to overcome the various dilemmas faced.¹¹ In terms of monetary policy, monetary policy complexity through the interest rate can be overcome by applying monetary policy quantitatively. When confronting capital flows, for instance, the exchange rate should be managed, albeit flexibly, to avoid misalignment from the currency's fundamental value because such conditions could threaten macroeconomic stability. Simultaneously, efforts to accumulate reserve assets are also imperative as part of the strategy "to prepare the umbrella before it rains" considering short-term capital flows are particularly vulnerable to the risk of reversal. Regarding capital flow policy, by maintaining a floating exchange rate regime, various macroprudential policy options are available to overcome excessive capital flow cycles, particularly short-term and volatile. In addition, macroprudential policy that aims to maintain financial system stability could mitigate bubble risks in the economy.

Second, the global crisis reinforced belief that price stability should remain the final goal of central bank monetary policy. Nevertheless, the global crisis also taught us that low inflation alone is not enough to achieve macroeconomic stability. Several crises that have occurred in recent decades have shown that macroeconomic stability primarily originates from the financial system sector. Financial markets are always inherently characterized by imperfections, which create excessive macroeconomic fluctuations. Therefore, the key to managing macroeconomic stability is not only success in terms of controlling goods (inflation) and external (balance of payments) imbalances, but also imbalances in the financial sector, such as excessive new loan growth, asset price bubbles, and procyclical risk-taking behavior in the financial sector that is vulnerable to changes in perception.

Third, the need to clearly position the role of the exchange rate in the ITF. Per the ITF monetary policy framework, a floating exchange rate system represents the optimal choice for an economy. This policy direction is required because the exchange rate system will act as an economic shock absorber. Nonetheless, exchange rates in an environment of globally integrated financial markets have been shown to move independently. In this case, the most dominant exchange rate dynamics will influence changes in investor risk perception on global financial markets, which would intensify short-term foreign capital flows more than fundamental factors.

Learning from experience, in general, all parties have agreed that the main goal of monetary policy should remain focused on achieving price stability or low inflation. The issue is that when faced with a complex challenge, the standard ITF-based monetary policy framework is ineffective. For example, according to standard ITF, the interest rate is the only monetary policy instrument that

¹¹A more detailed description concerning the central bank policy mix will be presented in the closing remarks of Chapter 15.

influences aggregate demand and the output gap, while anchoring inflation expectations to the target. Notwithstanding, in an open economy, hiking the interest rate is often ineffective due to the imminent deluge of foreign capital inflows that flood the economy with liquidity. Without sterilization, the additional liquidity would intensify inflationary pressures and create asset price bubbles that undermine financial system stability. Therefore, new instruments are required to avoid that dilemma faced by central banks.

In that context, the central bank can effectively maintain macroeconomic stability if able to control financial sector behavior. Consequently, ITF needs to be enriched to capture financial sector dynamics by stressing the importance of risk management in the macroeconomic and macroprudential policy frameworks as well as paying due consideration to the tail risks that would be detrimental to the economy.

Fundamentally, ITF can still be relied upon as a monetary policy strategy. Referring to a comprehensive evaluation of nine basic principles of monetary policy, including ITF, which were considered a general consensus prior to the crisis, Mishkin (2011) concluded that “none of the lessons from the financial crisis in any way undermines the nine basic principles of science of monetary policy.” The nine principles are as follows:

- (1) inflation is always and everywhere a monetary phenomenon;
- (2) price stability has important benefits;
- (3) there is no long-run trade-off between unemployment and inflation;
- (4) expectations play a crucial role in the determination of inflation and in the transmission of monetary policy to the macroeconomy;
- (5) real interest rates need to rise with higher inflation, that is, the Taylor principle;
- (6) monetary policy is subject to the time-inconsistency problem;
- (7) central bank independence helps improve the efficiency of monetary policy;
- (8) commitment to a strong nominal anchor is central to producing good monetary policy outcomes; and
- (9) financial frictions play an important role in business cycles.

Referring to those nine principles, Mishkin (2011) also indirectly stressed that, theoretically, ITF, which is oriented toward achieving low inflation with greater transparency, is still relevant to the monetary policy goal of achieving price stability.

8.6. Concluding Remarks

ITF proliferation throughout numerous central banks has been strengthened by its status as a new monetary policy framework that could viably be implemented in other countries, especially those confronting inflation problems to the detriment of economic advancement and the amelioration of public welfare. The strength of this new framework is not merely reflected in its goal-focused approach, coupled with avowed central bank commitment and monetary policy discipline, but also in terms of the institutional arrangements that reflect the

modern practices of an independent, accountable, and transparent central bank. The institutional arrangements are more in line with the spread of democracy around the world.

Various empirical studies have documented the benefits of ITF implementation to improve economic performance. ITF can help lower inflation to the levels achieved in advanced countries, reduce inflation persistence, anchor inflation expectations to the target as well as lower output volatility and the sacrifice ratio, while remaining flexible enough to overcome short-run economic shocks and improve the discipline, commitment, and credibility of the monetary policy instituted by the central bank. Moreover, ITF implementation can strengthen monetary policy independence, transparency, and accountability.

Several empirical studies have also provided evidence that ITF implementation is not always successful in terms of improving economic performance. Nevertheless, such studies tend to stress that the various success stories enjoyed by ITF countries were also enjoyed by non-ITF countries. In their conclusion, Ball and Sheridan (2003) also found empirical evidence that could not be interpreted as a rejection of ITF. *First*, ITF can provide benefits politically rather than the economically. As suggested by Bernanke et al. (1999), ITF implementation creates more open and transparent monetary policy formulation, thereby supporting a central bank's role that is more consistent with the principles of democracy.

Second, ITF implementation may improve economic performance in the future. Despite the number of economic shocks experienced in the past, many central banks still chose to apply an untested ITF to overcome the underlying shocks. In the future, policymakers may face supply-side shocks similar to what took place in the 1970s or strong political pressures for inflationary policies. At that time, we shall observe how well the central banks applying ITF are able to face the challenges compared to more effective central banks and those adopting a just-do-it policy approach.

Fundamentally, although ITF can still be relied upon as a sound monetary policy strategy, the specific issues that arose after the GFC made central bankers realize the need to strengthen the monetary policy framework through refinements to ITF strategy. To that end, assessments of ITF implementation in various developing countries have provided sufficient justification for the need to apply flexible ITF as an ideal format for a small open economy.¹²

¹²The term "flexible ITF" was first popularized by Svensson (1999) by juxtaposing strict ITF versus flexible ITF. Per strict ITF, or the so-called "inflation nutters," the central bank only focuses on the inflation gap (deviation between inflation and its target) (King, 1997). In contrast, per flexible ITF, the central bank pays due consideration to the inflation gap, as well as the output gap and/or interest rate smoothing. Nevertheless, no central banks currently apply strict ITF (refer to Svensson, 2010; Walsh, 2008).

Chapter 9

Inflation Targeting Framework: Implementation in Indonesia

9.1. Introduction

Enactment of the new Bank Indonesia (BI) Act, namely Act No. 23 of 1999, which was subsequently amended by Act No. 3 of 2004 and Act No. 6 of 2009, prompted a fundamental change underlying monetary policy formulation and implementation in Indonesia.¹ The new legislation contained a strong legal foundation for modern practices at an independent central bank, primarily concerning provisions on the goal of BI, monetary policy independence as well as public accountability and transparency consistent with the implementation of good governance. Act No. 23 of 1999 stipulated a clear mandate to achieve and maintain rupiah stability as the overarching goal of BI. Accordingly, BI's duties focused on three aspects, namely formulating and instituting monetary policy, regulating and supervising the banking industry, and regulating and operating the payment system.² The statutes also provided a strong legal basis for central bank independence, including goal independence, instrument independence, and institutional independence.

Implicitly, the regulations also mandated the implementation of monetary policy based on what was known in central banking theory and practice as the Inflation Targeting Framework (ITF) and, therefore, BI initiated various

¹Two amendments to the Bank Indonesia Act have been implemented, in 2004 and 2009. The first amendment, contained in Act No. 3 of 2004, strengthened several institutional aspects and had significant implications on the implementation of monetary policy, as described in Appendix 9.1.

²With the enactment of Act No. 21 of 2011 concerning the Financial Services Authority (OJK), the bank regulation and oversight function was transferred from Bank Indonesia to OJK. Bank Indonesia, as the macroprudential authority, in conjunction with other authorities (OJK, Ministry of Finance and Deposit Insurance Corporation (LPS)), plays its role in terms of maintaining financial system stability. A detailed description of the role central bank macroprudential policy plays in terms of maintaining financial system stability will be presented in Chapter 14.

implementation measures in Indonesia. As part of Act No. 23 of 1999 implementation, after the year 2000, BI set and announced its desired inflation target to be achieved through monetary policy. Furthermore, various endeavors were implemented to prepare the preconditions required for full ITF application in the future. Meanwhile, monetary policy per the operational target of base money under the auspices of the IMF was continued, but research was conducted to develop a new monetary policy framework consistent with ITF. Regular Board of Governors (RDG) meetings were held, as the highest decision-making forum at BI, to formulate monetary policy. For accountability and transparency, BI submitted periodic reports to the People's Representative Council and communicated routinely with the public regarding monetary policy implementation by BI.

The path to full ITF implementation in Indonesia was fraught with difficulties. Experience since 2000 has shown that inflation is constantly influenced by factors beyond the scope of monetary policy, while the monetary policy transmission mechanism to inflation and the economy has been impaired by several constraints. Some of the constraints were linked to prevailing economic and financial sector dynamics in Indonesia at that time, which were undergoing structural changes in the wake of the Asian Financial Crisis of 1997/1998 that blurred the monetary policy transmission process, making it difficult to predict. Other constraints stemmed from the monetary policy framework still under development consistent with ITF. Furthermore, support from the stakeholders, including the People's Representative Council, government and public, remained weak for BI in terms of building credibility for ITF implementation. The various economic and political dynamics prevalent at that time help to explain why ITF implementation in Indonesia was replete with difficulties.

After adequate preparations had been completed, BI officially introduced (full) ITF on July 1, 2005, using the interest rate as the operational target. Reorientation of the monetary policy target per ITF was also due in part to macroeconomic recovery policy to evade the devastating economic crisis. Nonetheless, this was not easily achieved considering the recent catastrophic economic crisis, the recovery from which was more expensive and prolonged than in other similarly affected countries in the region. Gradually, ITF implementation was accepted by the stakeholders after BI successfully built up adequate policy credibility in terms of controlling inflation in Indonesia.

During the past decade or so, when the economic situation had not fully recovered from the problems that emerged due to the global financial crisis (GFC), the challenges to implementing a new monetary policy paradigm based on the ITF have become more onerous and complex. This was particularly true concerning confidence in applying a credible ITF on one hand, and prioritizing price stability or output growth to maintain economic recovery momentum, which was blighted by ubiquitous uncertainty, on the other. Such developments compelled the central bank to strengthen its monetary policy framework by honing the future ITF implementation strategy.

This chapter will review ITF implementation in Indonesia since 2000.³ The discussion will focus on the institutional arrangements and operational frameworks found in the literature (Schaechter, Stone, & Zelmer, 2000).⁴ In terms of the institutional arrangements, the salient aspects of the new BI Act will be reviewed, along with inflation target formulation, accountability, and transparency. In terms of the operational framework, inflation projection techniques, monetary policy transmission, and monetary operations will be explored. The subsequent section will evaluate monetary policy performance in terms of controlling inflation and the various constraints faced, particularly the salient issues that emerged during the transition phase and during the early stages of ITF implementation. Before the concluding remarks, the substance and rationale behind strengthening the ITF implementation strategy in Indonesia after the GFC will be discussed.

9.2. Institutional Framework

As elaborated in the previous chapter, ITF-based monetary policy requires underlying institutional arrangements, including provisions in the central bank act, inflation target formulation as well as accountability and transparency. In the case of Indonesia, institutional arrangements are contained in the new BI Act. Furthermore, BI has also implemented incremental measures since 2000.

9.2.1. BI Act

The BI Act of 1999 provides a solid legal foundation concerning central bank goal clarity, monetary policy independence and restrictions on central bank financing of the government's fiscal deficit. The overarching purpose of BI is to achieve and maintain rupiah stability, encompassing price (inflation) stability and rupiah exchange rate stability. This represents a grand departure from the previous BI Act, which mandated that BI was tasked not only with maintaining rupiah stability but also with stimulating production activity and expanding employment opportunities.

With the floating exchange rate system embraced by Indonesia, this goal theoretically meant that price stability was the overriding objective, while exchange

³To review ITF implementation in Indonesia during the transition phase, refer to Warjiyo (2002), Alamsyah et al. (2000), and Sitorus et al. (2000). Moreover, to review ITF performance after the official introduction in July 2005, refer to Juhro et al. (2009) and Juhro (2015).

⁴A review of the institutional implications of ITF implementation will be presented in Part IV of this book, which explores the various institutional aspects of monetary policy in detail, including credibility and time consistency, independence, and accountability, as well as policy transparency and communication strategies.

rate stability was an intermediate target or indicator in monetary policy formulation.⁵ Such conditions were strengthened by laws that directed monetary policy toward achieving the inflation target. In this regard, the inflation target, pursuant to Act No. 23 of 1999, set by BI (goal independence), was amended by Act No. 3 of 2004, according to which the inflation target was set by the government after consultation with BI (goal dependence).

To achieve the inflation target, BI was given independence to formulate and implement monetary policy. In other words, the prerequisite of instrument independence demanded by ITF implementation was safeguarded by law. The monetary instruments include open market operations, the discount rate, reserve requirement and bank lending regulations. BI can also intervene on the FX market to stabilize the rupiah exchange rate as part of open market operations as well as provide short-term funding facilities as lender of last resort. In addition, considering the rapid development of sharia banking in Indonesia, monetary policy implementation is also in compliance with sharia principles.

The new BI Act also clearly stipulates that it is illegal for the central bank to finance the government deficit. Pursuant to Act No. 23 of 1999, BI was prohibited from extending loans to the government. The regulation, however, permits the purchase of government debt securities on the secondary market for open market operations. Thereafter, in accordance with the amendment, Act No. 3 of 2004, the purchase of government debt securities on the primary market was permitted to bolster stock for monetary instruments but limited to instruments of up to one year. The purchase of government securities on the primary market by BI to provide a financial safety net and, therefore, maintain financial system stability is only possible upon approval from the People's Representative. The regulations provide adequate restrictions to ensure no fiscal dominance, as required by sound monetary policy principles, especially ITF.

9.2.2. Inflation Target Formulation

Act No. 23 of 1999, as amended by Act No. 3 of 2004 and Act No. 6 of 2009, provides discretion to the Government and BI in terms of formulating the inflation target as the final goal of monetary policy, specifically in terms of the price indexes used, size of the target and horizon. Prevailing laws merely state that the inflation target must be determined: (1) based on a calendar year; (2) with consideration of price developments directly influenced by monetary policy; and (3) with consideration of the macroeconomic developments and outlook. Meanwhile, the announcement of the inflation target was changed from BI itself in pursuance of Act No. 23 of 1999 to the Government after consultation with BI in accordance with Act No. 3 of 2004.

⁵Nevertheless, a lack of understanding concerning this issue often places pressure on monetary policy implementation to achieve exchange rate stability as the final target. This was linked to rupiah exchange rate performance, which depreciated and fluctuated over time. With the relatively stable and appreciating rupiah of the past few years, however, such pressures have begun to ease.

As a country transitioning to full ITF implementation, inflation target formulation in terms of monetary policy in Indonesia has experienced a number of changes since the year 2000. At first, the inflation target was set by BI with a 2% range for CPI beyond the influence of government policy in terms of prices and incomes. For example, the inflation target was set at 3%–5% in 2000 and 4%–6% in 2001. In addition, BI also announced the projected impact of government policy in terms of prices and incomes on inflation, namely at 2% in 2000 and 2%–2.5% in 2001. Therefore, by adding the two, BI's projection of headline inflation was 5%–7% in 2000 and 6%–8.5% in 2001. Since 2002, however, the inflation target was set for all CPI components to facilitate greater public understanding. Furthermore, BI also began to announce a medium-term target, namely 6%–7% for achievement in 2006.

The inflation target, in the form of a value and horizon, was considered optimal, where the target could be oriented toward a low rate, the attainment of which would not impair the economic recovery process in Indonesia.⁶ In that context, taking into consideration the relatively high inflation in Indonesia in 2001, the central bank strived to meet the inflation target through disinflation by setting a lower target each year toward the medium-term target. Therefore, the indicative inflation target was set at 9% for 2002 and 2003, 8% for 2004, 7% for 2005, and 6% for 2006. In addition, bearing in mind the high level of uncertainty in terms of achieving the inflation target and the effectiveness of monetary policy to control inflation in the near term, the annual inflation target was accompanied by a reliable level of deviation, set at $\pm 1\%$. Currently, the government's inflation target,⁷ determined after coordination with BI, is set at $4\% \pm 1\%$ for 2016 and 2017, and $3.5\% \pm 1\%$ for 2018.

Using CPI as the inflation target ensures that core inflation can play more of a role as an indicator underlying monetary policy formulation, considering core inflation is a more accurate measure of the inflationary pressures that can be influenced by monetary policy. Therefore, BI has refined its methods to calculate core inflation using various techniques, including exclusion as a complement to core inflation with the trimmed method statistical technique. In this case, core inflation is calculated by omitting administered prices (AP) and volatile supply shocks. Using this technique, CPI inflation projections can be made by combining core inflation projections, which require a monetary policy response, with the projections of AP and volatile supply shocks. In addition to facilitating public transparency by explaining the various sources of inflationary pressures, this approach also helps to avoid an excessive monetary policy response, particularly in terms of AP and volatile supply shocks.

9.2.3. Accountability and Transparency

As explained in Chapter 8, accountability and transparency are integral elements of ITF implementation. One advantage of this monetary policy framework is

⁶Determining the BI Inflation Target, BI Annual Report, 2001.

⁷Minister of Finance Regulation (PMK) No. 93. PMK.011/2014 concerning the Inflation Target.

central bank commitment to use all instruments at its disposal to achieve the inflation target. That is why the central bank must enjoy instrument independence and be held accountable to the inflation target. Moreover, the central bank must remain transparent and routinely communicate with the public to convey the monetary policy measures instituted to achieve the inflation target. By building credibility in terms of achieving the inflation target, the central bank can fundamentally anchor public inflation expectations consistent with the inflation target and, therefore, ultimately achieve the inflation target.

To that end, BI has implemented several measures to enhance monetary policy accountability and transparency. At the beginning of each year, after the discussions and decisions of the Board of Governors meeting, BI publicly explains its appraisal of monetary policy performance in the previous year and announces the inflation target and direction of monetary policy for the upcoming year through press conferences, supplements published in the mass media, as well as through discussions with academics, bankers and the business community. Furthermore, a formal report on the assessment and direction of monetary policy is also submitted to the People's Representative Council as a performance appraisal of BI and the Board of Governors, as well as to the Government as a valuable form of information. BI also publishes an annual report that contains an evaluation of the economy and the monetary policy direction pursued.

Each quarter, BI also communicates its assessment of inflation and economic performance, along with the monetary policy direction for the upcoming period, through press releases, supplements published in the mass media and discussions with the market if required. Pursuant to prevailing laws, BI also submits a report to the People's Representative Council containing an evaluation of economic and inflation performance, monetary policy, macroprudential policy, and payment system policy. The report is also submitted to the government as a form of information and published as the Monetary Policy Statement to increase transparency. In terms of a monthly review, the decisions of the Board of Governors meeting are also communicated to the public through the Monetary Policy Review.

In addition to publishing and submitting periodic reports to the People's Representative Council, BI also constantly seeks to enhance monetary policy transparency through seminars, workshops, and other communication media for academics, the mass media and public to explain the latest economic developments and monetary policy instituted by BI. In addition, BI also collaborates with numerous universities to include the various policies taken by BI, including monetary, macroprudential, and payment system policies, along with the relevant theories and rationale into the curriculum to enrich student understanding of applying the theories learned in practice at the central bank.

9.3. Operational Framework

As explained in Chapter 8, ITF-based monetary policy credibility is determined by the success of the central bank in terms of attaining the inflation target and public trust in the central bank's commitment to its monetary policy. This clearly requires the support of a solid monetary policy operational framework. *First* is

the ability to project future inflation, thus current monetary policy can anchor inflation projections consistent with the inflation target. This is determined by the availability of various reliable techniques and models to produce inflation and macroeconomic projections. *Second* is a deep understanding of the monetary policy transmission mechanisms, which is important for the central bank to determine the transmission channel, the strength, and duration for monetary policy actions to be transmitted and influence the future direction of the economy and inflation within the target corridor. *Finally*, the central bank must strengthen monetary operations using the various instruments available to signal the markets, thus ensuring the operational target of monetary policy is consistent with the inflation target.

9.3.1. Inflation Projection Techniques

BI utilizes survey outcomes as well as formal economic models to project inflation and other monetary economic variables. In this context, due to the limited amount of surveys conducted by other institutions, BI performs various surveys to detect and capture public perceptions and expectations, including the banking industry, corporate sector, and households, regarding the direction of inflation and other key indicators. BI's surveys include the Business Survey (SKDU), Consumer Expectation Survey (SK), Retail Sales Survey, Property Survey, and Banking Survey. Inflation expectations are primarily captured by the Business Survey and Consumer Expectation Survey, which reflect public expectations at the corporate and consumer levels.

Research has also been conducted to develop formal economic models to project future inflation and other economic variables. Accordingly, the small-scale structural macroeconomic (SSM) model, medium-scale structural macroeconomic model (Short-Term Forecast Model for the Indonesian Economy [SOFIE]), and large-scale structural macroeconomic model (Macroeconomic Model of Bank Indonesia [MODBI]) have been developed. In addition, to strengthen the projection system and simultaneously analyze macroconsistent policy, BI has also adopted the Forecasting and Policy Analysis System (FPAS) developed by the IMF. In this regard, BI has also developed a dynamic "semi" general equilibrium macroeconomic model to simulate policy (ARIMBI) as well as specific *ad hoc* models to project certain macroeconomic variables, including inflation, exchange rates, and money supply.⁸ The various models are mutually complementary and adequate to deliver consistent future economic and inflation projections in the near and medium-long term. The projections generated by the various models

⁸For a brief explanation of each macroeconomic model used by BI, refer to the appendix. In practice, the various models are used together to cross-check the consistency of other projections. Furthermore, the results are benchmarked against the projections of other institutions. Besides, discussions are also held with various relevant parties, including government institutions as well as economists from other institutions, to garner greater understanding of the latest real economic developments.

can be used to determine the inflation target as well as evaluate and project future inflation and economic dynamics, internally at BI to assist in formulating monetary policy and as inputs for the government when preparing the state budget assumptions and medium-term macroeconomic scenarios in terms of the Annual Development Plan (Repeta) and National Development Program (Propennas).

Several reviews have also been conducted to prepare key indicators as required for monetary policymaking, including Leading Indicators for Inflation (LII) and Leading Economic Indicators (LEI). LII consist of real money supply, real credit, interest rates, stock prices and other variables, which provide critical information on the future direction of inflation to confirm the projections of the formal models. Meanwhile, the information contained in the LIE indicates the future direction of economic activity, particularly on the demand side. Considering that demand-side pressures will affect inflation, LIE also function as inputs to project inflation.

9.3.2. Monetary Policy Transmission

Monetary policy implementation in Indonesia during the period before the Asian Financial Crisis of 1997/1998 and during the IMF's program was fundamentally based on the money channel. Therefore, BI would annually project total money supply required for the economy in pursuance of the inflation target and assumptions of economic growth, interest rates, and the exchange rate in the demand for money function. Based on the projected total money supply as the intermediate target, BI would subsequently set the annual, weekly, quarterly, and monthly monetary base targets paying due consideration to seasonal public demand for money trends. The monetary base target was then used as the operational target of monetary operations at BI.⁹

As mentioned in Chapter 5, the monetary channel as the basis underlying monetary policymaking has been deemed less effective. Therefore, BI has performed various studies to review and comprehend the monetary policy transmission mechanisms in Indonesia. The latest comprehensive study conducted by BI showed various interesting changes, especially in the wake of the Asian Financial Crisis of 1997/1998. *First*, the credit channel was shown to significantly influence monetary policy transmission because of the bank disintermediation problem. Bank lending to the real sector is primarily affected by internal bank conditions on the supply side as well as corporate cash flow and leverage on the demand side. Differing from the money channel, imbalances on the credit market have impacted monetary policy transmission from the banking industry to the real sector.

The phenomenon of less effective monetary policy transmission through the credit channel has been reinforced by empirical evidence on transmission through

⁹Under the IMF Program, the quarterly and annual monetary base targets were used as performance criteria for monetary policy implementation in Indonesia. According to such a monetary policy framework, Indonesia could be said to be applying monetary targeting.

the interest rate channel. On one hand, changes in lending rates have had a negative and significant impact on investment (due to the cost of capital) and consumption (due to the substitution and income effects). On the other hand, however, the transmission of short-term interest rates, which can be influenced by monetary instruments, to term deposit rates and lending rates contains a lag and is asymmetric. The empirical evidence for the credit channel and interest rate channel demonstrates the importance of monetary policy, accompanied by efforts to overcome bank disintermediation and, therefore, enhance transmission effectiveness in terms of the economy and inflation.

It is also important to consider the empirical evidence for the exchange rate channel when formulating monetary policy. On one hand, exchange rate developments have a strong influence on inflation due to the direct impact of changing import prices of goods and services (direct pass-through effects) and the indirect impact through the output gap (indirect pass-through effects). On the other hand, however, the ability of monetary instruments to affect exchange rates is constrained by the impact of the risk premium on exchange rate developments. The empirical evidence demonstrates the importance of monetary policy to stabilize exchange rates through FX intervention, not to achieve a given exchange rate target but to minimize the impact on inflation.

The inflation expectations channel is also important to consider. The empirical evidence has shown that inflation expectations are influenced strongly by inflation realization in previous periods (inertia) and exchange rates, in addition to the inflation target. This not only corroborates the importance of exchange rate stabilization measures in the exchange rate channel, but also to enhance monetary policy credibility. Such efforts are required not only to garner commitment to achieving the inflation target, but also to improve the explanations given to the public concerning the monetary policy taken by BI to restore public confidence in the central bank's commitment to the inflation target.

Increasingly dynamic economic developments ultimately manifest in the domestic economy as more complex challenges that could potentially disrupt the monetary policy transmission mechanisms, particularly in terms of the asymmetric impact of monetary policy. Observations have shown that the asymmetric influence of monetary policy is linked to procyclical financial sector behavior as well as risk perception in terms of affecting the transmission mechanism.¹⁰ Under normal conditions, when the economic cycle is expansionary, monetary policy sensitivity to aggregate macroeconomic variables is consistent with the general concept. A reduction to the policy rate would be followed by lower interest rates and faster credit growth, which is in line with the hypothesis of the risk-taking channel, where financial institutions tend to lower their risk perception while simultaneously increasing tolerance to lending risk standards.

¹⁰Financial sector procyclicality is financial system behavior that drives stronger economic growth during an economic upswing and exacerbates the downturn during a contractionary phase. With procyclical behavior, the financial system exacerbates macroeconomic instability by creating output fluctuations.

Nevertheless, conditions changed during the GFC, when the sensitivity of the (lower) lending rate response to BI Rate (reductions) faded. Similar conditions occurred in the response of (increasing) bank lending to (lower) lending rates. Observations also showed that financial institutions became less sensitive to the transmission of lower interest rates. The banking industry tended to delay reductions to interest rates and ration credit despite significant easing of the policy rate (Juhro et al., 2009).

9.3.3. Monetary Operations

Monetary operations, as applied by BI from when Indonesia was still bound by the IMF Program until 2003, used base money as the operational target. Fundamentally, this strategy was initiated by setting the base money target at the beginning of the year in accordance with the projections of total money supply pursuant to demand for liquidity in the economy, which was calculated by projecting the demand for money function in line with the inflation target and other macroeconomic assumptions, particularly economic growth, interest rates, and exchange rates. The annual base money target was translated into quarterly, monthly, and weekly base money targets per the seasonal public demand for money. The weekly base money target was subsequently applied as the operational target of monetary operations at BI, primarily through open market operations, including weekly SBI auctions and other BI facilities to absorb market liquidity. Supporting exchange rate stabilization efforts, BI also conducted FX market intervention as necessary.

As mentioned previously, the monetary policy operational framework using base money as the operational target was eventually considered ineffective due primarily to the fading correlation between money supply and real sector activities and inflation, as well as the increasing complexity of controlling base money due to unstable and unpredictable demand for money. In July 2005, BI began to apply interest rates in its monetary operations. In addition to the dwindling effectiveness of base money in terms of monetary management to achieve the inflation target, the interest rate was required as the operational target of monetary policy in line with BI's efforts to officially implement an ITF-based monetary policy framework in full.

Various efforts have been undertaken to ensure more effective implementation of the interest rate as the operational target of monetary policy. To control inflation with the interest rate as the operational target, there must be a stable and predictable relationship between the current interest rate and future direction of inflation. In general, the central bank sets the policy rate to anchor the performance of the shortest tenor interest rate available on the market, and then depends on the financial sector to transmit the rate to the whole spectrum of interest rates on the market and, subsequently, to various real economic activities and, ultimately, to inflation. The operational target, therefore, must be set on the shortest tenor of the entire interest rate spectrum available. In the case of Indonesia, the overnight interbank rate or 1-month SBI rate was used. During preliminary ITF implementation in 2005, using the BI Rate as the policy rate,

BI continued to apply the 1-month SBI rate as the operational target. Since the middle of 2008, however, BI has used the overnight (O/N) interbank rate as the operational target of monetary policy.

There are at least two aspects for BI to consider when ensuring the interest rate is feasible as the operational target or monetary policy. The *first* is linked to the importance of forming the term structure of interest rates on the markets and the *second* is connected to efforts to overcome bank disintermediation. Regarding the term structure of interest rates, the secondary market for government bonds has begun to develop with rapidity after several government and BI initiatives to deepen the domestic financial markets. This market for government debt securities importantly underlies formation of the yield curve as a signal of BI's monetary policy. In addition, the availability of such securities also supports BI's ongoing efforts to diversify from SBI to government debt securities as monetary instruments.

Another critical BI effort is the use of models or other analytical tools that can simulate the direction of changes in the interest rate in response to deviations in future inflation projections from the target. Intensive studies have been performed in relation to this, primarily to formulate a forward-looking monetary policy rule. Further refinements of this model have been made. Relevant studies have also been conducted to develop a policy instrument mix and market signaling because policy practices that depended on SBI for open market operations were considered too expensive in terms of the monetary control costs incurred by BI. In addition to optimizing the use of government debt securities (SUN) as monetary instruments (supported by the nascent repo market), the utilization of other policy instruments to manage liquidity and FX flows was also a priority. Furthermore, the central bank is also required to build clearer and more regular communication channels regarding the direction of monetary policy as a form of market signaling, which is important not only to improve transparency but also to build BI monetary policy credibility.

9.4. Evolution Thus Far

9.4.1. Issues during the ITF Transition Period

As mentioned toward the beginning of this chapter, although Act No. 23 of 1999 provided a solid legal foundation for BI to introduce ITF, economic and financial conditions in Indonesia at that time, which had experienced seismic structural changes after the Asian Financial Crisis of 1997/1998, complicated full ITF implementation. Sources of inflationary pressures not only originated from the monetary sector but also from adjustments to AP as well as disruptions to the supply and distribution of goods and services in the economy, which are categorized as volatile foods (VF). The monetary policy transmission mechanism has also met numerous constraints, thereby undermining monetary policy effectiveness in terms of its influence on inflation and the real sector. The problem of bank disintermediation amidst conditions of growing excess liquidity has skewed the effect of changes in interest rate monetary instruments through the interest rate

and credit channels. Such conditions have been complicated further by a vulnerable exchange rate that occasionally threatens an adverse impact on inflation.

Several constraints and requirements must be overcome prior to full ITF implementation. Several opinions have been put forward concerning this topic, including Warjiyo (2002), Alamsyah et al. (2001), Sitorus (2000), Boediono (2000), and Felman (2000). In other words, a number of teething problems must be considered and overcome during the transition period for the successful implementation of full-fledged ITF in Indonesia.

First, the ongoing political transition process in Indonesia has undermined the effectiveness of various public policies, including monetary policy. Therefore, credible ITF implementation can only be achieved if conducted in conjunction with improvements in public perception of the government's general policy climate. Specifically, support from the government, People's Representative Council and the public to commitment for controlling inflation as the goal announced per the monetary policy of the central bank is also a key factor. The amendment to the BI Act, which stipulates that the inflation target is now determined by the Government after consultation with BI, is expected to enhance public selection, commitment, and coordination to controlling inflation by BI.

Second, there remain concerns over government financial sustainability, which affects monetary policy formulation at BI. As a consequence of the crisis, the Indonesian Government faced a deepening debt crisis, with domestic debt from issuances of bank recapitalization bonds amounting to around half of GDP. Around two-thirds of the recapitalization bonds were issued with variable interest rates adjusted to the three-month SBI rate. Consequently, if inflationary pressures accumulated, thus necessitating a hike to the SBI rate in order to achieve the inflation target, it would have a direct impact on the government servicing its domestic debt in the state budget. Although such conditions could happen in other countries, this issue cannot be neglected by BI when formulating monetary policy.

Third, fragilities remain in the Indonesian banking system, which is still in a state of recovery. Although credit growth has accelerated, the bank intermediation function remains a pertinent issue. The credit crunch is a proven phenomenon in Indonesia and the banking industry's interest rate response to the SBI rate is slow with a protracted lag. Considering the dominance of the banking system in Indonesia's financial system, monetary policy effectiveness has been undermined by constraints in the transmission mechanism through the banking sector, which has edged up the cost of monetary controls, meaning that a larger SBI interest rate response is required to prompt any movement in bank interest rates. Furthermore, this issue has been compounded by corporate sector restructuring and, therefore, bank liquidity tends only to circulate in the financial sector. Such conditions not only create decoupling between the financial and real sectors, but also expose the financial sector and complicate monetary policy implementation.

Fourth, housekeeping was still required by BI to hone its monetary policy framework, particularly from an operational perspective, to support ITF implementation. Details of this were explored in previous sections. The salient points, however, include the ability to accurately project inflation and other macroeconomic variables, understanding of the monetary policy transmission mechanism,

and determining the operational target required to achieve the inflation target. Various economic models are still under development, while economic and financial structural changes are also underway, which has complicated monetary policy implementation.

And *finally*, Indonesia's inclusion in the IMF Program also restricted space to implement the ITF. As mentioned previously, the monetary policy framework put in place by the IMF was based on the money channel, using base money as the operational target. Although using base money as the operational target was tried based on demand for money in the economy in line with the inflation target and other macroeconomic projections, the behavior of base money itself has become more difficult to control because of changes in public demand for money and seasonal factors, such as school holidays, Eid-ul-Fitr, Christmas, and New Year. Furthermore, it has become more complicated to control and link the base money target with the monetary policy framework to achieve the inflation target.

At the BOG meeting on March 4, 2004, the BI Board of Governors finally decided to gradually implement a monetary policy framework based on ITF in full with the interest rate as the operational target. This was a positive decision, not only to remove any existing doubts concerning BI's stance, but also to enhance monetary policy commitment and credibility moving forward. In addition, the amendment to the BI Act through Act No. 3 of 2004, which precipitated a number of opportunities and challenges, as well as the improving economic and financial conditions, were expected to support the full implementation of ITF.

There are several important observations that support successful ITF implementation as follows. *First*, pursuant to the amendment to the BI Act, the inflation target is now set by the Government after consultation with BI. On one hand, the amendment garnered government and public ownership and support toward the inflation target and provided a clear mandate for BI to direct monetary policy toward the inflation target. Such conditions are reinforced if BI and the Government can demonstrate tight coordination in terms of setting and announcing the inflation target, for instance through a Memorandum of Understanding (MoU) or monetary and fiscal policy implementation. On the other hand, however, the amendment also stoked demand for greater accountability and transparency at BI in terms of achieving the inflation target and implementing monetary policy. This is a logical consequence of a clear relationship between the principal (Government) and agent (BI) in terms of providing a mandate and independence to achieve the inflation target as the final goal of monetary policy in pursuance of the amendment to the BI Act.

Second, despite evidencing sound developments and relatively well maintained macroeconomic stability, the economic and financial structural changes are still far from complete. Consequently, funds tend to circulate in the financial sector, with little trickling down to the real sector due to decoupling, thus leaving monetary stability vulnerable, which requires monitoring and maintaining. It is also important for BI to observe various microaspects of the financial and real sectors to explore inflation processes and monetary policy transmission mechanisms. This is especially important in countries that have experienced a crisis, such as Indonesia, where various microproblems have emerged. For example, a study concerning

the price setting mechanisms in several major sectors is required or the behavior of important CPI components. Of no less importance would be a study into bank investment portfolio behavior or rupiah and FX money market conditions to explore interest rate and exchange rate behavior. In general, a variety of micro-side studies on economic and financial conditions would enrich and complement other studies on formal macroeconomic modeling approaches and, therefore, help monetary policy formulation and implementation effectiveness at BI.

Third, monetary policy must be strengthened to support the success of applying the interest rate as the operational target. This is not merely limited to the ability to formulate a rule mechanism to determine the interest rate target, but also how the instruments and monetary operations could support attainment of the interest rate operational target. This issue demands more attention in terms of allowing BI to purchase tradeable government securities (SBN) on the primary market to bolster its stock of monetary instruments and to provide the Government with emergency financing facilities. Despite the existing MoU between BI and the Government concerning this matter, which would be strengthened through additional legislation, the provision of emergency financing facilities would prompt monetary policy expansion and undermine the prerequisite of no fiscal dominance to support successful ITF implementation.

Last but not least is increased monetary policy transparency at BI through a more directed and continuous communication strategy. This encompasses communications with the academic community, markets, and public regarding the inflation targeting monetary policy framework to ensure complete understanding as the basis for forming inflation expectations or taking business decisions. A sound communication strategy would ultimately simplify and enhance the effectiveness of monetary policy implementation at BI.

9.4.2. An Evaluation of ITF Implementation since July 1, 2005

During the decade since ITF implementation on July 1, 2005, the Indonesian economy has faced various issues of no less import, linked primarily to the perfect storm of financial and economic crises that reappeared to blight the global and domestic economies just after Indonesia entered its fifth year of ITF implementation. With the current economic situation, which is yet to fully recover from the problems triggered by the GFC, the institutional challenges to implementation of an ITF-based monetary policy paradigm are becoming more complex, especially in terms of confidence in a credible ITF on one hand and determining the priority between price stability and output growth to maintain recovery momentum in an economy overshadowed by uncertainty on the other.

Consequently, the reviews of ITF application in Indonesia for nearly the first five years of implementation have already recorded several successes, including more disciplined monetary policy management, accompanied by enhanced quality in line with best practices, theoretical rationale and empirical conditions in Indonesia (Juhro et al., 2009). In this case, compared to conditions prior to ITF implementation, the successes include setting and announcing the inflation target, the institutional and operational frameworks, policy coordination as well

as policy analysis and research quality. In general, the positive assessment can be linked to the various aspects of business as usual. Specifically, however, ITF implementation has recorded a number of successes in the form of substantive improvements to economic fundamentals, which differentiates the benefits of ITF from other policy frameworks as follows: (1) maturation of institutional existence; (2) policy signal clarity; and (3) increased policy credibility.

Regarding the maturation of institutional existence, ITF application has successfully improved BI institutionally in terms of implementing symmetrical and structured monetary policy based on the principles of good governance when formulating public policy. Such developments are evidenced by transparent and independent policy-making processes and procedures, as well as public accountability to the inflation target. Consequently, BI has changed from an inward-oriented organization to more of an outward-oriented organization.

In terms of policy signal clarity, through a gradual learning process, supported by more intensive public communication, ITF has successfully strengthened monetary policy transmission through expectations. The public now understands the processes underlying monetary policy-making decisions and can better capture monetary policy signals, which strengthens and accelerates monetary policy transmission. This represents a clear departure from conditions prior to ITF implementation, when BI's policy signals, using base money, were not accurately detected by the markets, which, under certain circumstances, would alter or even exacerbate inflation expectations.

Congruent with the two important successes mentioned above, monetary policy credibility can slowly but surely be improved. Several indicators support that conclusion. *First*, observations using surveys and empirical testing showed a previous or ongoing shift in the formation of public inflation expectations from backward looking to forward looking, which has had a positive impact on reducing the degree of inflation persistence. *Second*, consistent with increasing monetary policy credibility, the regular announcement of BI's policy stance through the policy rate (BI Rate) became a crucial reference economic indicator for the markets and wider business community. In addition, regardless of the different perceptions regarding forecasting inflation and other macroindicators by BI and the Government, the public generally understood why BI and the Government used different calculation methods.

Nevertheless, monetary policy credibility requires constant improvement because attainment of the inflation target, as the overriding objective of the central bank, is not as easy to achieve as is generally thought. Several supply-side structural shocks have occurred in recent years, which pushed inflation beyond the target corridor in 2005, 2008, 2010, 2013, and 2014. Inflation in 2005, 2008, 2013, and 2014 soared beyond the upper bound of the target in line with the government's energy reform policy to hike fuel prices.¹¹ Inflation accelerated in 2010

¹¹Fuel prices were hiked twice in 2005, the first time by an average of 30% in March and then again by an average of 96% in October. Fuel prices were hiked again in May 2008 by around 33% and then again by 48% in July 2013. In 2014, the government raised fuel prices by 34% in November.

slightly beyond the target corridor after international commodity prices skyrocketed and inclement weather affected agricultural produce (Table 9.1).

By observing potential inflationary pressures in future periods, if no structural or integrated policy measures were taken, inflation would be difficult to steer within the long-term target of 3% in the immediate future, as occurred in advanced countries as well as neighboring countries in the ASEAN region. Consequently, many parties welcomed the government's energy reforms, primarily by reducing/removing fuel subsidies. Such policy consistency eased inflationary pressures in future periods in terms of AP because a change (increase) in the global oil price would no longer be such a massive strain on the government's budget,

Table 9.1: Actual and Target Inflation.

Year	Inflation Target	Actual Inflation	Core Inflation	SBI Rate (BI Rate)	Main Determinants
2005	6% $\pm$ 1%	17.1	9.7	9.17	Global shocks, fuel price hikes in March and October
2006	8% $\pm$ 1%	6.6	6.03	11.83	
2007	6% $\pm$ 1%	6.6	6.29	8.56	
2008	5% $\pm$ 1%	11.06	8.29	8.67	Fuel price hike (May)
2009	4.5% $\pm$ 1%	2.72	4.09	6.5	Rising international commodity prices, inclement weather
2010	4.5% $\pm$ 1%	6.39	4.29	6.5	
2011	4.5% $\pm$ 1%	3.8	4.1	6.0	
2012	4.5% $\pm$ 1%	4.3	4.4	5.8	Fuel price hike, inclement weather and rupiah depreciation
2013	4.5% $\pm$ 1%	8.4	5.0	7.5	
2014	4.5% $\pm$ 1%	8.4		7.5	Fuel price hike, inclement weather
2015	4% $\pm$ 1%	3.4		7.5	

Source: Bank Indonesia.

thereby obviating further energy/oil subsidy cuts, which would subsequently drive up domestic fuel prices. By naturally absorbing global oil price fluctuations through price adjustments to fuel consumed directly at home, a smoother inflation trend was expected, while minimizing any potential drastic inflation spikes. In the medium-long term, if the funds saved by reducing subsidies could effectively be allocated to infrastructure development and to support energy security, the impact on inflation would be propitious.

Difficulties achieving the inflation target also stemmed from structural issues on the supply side, which caused persistent and fluctuating inflation in Indonesia. Persistently high and fluctuating inflation was due to structural rigidity and the expanding role played by international commodity prices (during the economic boom) in the inflation structure of Indonesia. A body of research exists to corroborate such findings.¹² Relatively high inflation persistence in Indonesia was inextricably linked to structural problems, particularly on the supply side. Domestically, the issues were primarily related to microstructural constraints, such as inefficiency, infrastructure limitations, and the low-skilled labor market, which undermined supply-side responsiveness to policy stimuli on the supply side, the imperfect market structure and inefficient trade channel. Externally, however, international commodity price fluctuations influenced domestic inflationary pressures.

On top of the structural issues plaguing the supply side, difficulty achieving the inflation target also stemmed from the complexity of problems in the monetary sector faced by BI. As experienced during the past five years, BI has intervened on the FX market to overcome the deluge of capital inflows and avoid excessive appreciatory pressures on rupiah exchange rates, which boosted liquidity on domestic money markets. Such conditions have precipitated excess liquidity that must be reabsorbed by BI to curb future inflationary pressures. Of course, BI's efforts to maintain macroeconomic stability were not without consequence, raising the cost of monetary operations and, thus, affecting the financial performance of BI.

9.5. Strengthening the ITF Implementation Strategy after the Global Financial Crisis

In addition to the success stories outlined above, several fundamental refinements have been made to the ITF implementation strategy, including a behavioral change in the financial sector on one hand and calls for ITF to support the domestic economic recovery process after the GFC on the other. The issues culminated in an increasingly complex monetary policy management dimension

¹²Several research papers conducted by BI concluded a high degree of core inflation persistence. Alamsyah (2008) confirmed the findings for high inflation persistence. The degree of headline inflation persistence and a disaggregation based on groups of goods and services was recorded at 0.8–0.9, despite tracking a downward trend between the periods before and after the crisis.

in Indonesia, thereby requiring BI to more flexibly respond to the uncertainties appearing in the economy, beyond conventional wisdom.

Referring to the previous assessments, it can be concluded that ITF is still fundamentally reliable as the *de facto* monetary policy strategy in Indonesia. Notwithstanding, due to the issues explained above, BI must reinforce its monetary policy framework through further refinements to the ITF implementation strategy. To that end, an assessment of ITF implementation in Indonesia has also provided sufficient justification for the application of Flexible ITF as the ideal format for the Indonesian economy (Juhro et al., 2009).¹³

That rationale implies three fundamental consequences in terms of ITF design and the significance of flexibility, particularly in relation to the monetary policy preferences and response as follows:

- (1) In the near term and with maintained macroeconomic stability, there can be a potential shift in the focus of monetary policy toward greater emphasis on economic growth. This fundamentally confirms the idea that achieving price stability as the overriding goal does not imply that the central bank can neglect efforts to maintain economic growth momentum. In contrast, in the near term, the preference of monetary policy can be oriented toward specific considerations that maintain the economic recovery process, while striving toward price stability in the long term as a prerequisite for sustainable economic growth.
- (2) Congruent with some of the lessons garnered from the GFC, a more flexible monetary policy response is required when considering the strategic role and dynamics of the financial sector. To that end, an optimal and effective monetary and macroprudential policy instrument mix is required. Despite the need for greater consideration of the strategic role and dynamics of the financial sector, the strategic monetary policy framework in Indonesia maintains price stability as the main factor underlying the monetary policy response.
- (3) In terms of policy implementation, flexibility also contains institutional aspects considering that the root of the problems faced by the central bank is not always within the scope of monetary policy influence. Therefore, strong policy coordination between BI and the Government is imperative.

It should be reiterated that, in the context of ITF implementation, BI already implicitly applies Flexible ITF, meaning that in the near term, BI strives to stabilize inflation and the real economy together. ITF flexibility is reflected in the

¹³The term “Flexible ITF” was first popularized by Svensson (1999) by contrasting strict ITF versus flexible ITF. According to strict ITF implementation, otherwise known as inflation nutters, the central bank only focuses on the inflation gap (deviation between inflation and the target) (King, 1997). Conversely, per flexible ITF, the central bank not only focuses on the inflation gap but also the output gap and/or interest rate smoothing. Nevertheless, strict ITF is not currently applied in any country (refer to Svensson, 2010; Walsh, 2008).

orientation to achieve the overriding goal, namely inflation, but while also paying due consideration to the dynamic performance of other macroeconomic variables in the short term, such as economic growth, exchange rates, and financial sector developments. To that end, five important factors are always considered when determining the policy rate, namely: (1) the two-year inflation forecast and consistency with the inflation target; (2) the two-year economic growth forecast; (3) the exchange rate forecast and affecting factors (including capital flows), as well the impact on the inflation and economic growth forecasts; (4) bank credit and interest rates; and (5) asset valuation in the financial sector. The first two factors maintain BI Rate consistency with attainment of the inflation target, while considering the inflation-growth trade-off to maximize public welfare. The final three factors balance monetary stability and financial system stability, while also assessing the monetary policy transmission mechanism.

Similarly, from a policy perspective in terms of managing internal and external balance in line with economic dynamics after the GFC, BI has, over the past five years, also implemented a policy mix, encompassing interest rate policy, exchange rate policy and macroprudential policy.¹⁴ For example, the orientation of the policy mix in 2001 was as follows. *First*, the BI Rate was required to signal BI's avowed commitment to achieve the future inflation target. *Second*, to help ease inflationary pressures, BI embraced more appreciatory rupiah management, approaching the degree of appreciation found in neighboring countries, in line with the currency's fundamental path. Appreciatory rupiah management was aligned with an optimal accumulation of reserve assets to bolster self-assurance against the risk of a capital reversal. *Third*, to control the dynamic flow of foreign capital and simultaneously mitigate capital reversal risk, BI applied macroprudential policy to manage foreign capital flows, including a foreign currency reserve requirement along with a minimum holding period for BI Certificates (SBI) and mandatory hedging. *Fourth*, considering the excess liquidity in the banking system, BI prepared options to manage the excess liquidity, including a rupiah reserve requirement. *Fifth*, BI strengthened policy coordination with the Government to control inflation by providing recommendations to stabilize VF and regarding the impact of AP on inflation, while also optimizing the role of the National Inflation Task Force (TPI) and Regional Inflation Task Forces (TPID).

Based on the lengthy experiences and lessons of implementing a monetary policy strategy while exposed to a financial crisis, the in-depth literature and research on ITF per the characteristics of the Indonesian economy, as well as an assessment of ITF best practices at different central banks, the monetary policy strategy was strengthened in accordance with the domestic economy according to the format of Flexible ITF. In this context, the scope of flexibility not only refers to the policy preferences toward price and output stability, as contained in the literature, but also other strategic aspects of monetary policy. Therefore, based on the elements inherent to ITF, the fundamental components of flexible ITF were formulated per the following rationale (Juhro, 2015; Warjiyo, 2013c):

¹⁴A discussion on the central bank policy mix is presented specifically in Chapter 15.

(1) *Inflation targeting as the strategy underlying monetary policy:*

The main substance of inflation targeting is to control inflation per the target as the overriding objective of monetary policy. The trade-off between economic growth, exchange rate stability, and financial system stability is taken into consideration when formulating monetary policy but, if a conflict emerges, the inflation target is prioritized. In addition, the institutional elements of ITF must also be strengthened, including monetary policy independence, accountability, and transparency.

(2) *Monetary and macroprudential policy integration to strengthen policy transmission and support macroeconomic stability:*

The linkages between the monetary stability and financial system stability frameworks can be strengthened through monetary policy integration with macroprudential policy. The global crisis provided an invaluable lesson concerning the financial accelerator in terms of monetary policy. Financial system stability determines the effectiveness of monetary policy transmission. Likewise, the monetary policy response influences financial system stability. Therefore, a policy instrument mix is crucial. In that regard, the policy rate response, as the main policy stance, must be supported by macroprudential policy to manage foreign capital flows and domestic liquidity.

(3) *The role of policy to manage exchange rates and foreign capital flows in support of macroeconomic stability:*

Strengthening exchange rate policy to achieve price stability. In this case, exchange rate stability is managed to remain consistent with attainment of the inflation target and macroeconomic stability. An optimal solution to the impossible trinity is sought by observing the linkages between exchange rate stabilization policy and foreign capital flow management, as well as the implications on macroeconomic stability.

(4) *Strengthening policy coordination between BI and the Government to control inflation as well as maintain monetary and financial system stability:*

Strengthening the coordination framework is imperative considering that, in addition to the demand side, sources of inflationary pressure also originate from the supply side and strategic commodities. Furthermore, more limited economic capacity and constraints to the implementation of infrastructure programs necessitate an integrated policy strategy between the policy authorities. Besides, from a broader perspective, policy coordination can be applied to manage foreign capital/portfolio flows because such flows are vulnerable to short-term shocks and issues that could trigger a reversal.

(5) *Strengthening the policy communication strategy as part of the policy instruments:*

Communicating monetary policy is no longer merely a way to increase transparency and accountability but is becoming an important monetary policy instrument. The communication policy is designed to anchor public and market expectations, reduce uncertainty, decipher the noise, and increase predictability, thereby alleviating financial market volatility, while simultaneously providing greater public understanding of the goals of monetary policy, the monetary policy framework and operational framework as well as monetary policy transmission.

Referring to the five elements detailed above, in the achievement of the overriding objective, ITF and Flexible ITF are substantively the same, namely by targeting inflation. A new dimension that has emerged since the GFC is the central bank's role in maintaining financial system stability, which requires an integrated mandate to also achieve price stability. Furthermore, there is a need to strengthen the ITF implementation strategy itself; in this case in terms of embodying the element of flexibility, namely by providing adequate space to integrate monetary and financial system stability through a mix of monetary policy, macroprudential policy, exchange rate policy, and foreign capital flow management; as well as institutional reforms to optimize the role of policy coordination and communication.

9.6. Concluding Remarks

Over the past 15 years, with the economic situation yet to fully recover from the issues that emerged due to the GFC, the monetary policy challenges in terms of the ITF strategy in Indonesia have been onerous. Various challenges emerged as problems rooted in the dynamics of the strategic environment globally and nationally. Externally, the problems stemmed from the GFC along with short-term capital flow mobility, which had a strong impact on exchange rate developments. In addition, changing behavior in the financial system, which was linked to the growing role of risk perception and procyclicality, also exacerbated the complexity of monetary control and complicated decisionmaking. At home, the issues related to structural rigidity on the supply side, which fundamentally disrupted the policy transmission mechanism and placed additional pressures on monetary stability. Against such a dynamic backdrop, it was no longer a simple matter for BI to steer inflation toward the target and, therefore, build monetary policy credibility.

Such conditions were congruous with the ongoing academic debate and empirical studies that extolled the feasibility of ITF, especially in developing countries, which were typically undergoing structural change and in the early throes of transition or institutional development. Approaching its 25th anniversary, the legitimacy and compatibility of ITF has also been an arduous challenge since the GFC, which slashed global economic growth until the present day. No one could have predicted the rapidity and depth at which the financial crisis struck. In the quarterly and even monthly data, global growth projections by international organizations (IMF and World Bank) were continuously revised downwards beyond existing predictions. For example, in the middle of 2008, the IMF still projected global economic growth at around 3.9% but ubiquitous uncertainty and risk compelled the international institution to revise down its projection again and again. Currently, global economic growth is around 3% and expected to track a decelerating trend, primarily due to the deep contractions experienced in advanced countries. Weaker economic growth was coupled with an unforeseen decline in potential output, especially when compared to conditions before 2007, as characterized by robust economic growth along with relatively stable inflation.

In general, from the important lessons learned from those salient experiences, all parties concurred that the overriding goal of monetary policy must remain focused on achieving price stability or low inflation. Nevertheless, the substance of stability could be interpreted into a broader rationale, namely by considering financial system stability indicators, demonstrating that (flexible) ITF implementation was necessary but not sufficient. To that end, the successes of ITF implementation must be supported by a macroprudential policy regulatory framework. This would simultaneously overcome one of the prerequisites to successful ITF implementation, namely a sound and efficient financial system.

The fundamental substance must receive attention, namely that when formulating a post-crisis monetary policy strategy, the central bank must strengthen financial system stability to ensure a stable economy and financial system in terms of the macroeconomy and financial sector. The shift or emphasis in the mandate of the central bank to maintain financial system stability created the need for a new paradigm and a central bank policy mix.¹⁵ Therefore, the format of BI's mandate to concurrently maintain monetary and financial system stability needs to be formulated appropriately, especially in terms of formulating an optimal monetary and macroprudential policy mix. Chapter 15 discusses the new central bank policy mix paradigm and its implementation in Indonesia since 2010.

¹⁵Departing from the existing format and governance of monetary policy that was widely understood in terms of ITF implementation, the new format began development and implementation at central banks in the wake of the GFC in 2008/2009.

Appendix 9.1: Amendments to the Bank Indonesia Act and their Implications

Act No. 3 of 2004, as an amendment to the BI Act (No. 23) of 1999, contained several fundamental changes to the regulatory, institutional, and policy frameworks of BI. The amendment stipulated a total of four major changes in terms of monetary policy implementation, namely determining the inflation target, the purchase of government debt securities (SUN) on the primary market for monetary operations, the purchase of SUN on the primary market to provide government funds for emergency financing, as well as accountability and transparency.

Setting the Inflation Target by the Government

A major change was implemented with the introduction of Act No. 3 of 2004, namely that the inflation target formally set by BI (goal independence) was now determined by the government after consultation with BI (goal dependence). On one hand, the loss of independence to set the inflation target reduced BI's discretion in terms of determining the inflation target and formulating monetary policy. Per Act No. 23 of 1999, BI had full discretion to set the short- and long-term inflation targets, the type of inflation (headline or core) and the form of the target (point or range). The inflation target could also be adjusted to the ability of monetary policy to achieve the target. After an amendment to the BI Act, in the form of Act No. 3 of 2004, BI no longer enjoyed such discretion.

On the other hand, however, Act No. 3 of 2004 also contained a number of advantages. Several arguments were proposed in the monetary economics literature that emphasized the benefits of the government setting the inflation target. The main argument was supported by the institutional context of the economic authority under a democratic system, as well as the pragmatic consideration that an inflation target set by the government would provide additional credibility for the central bank's monetary policy. Mishkin (2000), for instance, recommended that the inflation target should be determined by an elected government because it allowed the public some oversight of government actions and, therefore, ensured accountability.¹⁶ In the context of democratic accountability, it is reasonable for the government to set the inflation target to guide the national economy because they are democratically charged with a public mandate.

Debelle and Fischer (1994) further revealed some other advantages if the government were to set the inflation target, with or without consulting the central bank. In general, the government's inflation target would be considered to

¹⁶The fifth guiding principle, that the public must be able to exercise control over government actions and that policymakers must be accountable, so basic to democracy, strongly suggests that the goals of monetary policy should be set by the elected government. In other words, the central bank should not be goal dependent. The corollary of this view is that the institutional commitment to price stability should come from the government in the form of an explicit, legislated mandate for the central bank to pursue price stability as its overriding, long-run goal.

represent the preferences of the public, while the inflation target set by the central bank would be considered overly tight.¹⁷ Furthermore, an inflation target set by the government would also bring another important “advantage” to the central bank, namely greater alignment of fiscal policy with monetary policy. If the government sets the inflation target as a mandate that must be achieved by the government, it would be harder to pressure the central bank into instituting expansive short-run monetary policy because it would be inconsistent with the achievement of price stability. Besides, “a government commitment to price stability also is a commitment to making monetary policy dominant over fiscal policy, ensuring a better alignment of fiscal policy with monetary policy.”

What must be avoided if the inflation target is set by the government is the possibility of a bias toward populist policy. In other words, a loose inflation target would be set to accommodate short-term economic growth, thus making it difficult to achieve low and stable inflation in the long run. In the case of developing countries, such concerns could be stoked further if government expertise – or experience at least – in terms of determining the inflation target is considered weak. If the government casually sets the inflation target for the monetary authority to strive toward, the economy could encounter unnecessary fluctuations that are counterproductive in the long term. Moreover, if the government applied a short-term timeframe when setting the inflation target, the change in policy stance required to achieve the inflation target would have to be of greater magnitude, which would amplify the economic gyrations further.

With the short-run viewpoint of the government, an inflation target set by the government could exceed the optimal long-run target socially. To reduce this difference, the inflation target should be set in conjunction or after consultation with the central bank as the most ideal situation. Of the 19 countries implementing ITF, 10 central banks set the inflation target in conjunction or after consultation with the government (Mishkin & Schmidt-Hebbel, 2001).

In the context of Indonesia, what became a challenge with the amendment, Act No. 3 of 2004, was how to align the views of BI and the Government regarding the importance of achieving low inflation in the medium-long term. The inflation rate is a crucial element of raising national economic competitiveness. What has the government achieved so far? The government successfully reduced the fiscal deficit to close to zero in 2006 and has maintained a surplus ever since, which is an important factor in the efforts to lower inflation. A long-term inflation target also helps the formation of inflation. In this regard, BI must be able to furnish the government with a comprehensive picture to announce a long-term inflation target, which has never been done previously.

Aligning the viewpoints of BI and the Government in terms of controlling inflation is a strength of the ITF. Through ITF, the government will have ownership in the program to control inflation. Congruently, government commitment to achieve the inflation target is also expected to increase. Considering the causes

¹⁷The democratic legitimacy of the government to set the inflation target would be strengthened if the leadership of the central bank was selected by parliament and not through other political processes.

of inflation beyond the control of the BI can be resolved through government policy, achieving the inflation target will be enhanced.

SUN Purchases on the Primary Market for Open Market Operations

Monetary policy effectiveness is determined by the monetary instruments available to the central bank. The more instruments available, the easier it is for the central bank to control monetary policy in order to achieve price stability. In that context, prior to the amendment of Act No. 23 of 1999, BI was only permitted to purchase securities on the secondary market for monetary operations. After the amendment, however, pursuant to Article 55 (4), BI may purchase short-term securities (up to one year) on the primary market for monetary operations. In addition, the amendment also paved the way for BI to purchase government debt securities (SUN) on the secondary market as an additional alternative monetary instrument in the future.

There are a number of advantages to BI purchasing SUN on the secondary market. *First*, purchasing SUN will boost its stock as a monetary instrument, as a replacement for Bank Indonesia Certificates (SBI). For use as a monetary instrument, BI must be able to trade SUN on the secondary market. *Second*, purchasing SUN will mitigate the risk of losses incurred by the central bank because the cost is passed on to the government through the state budget (currently, the SUN coupon is larger than the discounted SBI). *Third*, SUN purchases will expedite domestic secondary market development and, hopefully, create a benchmark yield for the financial markets. Nonetheless, Indonesia still faces limitations considering SUN supply depends on the government's commitment and financial posture.

SUN Purchases for Emergency Financing Facilities

Efforts to improve financial system stability also play an important role in terms of supporting monetary stability. Fundamentally, monetary policy is transmitted through the banking and financial sectors, therefore, financial system stability will determine the effectiveness of monetary policy implementation by the central bank. Consequently, if a bank should encounter financial difficulties, which is expected to have a systemic impact and could therefore trigger a crisis and threaten the financial system, pursuant to Act No. 3 of 2004, Article 11 (4), BI may provide an Emergency Funding Facility (FPD), funded by the state budget, something that was not regulated previously. Therefore, the government shall issue new SUN as required for emergency financing in accordance with the financial safety net to ensure the integrity of the overall financial system.¹⁸ To that end,

¹⁸The provision of emergency financing facilities differs from the short-term funding facility offered by Bank Indonesia as lender of last resort pursuant to Article 11 (1) of Act No. 3 of 2004. The emergency financing facility is provided to banks experiencing difficulties that threaten the integrity of the overall financial system, while the short-term funding facility is provided to banks to overcome short-term liquidity shortfalls that would not affect financial system stability in general.

BI is permitted to purchase SUN on the primary market, the proceeds of which can be used by the government to help banks experiencing financial difficulties.

An important aspect that demands attention is how SUN purchases on the primary market by BI could potentially create fiscal dominance, which would complicate policy management and undermine achievement of the inflation target. Furthermore, the time lag between BI mobilizing the funds and the Government issuing SUN also demands attention. BI disbursing the funds to the bank before receiving the SUN from the Government¹⁹ would raise the cost of open market operations. Therefore, a faster approval process by the People's Representative Council would be better for BI and for monetary control.

Increasing Transparency and Accountability

In terms of monetary policy management, emphasizing the importance of transparency and accountability has gained traction. Prior to the amendment, at the beginning of each fiscal year BI would be required to disclose information to the public through the mass media containing: (1) an evaluation of monetary policy implementation in the previous year and (2) the monetary policy plan and targets for the upcoming year, paying due consideration to the inflation target as well as economic and financial developments.

After the amendment, however, pursuant to Act No. 3 of 2004, Article 58, BI now reports its responsibilities in terms of task implementation to the People's Representative Council, which previously embodied the public in a broad scope. The report submitted to the People's Representative Council (and also the Government) at the beginning of each year contains: (1) task implementation and authority in the previous year and (2) the policy plan, targets and measures for the upcoming year, paying due consideration to the inflation trend as well as economic and financial developments. Furthermore, BI also submits a report on task implementation and authority to the People's Representative Council and Government on a quarterly basis. Both reports (annual and quarterly) are used for the annual performance appraisals of the Board of Governors and BI.²⁰

The report on task implementation and the policy plan, setting the targets and implementation measures is not only delivered to the People's Representative Council and Government, but also to the public (including the markets). This is important for the public to know the direction of policies to be taken as well as to discipline BI in terms of achieving the inflation target. Consequently, BI credibility is expected to improve.

¹⁹SUN issuances require time because People's Representative Council approval must first be sought.

²⁰The demand for greater accountability increased with the formation of the Supervisory Board, which has access to Bank Indonesia's budget, therefore the Board may also wish to question policy aspects. The duties of the Supervisory Board include: (1) reviewing the annual financial statements of Bank Indonesia; (2) reviewing the operational and investment budgets of Bank Indonesia; and (3) reviewing the decision-making procedures for operational activity that falls beyond the remit of monetary policy and asset management at Bank Indonesia.

Appendix 9.2: Bank Indonesia Macroeconomic Models

In-depth understanding of the overall macroeconomy is crucial when formulating monetary policy. The ability of macroeconomic modeling will determine the accuracy of macroeconomic projections, including inflation, economic growth and credit growth, as well as the monetary policy response required in line with the inflation target. In this regard, BI has developed structural macroeconomic models to analyze and project future economic and inflation trends. One of the large-scale structural macroeconomic models developed by BI is known as MODBI, which currently consists of five blocks/sectors (aggregate demand and supply (GDP), price, monetary, fiscal and external) and 87 equations, comprised of 57 behavioral equations and 30 identity equations. MODBI, which was first developed at the beginning of the 1990s, is often used to analyze different macroeconomic variables and their projections in the medium-long run (five-year horizon).

The passing of Act No. 23 of 1999, which laid the groundwork for ITF implementation in Indonesia, created greater demand to develop larger macroeconomic models, primarily to support the implementation of ITF-based monetary policy. Larger macroeconomic models were needed for inflation and economic analysis and projections as the materials underlying monetary policymaking at the Board of Governors meeting (RDG). Consequently, BI began to develop and strengthen several macroeconomic models for short-term analysis and projections, the first of which was the SSM.²¹ As an inflation projection model, the basic model reflects an ITF-based monetary policy framework. The model consists of seven equations, namely the short-term Phillips curve, aggregate demand, output gap, demand for money function, traded inflation, exchange rate, and Taylor rule to determine the operational target.²²

For quarterly macroeconomic analysis and projections, BI also developed a medium-scale macroeconomic structural model known as SOFIE. The SOFIE model consists of six blocks/sectors (demand/GDP, price, monetary, fiscal, banking, and external), with 279 equations comprised of 170 behavioral equations and 79 identity equations. The basic construction of the model also reflects the theoretical and practical implementation of ITF. The model also encompasses the short-term Phillips curve, aggregate demand, aggregate supply, the monetary sector, interest rate, and exchange rate behavior as well as a type of Taylor rule. The SOFIE model is more disaggregated than the SSM model because the goal is to illustrate the behavior of several macroeconomic variables consistent with the inflation projections, as well as use an error-correction technique as the econometric approach. The model is used to help researchers analyze and project

²¹The preliminary version of this model is presented in Appendix 9.2, The Inflation Forecasting Model, of the paper by Alamsyah et al. (2001).

²²For exchange rate analysis and projections, a simple model was developed known as the Behavioral Model of Effective Exchange Rate (BEER). The model consists of several equations to illustrate exchange rate behavior as influenced by the terms of trade and risk premium (measured using the spread between Government Bond Yields issued on the New York market with US Treasury bills).

macroeconomic variables in the short term (two years) on a quarterly basis, which serves as materials for the Board of Governors meetings. The model is currently under further development to detect clearer behavioral trends concerning monetary policy transmission and determining the interest rate operational target in line with the Taylor rule. In 2010, BI strengthened SOFIE through integration with the SSM model.

For simulations, a dynamic stochastic macroeconomic model was developed known as Dynamic General Equilibrium Model of Bank Indonesia (GEMBI). In a departure from the structural models, GEMBI uses a modern approach based on dynamic stochastic macroeconomic theory. By using certain stylized facts, the model can be calibrated to produce long-term projections of macroeconomic variables, including economic growth, inflation, exchange rates, and so on. The goal is more simulation-based to illustrate the medium-long term behavior of various macroeconomic variables in response to a number of alternative monetary policy scenarios. Nevertheless, using the GEMBI model on too large of a scale to accommodate the economic system in its entirety would ultimately encounter numerous constraints/challenges, particularly in terms of recalibrating the behavioral parameters (parameterization) and the linkages between certain variables without a firm theoretical or empirical footing. The GEMBI model was eventually simplified and is currently no longer in use.

During the early days of ITF implementation, the four macroeconomic models mentioned above were mutually complementary in terms of analyzing and projecting inflation along with other macroeconomic variables. The SOFIE and MODBI models produced quarterly and annual projections, which were subsequently used as the initial values in the GEMBI model to simulate the dynamic long-term behavior of key macroeconomic variables in the medium-long term according to several alternative monetary policy scenarios. Meanwhile, the SSM model was more commonly used to analyze the behavior of inflation specifically.

To simultaneously strengthen the projection and policy analysis system consistent with post-global crisis conditions, BI adopted the FPAS approach recommended by the IMF in 2010. Consequently, BI developed a dynamic “semi” general equilibrium macroeconomic model for policy simulations, namely the ARIMBI model (Aggregate Rational Inflation – Targeting Model for Bank Indonesia). The ARIMBI model was developed based on New Keynesian theory and consists of six blocks/sectors, namely aggregate demand, aggregate supply, price, monetary, financial, and external.

Over the past three years, congruent with the increasingly complex policy challenges, the ARIMBI model has been strengthened as the main modeling system in the FPAS, using other existing models, such as SOFIE and various *ad hoc* models to expand the scope of policy analysis through the integration of external or balance of payments (BOP) variables (including the current account as well as capital and financial account), the financial sector (including credit and default risk) and the endogenous behavior of risk variables. Consequently, it is now possible to comprehensively simulate BI’s policy mix using monetary instruments (interest rate) and macroprudential instruments (RR and LTV).